

# Millennium Cohort Study

First, Second, Third and Fourth Surveys

A Guide to the Datasets (Seventh Edition)

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## **Introduction**

## **Acknowledgements**

We are grateful for the entirely voluntary co-operation of the children who form the Millennium Birth Cohort and their mothers, fathers and other family members.

We wish to acknowledge the initiation and funding of the survey by the Economic and Social Research Council, and also thank the Office for National Statistics, who lead the consortium of Government Departments (Work and Pensions; Children, Schools and Families; and Health) and the governments of Wales, Scotland and Northern Ireland.

The National Evaluation of the Children's Fund offered the opportunity to enhance the second survey by funding data collection on older siblings and neighbourhood observations.

The work could not have been accomplished without the involvement of a large number of advisers drawn from academe, policy-makers and funders, who we consulted throughout the design of the surveys. We offer particular thanks to those who served on the MCS Advisory Committee, the MCS Advisory Groups and the British Birth Cohorts Scientific Committee.

The surveys also benefited from the enthusiasm and expertise of:

Research programmers, researchers, field-managers and interviewers at the National Centre for Social Research (NatCen) (sweeps 1, 3 and 4) and National Opinion Polls (NOP) (sweep 2).

The staff of the Information Centre at Newcastle HM Revenue and Customs (formerly the Department of Social Security).

The members of the Centre for Longitudinal Studies (CLS) tracing team, IT team, database, survey and research teams.

The MCS internal team who contributed to the deposit and documentation of MCS1, 2, 3 and 4, as listed below:

|                   |                                                        |
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| Anitha George     | Research Officer                                       |
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| Dr Elizabeth Jones  | Research Officer                   |
| Prof. Heather Joshi | Study Director (2000-2011)         |
| Sosthenes Ketende   | Research Officer                   |
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| Prof. John McDonald | Director of Methodology            |
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| Dr Alice Sullivan   | Research Fellow                    |
| Mina Thompson       | Administrator / Personal Assistant |
| Kelly Ward          | Research Officer                   |

We are also deeply grateful to the late Prof. Neville Butler who made an invaluable contribution to the Millennium Cohort Study until his final year (2007).

## Preface

This document describes the MCS data sweeps 1-4.

**Table 1: MCS Deposit – Elements**

| Title                                          | Format |
|------------------------------------------------|--------|
| <b>MCS4 Data:</b>                              |        |
| Fourth Survey Main and Partner Dataset         | SPSS   |
| Fourth Survey Main and Partner Dataset         | STATA  |
| Fourth Survey Household Grid Dataset           | SPSS   |
| Fourth Survey Household Grid Dataset           | STATA  |
| Fourth Survey Child Assessment Dataset         | SPSS   |
| Fourth Survey Child Assessment Dataset         | STATA  |
| Fourth Survey Our Adventures (Wales) Dataset   | SPSS   |
| Fourth Survey Our Adventures (Wales) Dataset   | STATA  |
| Fourth Survey Child Measurement Dataset        | SPSS   |
| Fourth Survey Child Measurement Dataset        | STATA  |
| Fourth Survey Teacher Survey Dataset           | SPSS   |
| Fourth Survey Teacher Survey Dataset           | STATA  |
| Fourth Survey Child Self Completion Dataset    | SPSS   |
| Fourth Survey Child Self Completion Dataset    | STATA  |
| Fourth Survey Main Bracketed Income Dataset    | SPSS   |
| Fourth Survey Main Bracketed Income Dataset    | STATA  |
| Fourth Survey Partner Bracketed Income Dataset | SPSS   |
| Fourth Survey Partner Bracketed Income Dataset | STATA  |

| <b>Title</b>                                  | <b>Format</b> |
|-----------------------------------------------|---------------|
| Fourth Survey Proxy Bracketed Income Dataset  | SPSS          |
| Fourth Survey Proxy Bracketed Income Dataset  | STATA         |
| Fourth Survey Derived Variables Dataset       | SPSS          |
| Fourth Survey Derived Variables Dataset       | STATA         |
| <b>MCS3 Data:</b>                             |               |
| Third Survey Main and Partner Dataset         | SPSS          |
| Third Survey Main and Partner Dataset         | STATA         |
| Third Survey Household Grid Dataset           | SPSS          |
| Third Survey Household Grid Dataset           | STATA         |
| Third Survey Child Assessment Dataset         | SPSS          |
| Third Survey Child Assessment Dataset         | STATA         |
| Third Survey Older Siblings Dataset           | SPSS          |
| Third Survey Older Siblings Dataset           | STATA         |
| Third Survey Foundation Stage Profile Dataset | SPSS          |
| Third Survey Foundation Stage Profile Dataset | STATA         |
| Third Survey Teacher Survey Dataset           | SPSS          |
| Third Survey Teacher Survey Dataset           | STATA         |
| Third Survey Geographically Linked Dataset    | SPSS          |
| Third Survey Geographically Linked Dataset    | STATA         |
| Third Survey Derived Variables Dataset        | SPSS          |
| Third Survey Derived Variables Dataset        | STATA         |
| <b>MCS2 Data:</b>                             |               |
| Second Survey Main and Partner Dataset        | SPSS          |

| <b>Title</b>                                     | <b>Format</b> |
|--------------------------------------------------|---------------|
| Second Survey Main and Partner Dataset           | STATA         |
| Second Survey Household Grid Dataset             | SPSS          |
| Second Survey Household Grid Dataset             | STATA         |
| Second Survey Child Assessment Dataset           | SPSS          |
| Second Survey Child Assessment Dataset           | STATA         |
| Second Survey Child Measurement Dataset          | SPSS          |
| Second Survey Child Measurement Dataset          | STATA         |
| Second Survey Older Siblings Dataset             | SPSS          |
| Second Survey Older Siblings Dataset             | STATA         |
| Second Survey Neighbourhood Observations Dataset | SPSS          |
| Second Survey Neighbourhood Observations Dataset | STATA         |
| Second Survey Geographically Linked Dataset      | SPSS          |
| Second Survey Geographically Linked Dataset      | STATA         |
| Second Survey Derived Variables Dataset          | SPSS          |
| Second Survey Derived Variables Dataset          | STATA         |
| <b>MCS1 Data:</b>                                |               |
| First Survey Parent Interview Dataset            | SPSS          |
| First Survey Parent Interview Dataset            | STATA         |
| First Survey Household Grid Dataset              | SPSS          |
| First Survey Household Grid Dataset              | STATA         |
| First Survey Geographically Linked Dataset       | SPSS          |
| First Survey Geographically Linked Dataset       | STATA         |
| First Survey Derived Variables Dataset           | SPSS          |

| <b>Title</b>                                               | <b>Format</b> |
|------------------------------------------------------------|---------------|
| First Survey Derived Variables Dataset                     | STATA         |
| <b>MCS1, MCS2, MCS3 and MCS4 Data:</b>                     |               |
| Longitudinal Family Level Dataset                          | SPSS          |
| Longitudinal Family Level Dataset                          | STATA         |
| <b>Documentation:</b>                                      |               |
| First Survey Questionnaire                                 | PDF           |
| First Survey Derived Variables                             | PDF           |
| First Survey: Technical Report on Fieldwork (NatCen) 2004  | PDF           |
| Second Survey Questionnaire                                | PDF           |
| Second Survey Derived Variables                            | PDF           |
| Second Survey: Technical Report on Fieldwork (NOP) 2006    | PDF           |
| Second Survey: Technical Report on Response (2006)         | PDF           |
| Third Survey Questionnaire                                 | PDF           |
| Third Survey Derived Variables                             | PDF           |
| Third Survey Guide to the School Assessment Datasets       | PDF           |
| Third Survey Technical Report on Fieldwork (NatCen) 2007   | PDF           |
| Third Survey Teacher Survey Technical Report               | PDF           |
| Fourth Survey Questionnaire                                | PDF           |
| Fourth Survey Derived Variables                            | PDF           |
| Fourth Survey Technical Report on Fieldwork (NatCen) 2010  | PDF           |
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## **Overview of document**

1. Part One provides an introduction to the study, its history and objectives.
2. Part Two explains the sample.
3. Part Three provides information about the survey development.
4. Part Four discusses the survey content.
5. Part Five explains the fieldwork.
6. Part Six discusses the achieved sample.
7. Part Seven provides detailed information about the data.
8. Part Eight discusses the ethical considerations.

## **PART ONE: BACKGROUND**

### **1. Introduction**

The Millennium Cohort Study (MCS) is the fourth of Britain's world-renowned national longitudinal birth cohort studies. Each follows a large sample of individuals born over a limited period of time through the course of their lives, charting the effects of events and circumstances in early life on outcomes and achievements later on. They show how histories of health, wealth, education, family and employment are interwoven for individuals and vary between them.

The MCS offers large-scale information about the New Century's children, and the families who are bringing them up, for the four countries of the United Kingdom. Its first sweep, carried out during 2001-2002, laid the foundations for a major new longitudinal research resource, taking a new "year long" cohort of around 19,000 babies. In England and Wales they were born over the 12 months starting in September 2000, and over 13½ months from late November 2000 in Scotland and Northern Ireland. Information was collected from parents when the babies were aged 9 months. The sample design allowed for disproportionate representation of families living in areas of child poverty, and in areas of England with high ethnic minority populations. The first survey recorded the circumstances of pregnancy and birth, as well as those of the all-important early months of life, and the social and economic background of the families into which the children have been born. These multidisciplinary baseline data reveal the diversity of starting points from which the 'Children of the New Century' have set out. The second survey data add a longitudinal element to the dataset and will allow researchers for the first time to chart the changing circumstances of children and their families, and relate age 3 outcomes to earlier circumstances and experiences. The third survey visited the families when the children reached age 5 and had started primary school, and the fourth returned to the families when the children were age 7.

This document reviews the background to the data collection for the first four surveys of the Millennium Cohort, from its commissioning by the Economic and Social

Research Council (ESRC) in early 2000 to the latest deposit in the ESRC Data Archive in March 2010. It covers the development and design of the project, the pilot surveys and the main fieldwork in all four UK countries, the subsequent coding and data preparation, along with an explanation of the datasets.

## 2. History

In 1999, the British government decided to mark the coming millennium with a new birth cohort study. The ESRC commissioned a feasibility study which resulted in an invitation to tender for the Principal Investigator role in a new study and to submit expressions of interest to carry out its fieldwork. These invitations were issued when the new millennium was already 2 months old on 23 February 2000.

The original ESRC funding was for a study which in some ways resembled that of the previous cohorts, but which departed from the previous designs in the following respects:

- The sample was to include children born within a full 12 month rather than one week period, to include births in all seasons.
- The sampled birth dates were to include births in the year 2000, even though it might not be possible to start with births in January.
- The geographical coverage of the study was to cover the whole of the UK.
- The content was to emphasise the social and economic circumstances of the families.
- The target sample size was to be 15,000.
- The target interview length was constrained by a fieldwork budget of £1.7m.
- The academic contractors were to consider sample designs that allowed for the over-representation of ethnic and national minorities.
- The first survey was to be carried out when the children were at about the same age in months, as near as possible to 6 months old.

A further contrast with the earlier birth cohort surveys concerns the mode of data collection. Most of these conditions pointed towards fieldwork being carried out by a professional fieldwork agency, with experience of probability sampling and social scientific questionnaires, administered by computer, rather than the health visitors who carried out the interviews shortly after the births of the previous cohorts. The limited number of professional interviewers, and the consequent need to spread interviewing over time, also favoured a wider span of birth dates than a single week.

## Objectives

The objectives of the first MCS survey were laid down in the Centre for Longitudinal Studies' proposal to the ESRC in March 2000:

- To chart the initial conditions of social, economic and health advantages and disadvantages facing new children in the new century, capturing information that the research community of the future will require.

- To provide a basis for comparing patterns of development with the preceding cohorts.
- To collect information on previously neglected topics, such as fathers' involvement in the children's care and development.
- To focus on the children's parents as the most immediate elements of the child's 'background', charting their experience as mothers and fathers of this year's babies, to record how they (and any other children in the family) are adapting to the newcomer, and what their aspirations for her/his future may be.
- To emphasise intergenerational links including those back to the parents' own childhood.
- To investigate the wider social ecology of the family, including social networks, civic engagement, community facilities and services; splicing in geo-coded data when available.

The later addition of co-funding from government departments broadened the objectives of the study to include boosted samples in Wales, Scotland, Northern Ireland and disadvantaged areas of England. It also made possible linkage to administrative records and other data enhancements as well as longer interviews and analysis opportunities.

With the second survey at age 3, MCS was established as a longitudinal multipurpose dataset, tracking the diverse lives of the 'Children of the New Century' out of infancy and into early childhood and laying the foundation for future follow-ups.

Specific objectives included:

- To assess key aspects of the child's physical, cognitive, social and emotional development.
- To chart continuity and change since the age of 9 months in the child's family and wider environment.
- To collect data which would foster comparisons with other cohort studies, in the UK and elsewhere.
- To provide prospective data for comparative purposes in the national evaluation of Sure Start.
- To 'recapture' information which, due to constraints on interview time, was not collected at the first sweep.
- To collect information from the cohort member's older siblings, for the National Evaluation of the Children's Fund.
- To collect information from 'new families' who had moved into MCS survey areas too late to be included by the Department for Work and Pensions (DWP) in the original sample.

The objectives of the third and fourth surveys were:

- To continue tracking the child's physical, cognitive and behavioural development.
- To chart continuity and change in the child's family circumstances and physical environment up to age 7.
- To record the child's transition to primary school and their experience of the first years at school.
- To track their previous experience of early education and day-care, along with current out-of-school care arrangements.
- To record the progress of their older siblings.
- To re-contact families who had participated in at least one of the earlier surveys, but who may not have participated in all sweeps.
- To ask the children directly about their thoughts and experiences at age 7.

## PART TWO: SAMPLING

*For a more comprehensive discussion of the sampling procedure used, please refer to the MCS Technical Report on Sampling (4th Edition) (Plewis 2007).*

### 1. Sample Design of MCS1

The sample is clustered geographically and is disproportionately stratified to over-represent areas with high proportions of: ethnic minorities in England, areas of high child poverty and the three smaller countries of the UK. The geography of electoral wards was used as a sampling frame. The child poverty component of the Index of Deprivation 2000, using 1998 ward boundaries in England, provided an indicator of wards with high proportions of children in families receiving means-tested benefits.

The sample for the first sweep included babies born between 1 September 2000 and 31 August 2001 in England and Wales, who would form an academic year cohort. In Scotland and Northern Ireland the start date of the birthdays was delayed to 23 November 2000 in order to avoid an overlap with an infant feeding survey being carried out in September and October. In the event, the sampled cohort was extended to 59 weeks of births to make up for a shortfall in numbers which became apparent during fieldwork. The last eligible birth date in these countries was 11 January 2002.

Children with sample birth dates were eligible for the survey if they were living in one of approximately 400 electoral wards across the whole of the UK when they were 9 months old.

The disproportionately stratified design of the survey was to ensure adequate representation of:

- All UK countries.

- Areas in England with higher minority ethnic populations in 1991 (where at least 30 per cent of the population was Black or Asian).
- Disadvantaged areas (apart from the ethnic minority wards above, drawn from the poorest 25 per cent of wards based on the Child Poverty Index).

## **2. Sampling Procedures for MCS1**

### **The Use of Child Benefit Records**

Cohort children were found using Child Benefit records provided by the Department of Social Security (subsequently Department for Work and Pensions and then HM Revenue and Customs). The alternative might have been to use birth registration records, but this was not chosen because of concern about non-response to the postal opt-in required with this method. The DWP asked people to opt out rather than in, which is more inclusive of marginal and low literacy respondents.

The DSS/DWP withdrew 'sensitive cases' from the issued sample. These included parents in families where children had died or been taken into care, or where there was an investigation into benefit fraud. But this represents less than 3 per cent of cases. Another reason for exclusion was to avoid families who had also taken part in the Survey of Low Income Families (now the Families and Children Survey, FACS). This affected only 40 cases (out of over 27,000).

### **Recruitment via Health Visitors**

Because the Child Benefit records would not reveal all families who had moved into the sample wards as the child approached 9 months of age, we attempted to find in-movers with the help of Health Visitors. These local community health professionals were expected to be aware of families transferring into these areas. They were asked to see if families moving into survey wards were willing to be recruited.

Health Visitors reported 220 cohort in-moving families with children over 6 months of age, however only 56 had not also been found by DSS/DWP. We also distributed a postal survey on local services to Health Visitors in the sample areas. There were several problems which may explain the rather disappointing result of this exercise. First, helping with the survey was not part of the Health Visitors' already demanding normal duties. Second, Health Visitors' caseloads do not neatly coincide with electoral wards. Third, there is no central list of Health Visitors for easy contact. Tracking down the relevant Health Visitors or Supervisors proved time consuming, particularly in view of the reorganisation of the Community Care Trusts into Primary Care Trusts during 2001. The exercise benefited vitally from the help and experience of the late Neville Butler and the financial support of the International Centre for Child Studies (ICCS). It was aided by additional ESRC funds for maps. We eventually distributed 3,330 packs to Health Visitors with the help of 306 supervisors.

### **3. Timing of Cohort Births and Age at Interview**

As described above, it was originally planned that the Study would survey children who had been born during 2000 around 6 months of age. Given the need to put the fieldwork contract out to tender, this timetable proved unsustainable. The resulting shifting of birth dates meant that the age at which the first interview should take place was settled at 9 months with fieldwork starting in June 2001, postponed from May 2001, in view of the census and the anticipated General Election. This meant that projected deposit of the dataset was also postponed one month to the end of March 2003.

#### **Intensification and Extension of the Sample**

At their meeting in September 2001 the Advisory Committee expressed concern over a projected shortfall in sample numbers. This was occasioned at least partly by actual births falling below the anticipated level. The view of advisers and most government departments was to maintain or raise response rates rather than extend the sample. To this end, the National Centre for Social Research (NatCen) re-issued some extra cases in difficult areas of Great Britain.

The DWP also supplemented their operations to look for families moving into sample addresses when the babies were between 7 and 8 months old. Such families were offered the opportunity to opt out when the children were 9 months old. If they did not, they were issued for interview at 10 months of age. This took place from November 2001 onwards covering children born after 16 March 2001. This exercise found a total of 518 families with new addresses in the survey areas, but only 293 could be issued to field, as these cases involved a disproportionate number of cases excluded by the DWP as 'sensitive'.

The Executives of Scotland and Northern Ireland expressed concern at the May 2002 Advisory Committee about the expected shortfall in sample numbers in these countries. Instead of the target of 2,500 children in Scotland and 2,000 in Northern Ireland, our projections for 53 weeks of sampling were for 2,173 and 1,659 children in the achieved samples, respectively. The DWP offered to do one more wave of sampling, covering children living in Scotland and Northern Ireland who were born in the 56 weeks starting 1 December 2001 and finishing 11 January 2002.

The extension of birth dates by 6 weeks in samples in Scottish and Northern Irish wards was expected, on simplified assumptions, to increase the total of Scottish children towards 2,419 and Northern Irish children towards 1,847. It would have taken at least another 2 weeks of births and extra expenditure to reach the original targets of 2,500 and 2,000 children, respectively. This extension delayed the delivery of the dataset by a further 6 weeks and increased the number of months of fieldwork that would be required to interview the whole cohort at the same exact age in future sweeps.

## **4. Sampling at MCS2**

The survey attempted to follow all the 18,553 families who took part in MCS1 where the child was still alive and living in the UK. It also attempted to make contact with another 1,389 'New Families' in England who appeared to have been living in sample wards at the time of MCS1, but whose addresses reached DWP records too late to be included in the first survey.

## **5. Sampling at MCS3**

The sample issued for MCS3 comprised all those who had responded to the survey at least once, i.e. to MCS1 or the 631 additional cases who had responded to MCS2 in the New Families, less those known to have become ineligible through the death or emigration of the cohort child, and also less those deemed to have made a permanent refusal (also excluding the one case in the original MCS1 total of 18,533 subsequently discovered to have been invalid). Thus nearly all non-respondents to the second survey who had been interviewed in the first survey were given the opportunity to rejoin the survey at age 5. The non-respondents to the New Families sample were not reissued.

## **6. Sampling at MCS4**

There were 19,244 families potentially eligible for inclusion in the issued sample at MCS4. However, 2,213 families were not issued to the field due to ineligibility due to death or emigration (n=362), permanent refusal (n=1,705), permanent untraceability (n=136) and sensitive family circumstances (n=10). Their outcomes were known and recorded before the start of the fieldwork. The families not issued due to sensitive family circumstances are recorded as 'unproductive other'. Therefore, the MCS4 issued sample was 17,031 (19,244 less 2,213).

## **PART THREE: SURVEY DEVELOPMENT**

*For a more comprehensive discussion of survey development, please refer to the MCS1 Technical Report on Fieldwork (NatCen 2004) or the MCS2 Technical Report on Fieldwork (NOP 2006) or the MCS3 Technical Report on Fieldwork (NatCen 2007) or the MCS4 Technical Report on Fieldwork (NatCen 2010).*

### **1. Development and Piloting of MCS1**

The questionnaire was developed by the CLS team with input from 55 potential users of the dataset from academe and government departments who attended a consultation meeting on 11 October 2000. An instrument was initially piloted in January 2001 and redeveloped into a shorter version for the second Dress Rehearsal Pilot in April 2001.

#### **First Pilot**

The first pilot in January 2001 was conducted as a paper interview and computer-aided self-completion interview (CASI) in order to assess the timing of the instrument before the major work to convert the interview schedule into computer-aided personal interview (CAPI) format. The sample size was boosted from 30 to 60 thanks to the ONS consortium funding. Further details are in the NatCen Technical Report on Fieldwork (NatCen 2004).

The overall response by parents to the pilot survey was positive, and interviewers reported that the gift vouchers included in this pilot did not seem to be necessary to persuade people to talk about their babies. However, the instrument was substantially over length, averaging over 2 hours, the target being 105 minutes. Reductions to the questionnaire were made in consultation with collaborators, and using the results of factor analysis of the CASI responses on attitude and behaviour.

#### **Dress Rehearsal Pilot**

The second pilot took place during April 2001 and was fully computer-based (CAPI and CASI). As a 'dress rehearsal' for the main stage, all the contact and administrative processes were tested as well as the near final form of the survey instruments. Thirteen wards were selected for this pilot, including one in each of Wales and Scotland. The wards in England and Wales were chosen from those that were to be used in the main stage. As the Scottish wards had not yet been selected, a large deprived ward was purposively picked.

The DWP sampling route was tested with letters sent from the DWP at Newcastle to parents of babies born between 12 June and 22 July 2000 on the Child Benefit register in the chosen wards. The use of an advance letter sent by interviewers was also piloted.

In addition, Health Visitors (HVs) were approached in the 12 English and Welsh wards in order to pilot their contribution. Two HV supervisors declined to help, as we had not received Multi-centre Research Ethics Committee (MREC) approval at that time.

The main result of the pilot was that respondents and interviewers had, on the whole, enjoyed the experience. The interview and particularly the self-completion, however, were still too long, especially where translation was needed. Of the two experiments which were part of the Department of Health-financed feasibility studies – including one on transcribing data from the Personal Child Health Record ('Red Book') – was not sufficiently successful to take into main fieldwork; but the pilot consent for health record linkage went well.

### **Structure and Content of Final Instrument**

The main instrument was cut by 8 minutes and the partner's by 5 minutes. Refinements were made to individual items to improve their comprehension and the flow of the questions, routings were checked and answer codes revised. Checks were also imposed to improve the automatic edits written into the CAPI program to ensure the quality of data collected. A few changes were made to questions for use in Scotland and Northern Ireland – on qualifications, religion and ethnicity – to make them country-relevant.

The content of the sweep 1 instruments is summarised in Table 2. The module lettering reflects the order of each interview, with the self-completion inserted between interview questions on health and employment. The lettering of the modules appears in the CAPI document and in the labelling of variables in the dataset, described below.

## **2. Development and Piloting of MCS2**

The questionnaire was developed by the CLS team with input from a team of external MCS2 collaborators. The questionnaire development was discussed at a consultative meeting on 22 April 2002. An instrument was initially piloted in May 2003, and redeveloped for the second Dress Rehearsal Pilot in June 2003.

### **First Pilot**

The first pilot in May 2003 was carried out as CAPI and CASI interviews of around 30 families in order to establish the time taken to carry out the early drafts of the interview, self-completion and child assessments. It was also designed to identify other problems such as flow, question wording recall and filtering.

The interviewers reported few major problems with any of the survey instruments apart from the length. The main interview and self-completion was 20 minutes over

the target length of 70 minutes, with the self-completion taking nearly twice as long as budgeted.

Subsequent reductions were made to the instruments in consultation with collaborators.

### **Dress Rehearsal Pilot**

The dress rehearsal for the study took place in June 2003. All of the procedures planned for main-stage sampling and fieldwork were tested, including the taking of saliva samples from the children; home and neighbourhood observations; and the self-completion questionnaire for older siblings. The sample used for the MCS2 dress rehearsal consisted of respondents from the MCS1 dress rehearsal. Forty-eight families were interviewed in 13 wards in England, Wales and Scotland.

## **3. Development and Piloting of MCS3**

The questionnaire was developed by the CLS team with input from a team of external MCS3 collaborators. The questionnaire development was discussed at a consultative meeting in July 2004. An instrument was initially piloted in May 2005, and redeveloped for the second Dress Rehearsal Pilot in September/October 2005.

### **First Pilot**

The first pilot in May 2005 was carried out as CAPI and CASI interviews of 49 families in order to establish the time taken to carry out the early drafts of the interview, self-completion and child assessments and measurements. It was also designed to identify other problems such as flow, question wording recall and filtering. The sample was a quota sample recruited by interviewers.

The interviewers reported few major problems with any of the survey instruments apart from the length. Both the main and partner interviews and the child cognitive assessments were significantly longer than the budgeted length.

Subsequent reductions were made to the instruments in consultation with collaborators. One of the cognitive assessments (British Picture Vocabulary Scale) was dropped.

### **Dress Rehearsal Pilot**

The dress rehearsal for the study took place in September/October 2005. All of the procedures planned for main-stage sampling and fieldwork were tested.

The sample used for the MCS3 dress rehearsal consisted, in England, Scotland and Wales, of respondents from the MCS1 dress rehearsal and additional families sampled for MCS3. Northern Ireland was included in the dress rehearsal for the first time at MCS3; and all families in Northern Ireland were newly sampled for MCS3. The dress rehearsal sample was drawn from Child Benefit records in 14 wards of the UK and 109 families were interviewed.

The dress rehearsal also included a postal teacher survey in Wales, Scotland and Northern Ireland. This was in order to collect data equivalent to the Foundation Stage Profile in England (which was obtained through data linkage for consenting families).

#### **4. Development and Piloting of MCS4**

The data collection instruments were developed by the CLS team with input from a team of external MCS4 advisors. The development work started with a consultative conference in July 2008 at which the convenors of the MCS4 advisory groups presented their recommendations. A consultation on the first draft questionnaire for parents and cohort members took place in January/February 2007 and on the first draft teacher questionnaire in February/March 2007. The first pilot took place in March-June 2007 and the Dress Rehearsal Pilots for families and teachers in July-August 2007 and October-December 2007, respectively.

##### **First Pilot**

The first pilot in March/April 2007 was carried out as CAPI and CASI interviews of 38 families in order to establish the time taken to carry out the early drafts of the parent interviews and self-completion; child self-completion; and child assessments and measurements. It was also designed to identify other problems such as flow, question wording recall and filtering. Of the 38 interviewed families, 26 had previously been interviewed at MCS3 pilot 1, and 12 were newly recruited by interviewers. It was a quota sample and covered Great Britain only. The target sample of 50 families was not met. This was attributed to the fact that the fieldwork period was short (2 weeks) and took place in the run up to Easter.

Overall, the pilot was very successful, including the child self-completion and leaflet for the child which were both new for MCS4 and the physical activity monitoring (which was administered by the Institute of Child Health).

The two main exceptions were:

- Measurement of body-fat: The scales proved unsuitable due primarily to their weight and lack of portability.
- Assessment on numeracy (using BAS Number Skills): This proved problematic.

The teacher survey pilot took place in May-June 2007. Of the families who took part in the main pilot, 32 gave consent for their child's teacher to be approached. Of these, 23 returned a questionnaire after 2 reminders, giving a response rate of around 72 per cent.

### **Dress Rehearsal Pilot**

The dress rehearsal for the study took place in July/August 2007. All of the procedures planned for main-stage sampling and fieldwork were tested.

Following the findings of the first pilot, different weighing scales were used (NatCen 2007) for the body-fat measurement and a different assessment for numeracy adopted. This was NFER's Progress in Maths. The dress rehearsal also included another new element: consent to link to education and health records for cohort members and siblings, and to economic and health records for parents.

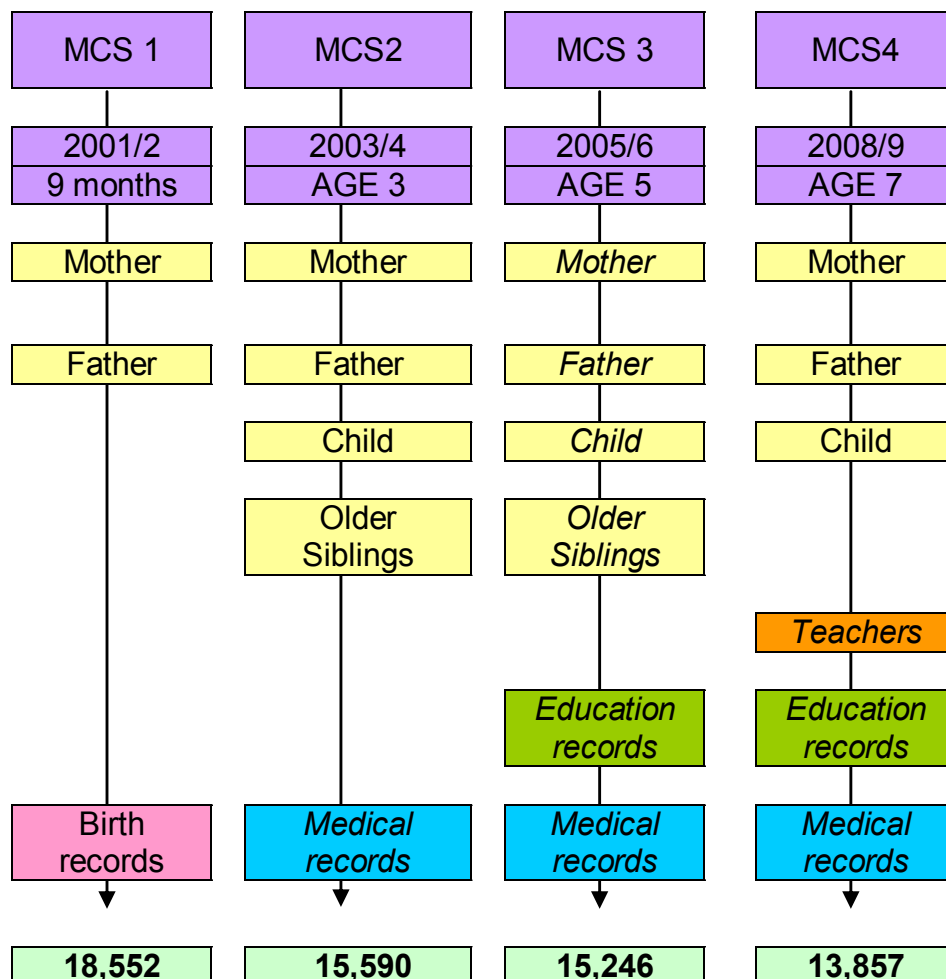
The longitudinal dress rehearsal sample, drawn from Child Benefit records in 14 wards of the UK, consisted, in Great Britain, of respondents sampled for the MCS1 dress rehearsal and additional families sampled for MCS3. In Northern Ireland it consisted of respondents sampled at MCS3, and 102 families were interviewed. This was in excess of the target sample of 100 families.

The dress rehearsal also included a postal teacher survey which was carried out in October-December 2007. In all, 84 teachers were approached (consenting families in the main dress rehearsal) and 38 questionnaires were returned after 2 reminders, giving a response rate of 45 per cent. This relatively low response rate was attributed to the fact that the teacher survey pilot took place several months after the main pilot, in a different academic year and in the run up to Christmas.

Overall, the pilot was very successful. The main issue was that the Progress in Maths assessment was too long. Following the dress rehearsal, a revised version of Progress in Maths was developed specifically for the study by NFER, which introduced routing in order to reduce the average time taken.

## PART FOUR: SURVEY CONTENT

The chart below shows the content of the MCS surveys at a glance.



Tables 2-5 below show in detail elements included at each sweep of the MCS. For more details of the content for all surveys, please refer to the respective questionnaires.

**Table 2: MCS1 – Summary of MCS1 Survey Elements**

| <b>Respondent</b>                                                                                                                                                                             | <b>Mode</b>     | <b>Summary of content</b>                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------------------------------------------------|
| <b>Mother/Father</b>                                                                                                                                                                          | Interview       | Household Module                                   |
| <b>Mother/Main*</b>                                                                                                                                                                           |                 | Module A: Non-resident parents                     |
|                                                                                                                                                                                               |                 | Module C: Pregnancy, labour and delivery           |
|                                                                                                                                                                                               |                 | Module D: Baby's health and development            |
|                                                                                                                                                                                               |                 | Module E: Childcare                                |
|                                                                                                                                                                                               |                 | Module F: Grandparents and friends                 |
|                                                                                                                                                                                               |                 | Module G: Parental health                          |
|                                                                                                                                                                                               | Self-completion | Module H:                                          |
|                                                                                                                                                                                               |                 | - Baby's temperament & behaviour                   |
|                                                                                                                                                                                               |                 | - Relationship with partner                        |
|                                                                                                                                                                                               |                 | - Previous relationships                           |
|                                                                                                                                                                                               |                 | - Domestic tasks                                   |
|                                                                                                                                                                                               |                 | - Previous pregnancies                             |
|                                                                                                                                                                                               |                 | - Mental health                                    |
|                                                                                                                                                                                               |                 | - Attitudes to relationships, parenting, work, etc |
|                                                                                                                                                                                               | Interview       | Module J: Employment, income, education            |
|                                                                                                                                                                                               |                 | Module K: Housing and local area                   |
|                                                                                                                                                                                               |                 | Module L: Interests                                |
| <b>Father/Partner*</b>                                                                                                                                                                        | Interview       | Module B: Father's involvement with baby           |
|                                                                                                                                                                                               |                 | Module C: Pregnancy, labour and delivery (where    |
|                                                                                                                                                                                               |                 | Module F: Grandparents and friends                 |
|                                                                                                                                                                                               |                 | Module G: Parental health                          |
|                                                                                                                                                                                               | Self-completion | Module H: Self-completion                          |
|                                                                                                                                                                                               |                 | - Baby's temperament & behaviour                   |
|                                                                                                                                                                                               |                 | - Relationship with partner                        |
|                                                                                                                                                                                               |                 | - Previous partners                                |
|                                                                                                                                                                                               |                 | - Previous children                                |
|                                                                                                                                                                                               |                 | - Mental health                                    |
|                                                                                                                                                                                               |                 | - Attitudes to marriage, parenting, work, etc      |
|                                                                                                                                                                                               | Interview       | Module J: Employment and education                 |
|                                                                                                                                                                                               |                 | Module L: Interests                                |
| <p><i>* In the majority of cases, the Main interview was undertaken by the mother/mother figure while the Partner interview was undertaken by the father/father figure. See Table 20.</i></p> |                 |                                                    |

**Table 3: MCS2 – Summary of MCS2 Survey Elements**

| <b>Respondent</b>                                                                                                                                                                                            | <b>Mode</b>       | <b>Summary of content</b>                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------|
| <b>Mother/Father</b>                                                                                                                                                                                         | Interview         | Household Module                                            |
| <b>Mother/Main*</b>                                                                                                                                                                                          |                   | Module A: Non-resident parents                              |
|                                                                                                                                                                                                              |                   | Module C: Pregnancy, labour and delivery                    |
|                                                                                                                                                                                                              |                   | Module D: Baby's health and development                     |
|                                                                                                                                                                                                              |                   | Module E: Childcare                                         |
|                                                                                                                                                                                                              |                   | Module F: Grandparents and friends                          |
|                                                                                                                                                                                                              |                   | Module G: Parental health                                   |
|                                                                                                                                                                                                              | Self-completion   | Module H:                                                   |
|                                                                                                                                                                                                              |                   | - Child's temperament & behaviour                           |
|                                                                                                                                                                                                              |                   | - Relationship with partner                                 |
|                                                                                                                                                                                                              |                   | - Previous relationships                                    |
|                                                                                                                                                                                                              |                   | - Domestic tasks                                            |
|                                                                                                                                                                                                              |                   | - Previous pregnancies                                      |
|                                                                                                                                                                                                              |                   | - Mental health                                             |
|                                                                                                                                                                                                              |                   | - Attitudes to relationships, parenting, work, etc          |
|                                                                                                                                                                                                              | Interview         | Module J: Employment, income, education                     |
|                                                                                                                                                                                                              |                   | Module K: Housing and local area                            |
|                                                                                                                                                                                                              |                   | Module L: Interests and time with baby                      |
|                                                                                                                                                                                                              |                   | Module N : Older siblings                                   |
| <b>Father/Partner*</b>                                                                                                                                                                                       | Interview         |                                                             |
|                                                                                                                                                                                                              |                   | Module B: Father's involvement with baby                    |
|                                                                                                                                                                                                              |                   | Module C: Pregnancy, labour and delivery (where applicable) |
|                                                                                                                                                                                                              |                   | Module F: Grandparents and friends                          |
|                                                                                                                                                                                                              |                   | Module G: Parent's health                                   |
|                                                                                                                                                                                                              | Self-completion   | Module H: Self-completion                                   |
|                                                                                                                                                                                                              |                   | - Baby's temperament & behaviour                            |
|                                                                                                                                                                                                              |                   | - Relationship with partner                                 |
|                                                                                                                                                                                                              |                   | - Previous partners                                         |
|                                                                                                                                                                                                              |                   | - Previous children                                         |
|                                                                                                                                                                                                              |                   | - Mental health                                             |
|                                                                                                                                                                                                              |                   | - Attitudes to marriage, parenting, work, etc               |
|                                                                                                                                                                                                              | Interview         | Module J: Employment and education                          |
|                                                                                                                                                                                                              |                   | Module L: Interests                                         |
| <b>Interviewer</b>                                                                                                                                                                                           | Observations      | Home environment                                            |
|                                                                                                                                                                                                              |                   | Neighbourhood                                               |
| <b>Child</b>                                                                                                                                                                                                 | Assessments       | BAS Naming Vocabulary                                       |
|                                                                                                                                                                                                              |                   | Bracken Basic Concept Scale                                 |
|                                                                                                                                                                                                              |                   | Height and weight                                           |
|                                                                                                                                                                                                              |                   | Oral fluids                                                 |
| <b>Older sibling</b>                                                                                                                                                                                         | Self-completion** |                                                             |
| <p>* In the majority of cases, the Main interview was undertaken by the mother/mother figure and the Partner interview was undertaken by the father/father figure. See Table 21.</p> <p>** England only.</p> |                   |                                                             |

**Table 4: MCS3 – Summary of MCS3 Survey Elements**

| <b>Respondent</b>      | <b>Mode</b>     | <b>Summary of content</b>                                                 |
|------------------------|-----------------|---------------------------------------------------------------------------|
| <b>Mother/Father</b>   | Interview       | Module HD: Household demographics                                         |
| <b>Mother/Main*</b>    |                 | Module FC: Family context                                                 |
|                        |                 | Module ES: Early education, schooling and childcare                       |
|                        |                 | Module AB: Child and family activities and child behaviour                |
|                        |                 | Module PA: Parenting activities                                           |
|                        |                 | Module CH: Child health                                                   |
|                        |                 | Module PH: Parental health                                                |
|                        |                 | Module EI: Employment, education and income                               |
|                        |                 | Module HA: Housing and local area                                         |
|                        |                 | Module OM: Other matters                                                  |
|                        | Self-completion | Module SC: Self-completion                                                |
|                        |                 | - Child's temperament and behaviour                                       |
|                        |                 | - Child's relationship with siblings                                      |
|                        |                 | - Parenting and parent-child relationship                                 |
|                        |                 | - Mental health and drug-taking                                           |
|                        |                 | - Relationship with partner                                               |
|                        |                 | - Previous relationships, children living elsewhere, non-resident parents |
|                        |                 | - Attitudes and ethnic identity                                           |
|                        |                 | - Racial harassment and discrimination                                    |
|                        |                 | - Work-life balance and life satisfaction                                 |
|                        |                 | - Older Siblings' temperament and behaviour                               |
|                        | Interview       | Module OS: Older siblings                                                 |
|                        |                 | Module Z: Consents and contact information                                |
| <b>Father/Partner*</b> | Interview       | Module FC: Family context                                                 |
|                        |                 | Module ES: Early education, schooling and childcare (some)                |
|                        |                 | Module PA: Parenting activities                                           |
|                        |                 | Module PH: Parental health                                                |
|                        |                 | Module EI: Employment, education and income                               |
|                        |                 | Module OM: Other Matters                                                  |
|                        | Self-completion | Module SC: Self-completion                                                |
|                        |                 | - Parenting and parent-child relationship                                 |
|                        |                 | - Mental health and drug-taking                                           |
|                        |                 | - Relationship with partner                                               |
|                        |                 | - Previous relationships, children living elsewhere                       |
|                        |                 | - Attitudes and ethnic identity                                           |
|                        |                 | - Racial harassment and discrimination                                    |
|                        |                 | - Work-life balance and life satisfaction                                 |
|                        | Interview       | Module Z: Consents and contact information                                |

| Respondent                                                                                                                                                                                                                                                         | Mode               | Summary of content                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------|
| Interviewer                                                                                                                                                                                                                                                        | Observations       | Cognitive assessment                                        |
| Child                                                                                                                                                                                                                                                              | Assessments        | Story of Sally and Anne                                     |
|                                                                                                                                                                                                                                                                    |                    | British Ability Scales: Picture Similarities                |
|                                                                                                                                                                                                                                                                    |                    | British Ability Scales: Naming Vocabulary                   |
|                                                                                                                                                                                                                                                                    |                    | British Ability Scales: Pattern Construction                |
|                                                                                                                                                                                                                                                                    | Measurements       | Height, weight and waist circumference                      |
| Older sibling                                                                                                                                                                                                                                                      | Self-completion**  |                                                             |
| Teacher                                                                                                                                                                                                                                                            | Self-completion*** | Questions equivalent to Foundation Stage Profile in England |
| <p>* In the majority of cases, the Main interview was undertaken by the mother/mother figure and the Partner interview was undertaken by the father/father figure. See Table 22.</p> <p>** England only.</p> <p>*** Wales, Scotland and Northern Ireland only.</p> |                    |                                                             |

**Table 5: MCS4 – Summary of MCS4 Survey Elements**

| Respondent      | Mode            | Summary of content                                                        |
|-----------------|-----------------|---------------------------------------------------------------------------|
| Mother/Father   | Interview       | Module HD: Household demographics                                         |
| Mother/Main*    |                 | Module FC: Family context                                                 |
|                 |                 | Module ES: Early education, schooling and childcare                       |
|                 |                 | Module AB: Child and family activities and child behaviour                |
|                 |                 | Module PA: Parenting activities                                           |
|                 |                 | Module CH: Child health                                                   |
|                 |                 | Module PH: Parental health                                                |
|                 |                 | Module EI: Employment, education and income                               |
|                 |                 | Module HA: Housing and local area                                         |
|                 |                 | Module OM: Other matters                                                  |
|                 | Self-completion | Module SC: Self-completion                                                |
|                 |                 | - Child's temperament and behaviour                                       |
|                 |                 | - Child's relationship with siblings                                      |
|                 |                 | - Parenting and parent-child relationship                                 |
|                 |                 | - Mental health                                                           |
|                 |                 | - Relationship with partner                                               |
|                 |                 | - Previous relationships, children living elsewhere, non-resident parents |
|                 |                 | - Attitudes, ethnic identity, racial harassment and discrimination        |
|                 |                 | - Personality (OCEAN)                                                     |
|                 |                 | - Life satisfaction                                                       |
|                 | Interview       | Module Z: Consents and contact information                                |
| Father/Partner* | Interview       | Module FC: Family context                                                 |

| Respondent                                                                                                                                                                          | Mode            | Summary of content                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------------------------------------------------------------------------|
|                                                                                                                                                                                     |                 | Module ES: Early education, schooling and childcare (some)                        |
|                                                                                                                                                                                     |                 | Module PA: Parenting activities                                                   |
|                                                                                                                                                                                     |                 | Module PH: Parental health                                                        |
|                                                                                                                                                                                     |                 | Module EI: Employment, education and income                                       |
|                                                                                                                                                                                     |                 | Module OM: Other Matters                                                          |
|                                                                                                                                                                                     | Self-completion | Module SC: Self-completion                                                        |
|                                                                                                                                                                                     |                 | - Parenting and parent-child relationship                                         |
|                                                                                                                                                                                     |                 | - Mental health                                                                   |
|                                                                                                                                                                                     |                 | - Relationship with partner                                                       |
|                                                                                                                                                                                     |                 | - Previous relationships, children living elsewhere                               |
|                                                                                                                                                                                     |                 | - Attitude, ethnic identity and racial harassment and discrimination              |
|                                                                                                                                                                                     |                 | - Personality (OCEAN)                                                             |
|                                                                                                                                                                                     |                 | - Life satisfaction                                                               |
|                                                                                                                                                                                     | Interview       | Module Z: Consents and contact information                                        |
| Interviewer                                                                                                                                                                         | Observations    | Cognitive assessment                                                              |
| Child                                                                                                                                                                               | Assessments     | Story of Sally and Anne                                                           |
|                                                                                                                                                                                     |                 | British Ability Scales: Word Reading                                              |
|                                                                                                                                                                                     |                 | British Ability Scales: Pattern Construction                                      |
|                                                                                                                                                                                     |                 | Progress in Maths (Millennium Cohort Study edition)                               |
|                                                                                                                                                                                     | Measurements    | Height, weight, body-fat and waist circumference and physical activity monitoring |
|                                                                                                                                                                                     | Self-completion | Hobbies, friends and family, feelings, school                                     |
| Teacher                                                                                                                                                                             | Self-completion | Child's abilities and behaviour                                                   |
|                                                                                                                                                                                     |                 | Suspensions and exclusions                                                        |
|                                                                                                                                                                                     |                 | Language of schooling and language needs                                          |
|                                                                                                                                                                                     |                 | Special Educational Needs/Additional support needs                                |
|                                                                                                                                                                                     |                 | Parental interest in education                                                    |
|                                                                                                                                                                                     |                 | Setting and streaming                                                             |
|                                                                                                                                                                                     |                 | Teacher demographics                                                              |
|                                                                                                                                                                                     |                 | Study child's class                                                               |
| * In the majority of cases, the Main interview was undertaken by the mother/mother figure and the Partner interview was undertaken by the father/father figure. See Table 23 below. |                 |                                                                                   |

## PART FIVE: FIELDWORK

*For a more comprehensive discussion of fieldwork please refer to the MCS1 Technical Report on Fieldwork (NatCen 2004) or the MCS2 Technical Report on Fieldwork (NOP 2006) or the MCS3 Technical Report on Fieldwork (NatCen 2007) or the MCS4 Technical Report on Fieldwork (NatCen 2010).*

### 1. Fieldwork for MCS1

Following a competitive tender process NatCen was appointed to carry out the fieldwork for MCS1. The fieldwork in Northern Ireland was sub-contracted by NatCen to the Central Survey Unit of NISRA (the Northern Ireland Statistics and Research Agency). For the most part it took place in 2002, having started in England and Wales in June 2001, and in Scotland and Northern Ireland in September 2001. It finished in January 2003.

#### Briefings

Briefings for the 232 interviewers who were to work in England and Wales were held in 17 regional one-day meetings between 31 May and 15 June 2001. A further 42 interviewers working in Scotland were briefed at 4 sessions between 29 August and 6 September. These training sessions were conducted jointly by researchers from NatCen and CLS. In Northern Ireland, some 50 interviewers were briefed at 4 sessions between 17 and 28 August.

#### Fieldwork Timetable

The fieldwork for MCS1 (and MCS2) was carried out in 17 consecutive waves. Each issued wave of fieldwork contained babies born in a 4-weekly birth cycle (apart from the last), with the first wave covering the births between 1 and 28 September 2000 in England and Wales. This rhythm of recruiting the sample was dictated by the cycle of DWP procedures, scanning the Child Benefit database every 4 weeks.

Interviewers arranged interviews as soon as possible after the addresses were issued, aiming to reach the families while the baby was as close as possible to 9.5 months of age. Interviews with partners could be delayed until the child's first birthday (as were some main interviews where the address had been issued late).

The process for drawing each wave of the DWP sample is as follows:

Prior to fieldwork, the DWP sent opt-out letters to all parents of children with an eligible birth date who were registered (for Child Benefit purposes) as living within one of the sampled wards, apart from any cases flagged as sensitive. Batches of letters, including an information leaflet, were sent every 4 weeks to families whose babies were approximately 7 months old. The letter invited parents to take part in the study and gave them the opportunity to opt out of the study by telephoning or writing to the DWP. Any parents who opted out of the study were then removed from the sample.

The final stage was for the DWP to remove cases which they discovered had subsequently moved out of the sampled wards and to update the addresses for cases which had moved within or between sampled wards. At this stage any late opt-outs or newly sensitive cases were also removed.

The data were sent by the DWP to CLS in two stages, a week apart, in order to ensure that any late opt-outs or changes of addresses could be notified as near to the start of fieldwork as possible. After the final data were received serial numbers were assigned to each valid case and the data were sent to NatCen, for issue to the field.

The fieldwork timetable for the project detailing the dates of birth and fieldwork is shown in the table below.

**Table 6: Fieldwork timetable for MCS1**

| <b>Fieldwork Wave</b> | <b>Baby's Date of Birth</b> | <b>Fieldwork Period</b>   |
|-----------------------|-----------------------------|---------------------------|
| Wave 1                | 1 – 28 Sep 2000             | 11 June – 8 Jul 2001      |
| Wave 2                | 29 Sep – 26 Oct 2000        | 9 Jul – 5 Aug 2001        |
| Wave 3                | 27 October – 23 Nov 2000    | 6 Aug – 2 Sep 2001        |
| Wave 4                | 24 Nov – 21 Dec 2000        | 3 Sep – 30 Sep 2001       |
| Wave 5                | 22 Dec 2000 – 18 Jan 2001   | 1 Oct – 28 Oct 2001       |
| Wave 6                | 19 Jan – 15 Feb 2001        | 29 Oct – 25 Nov 2001      |
| Wave 7                | 16 Feb – 15 Mar 2001        | 26 Nov – 23 Dec 2001      |
| Wave 8                | 16 Mar – 12 Apr 2001        | 24 Dec 2001 – 20 Jan 2002 |
| Wave 9                | 13 Apr – 10 May 2001        | 21 Jan – 17 Feb 2002      |
| Wave 10               | 11 May – 7 June 2001        | 18 Feb – 17 Mar 2002      |
| Wave 11               | 8 June – 5 Jul 2001         | 18 Mar – 14 Apr 2002      |
| Wave 12               | 6 Jul – 2 Aug 2001          | 15 Apr – 12 May 2002      |
| Wave 13               | 3 Aug – 30 Aug 2001         | 13 May – 9 June 2002      |
| Wave 14               | 31 Aug – 27 Sep 2001        | 10 June – 7 Jul 2002      |
| Wave 15               | 28 Sep – 25 Oct 2001        | 8 Jul – 4 Aug 2002        |
| Wave 16               | 26 Oct – 23 Nov 2001        | 5 Aug – 22 Sep 2002       |
| Wave 17               | 24 Nov 2001-11 Jan 2002     | 23 Sep –10 Jan 2003       |

*Note: NatCen numbered these waves 2-18 as they counted the Dress Rehearsal Pilot as Wave 1.*

Waves 1-13 of fieldwork took place in England and Wales from June 2001 to July 2002. The last wave in England and Wales, wave 13, which included babies born on 31 August, was delayed by 4 weeks for operational reasons, so this wave contained interviews mostly conducted at 10 rather than 9 months for these 2 countries. The last wave in Scotland and Northern Ireland, wave 17, was the extended sample spanning 7 weeks of births. The latest interview (with a partner) took place in Northern Ireland on the last-but-one eligible day, 10 January 2003. Fieldwork in Scotland (and with all main informants) finished before the end of 2002.

The aim was that the fieldwork for each wave should be as self-contained as possible, with the minimum amount of overlap. Interviewers were briefed to interview families when the baby was 9 months and 15 days old, ideally, in order to standardise the data being collected as far as possible. Allowing for delayed interviewing due to tracing problems, the window of opportunity to interview was brief, up to 11 months of the babies' age for the main interview and up to 12 months for the partner.

Seventy-five per cent of main interviews took place while the baby was aged 9 months – 3,579 (19 per cent) at 10 months with 541 (3 per cent) at 8 months – representing babies born towards the end of the 4-week span interviewed early in the fieldwork period. However, 479 interviews took place late, 475 at 11 months and only 4 in months 12-13. Seventeen were not interviewed because the time window had expired by the time they were found. They are included in the 'other ineligible', Table 7.2 in the Technical Report on Sampling (4th Edition) (Plewis 2007).

## Languages

In order to comply with the recommendations made by the Multi-Centre Research Ethics Committee (MREC), a simplified leaflet was produced for interviewers to give to respondent families on the doorstep. This leaflet, the advance letter and the thank-you letter were translated into the most common non-English languages spoken in the 19 selected 'ethnic' wards. The languages appropriate for translation were: Bengali, Gujarati, Kurdish, Punjabi, Somali, Turkish and Urdu. The first leaflet had already been translated into Welsh. Some interviews were carried out in verbal translation (in these and other languages) by relatives or friends. In certain circumstances where no one was available to translate into English, NatCen provided translator interviewers. Other languages encountered in non-trivial numbers included Arabic, Hindi and Tamil. Two hundred and twenty-six (1 per cent) main interviews were carried out in a language other than English and a further 547 (3 per cent) were done in a mix of English and another language. For partners the corresponding figures were 306 (2 per cent) and 94 (1 per cent).

## In-field Tracing

On the whole, the addresses supplied by DWP proved to be current. Unfortunately, in a proportion of those issued to the field, the families had moved, either after the baby was aged 7 months or else before the baby reached that age, but had not informed the DWP of their move. Where a family was not living at the issued address and the interviewer could not establish a new local address, cases were returned to CLS for tracing. Where a new address was found within a selected ward, cases were re-issued to the field. Where a family had moved to a non-selected area, but were resident at their old address when the baby was aged 9 months, they could be interviewed at the new address.

## Data Collection Errors

In a number of cases, interviewers made errors in data collection which were identified by the fieldwork agency during the data preparation stage. Where possible, the data were cleaned to correct these errors. In a small number of cases (identified below) this has not been possible and users should exercise caution when using data for these cases. These cases mostly involved incorrect application of the proxy module and are identified on the variable 'errtype'.

**Table 7: MCS1 Data Collection Errors**

|   | Error Type                                                                                                                                                                                                                                                                                                               | N   | Action taken                                                                                                                                                                                                                    |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Proxy module done in error, i.e. the proxy section of the Main interview was completed about a partner who was not eligible to be interviewed by proxy.                                                                                                                                                                  | 117 | Data deleted from proxy module, household outcome code re-classified to 'partial household' and partner outcome code re-classified to unproductive.                                                                             |
| 2 | Partner interview done by proxy in error, i.e. the main respondent has completed the partner interview on behalf of partner. Partner should have done the interview him/herself.                                                                                                                                         | 42  | Data deleted from partner interview, household outcome code re-classified as 'partial household' and partner outcome code re-classified to unproductive.                                                                        |
| 3 | Partner answered proxy in person, should have done normal partner interview, i.e. the partner completed the proxy module in person (about him/herself).                                                                                                                                                                  | 6   | Data transferred from proxy section to equivalent variables in partner interview, household outcome code re-classified as 'main and partner in person' and partner outcome code re-classified to 'partial interview in person'. |
| 4 | Main interview done by father, partner interview by mother, i.e. the data indicate that the mother did the main interview and the father did the partner interview but the main interview was actually conducted with the father (in error) and the partner interview was actually conducted with the mother (in error). | 2   | NONE                                                                                                                                                                                                                            |

|   | Error Type                                                                                                                                                                                                                                                                                  | N | Action taken                  |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------------------------------|
| 5 | Father did both main and partner interviews, i.e. the data indicate that the mother completed the main interview and the father completed the partner interview but actually the father conducted both interviews (should have only done the partner interview).                            | 1 | NONE                          |
| 6 | Main interview done by partner, no other interview, i.e. the data indicate that the mother completed the main questionnaire and the father did not respond to the partner questionnaire but actually the father completed the main interview (in error) and there was no partner interview. | 1 | NONE                          |
| 7 | Grandmother (person 1 in household) was incorrectly coded as natural mother. The actual natural mother (who was person 3) completed the main interview.                                                                                                                                     | 1 | Relevant variables corrected. |

## 2. Fieldwork for MCS2

Following a competitive tender process, the fieldwork for MCS2 was carried out by NOP Research. The work in Northern Ireland was sub-contracted to Millward Brown Ulster. This survey was conducted mainly during 2004. The main-stage started in England and Wales in September 2003, and in Scotland and Northern Ireland in December 2003. Fieldwork finished in early 2005.

### Briefings

Interviewers who were to work in England and Wales were briefed before the start of fieldwork in 13 regional 3-day meetings. Interviewers working in Scotland were briefed at 3 additional sessions. These training sessions were conducted jointly by researchers from NOP and CLS. In Northern Ireland, some interviewers were briefed in just one session by Millward Brown and CLS researchers. There were 5 further briefings during the course of fieldwork as new interviewers were added.

Some 150 interviewers were initially briefed to work on the survey; but by the time fieldwork was complete around 200 interviewers had worked on the survey. Further details may be found in the NOP Technical Report on Fieldwork (NOP 2006).

## Fieldwork Timetable

Fieldwork started in September 2003 in England and Wales finished in April 2005. In Scotland and Northern Ireland, fieldwork started in December 2003 and finished in January 2005.

**Table 8: Fieldwork timetable for MCS2**

| Fieldwork Wave | Baby's Date of Birth      | Fieldwork Period |
|----------------|---------------------------|------------------|
| Wave 1         | 1 – 28 Sep 2000           | September 2003   |
| Wave 2         | 29 Sep – 26 Oct 2000      | October 2003     |
| Wave 3         | 27 Oct – 23 Nov 2000      | November 2003    |
| Wave 4         | 24 Nov – 21 Dec 2000      | December 2003    |
| Wave 5         | 22 Dec 2000 – 18 Jan 2001 | January 2004     |
| Wave 6         | 19 Jan – 15 Feb 2001      | February 2004    |
| Wave 7         | 16 Feb – 15 March 2001    | March 2004       |
| Wave 8         | 16 Mar – 12 April 2001    | April 2004       |
| Wave 9         | 13 April – 10 May 2001    | May 2004         |
| Wave 10        | 11 May – 7 Jun 2001       | June 2004        |
| Wave 11        | 8 Jun – 5 Jul 2001        | July 2004        |
| Wave 12        | 6 Jul – 2 Aug 2001        | August 2004      |
| Wave 13        | 3 Aug – 30 Aug 2001       | September 2004   |
| Wave 14        | 31 Aug – 27 Sep 2001      | October 2004     |
| Wave 15        | 28 Sep – 25 Oct 2001      | November 2004    |
| Wave 16        | 26 Oct – 23 Nov 2001      | December 2004    |
| Wave 17        | 24 Nov 2001-11 Jan 2002   | January 2005     |

## **Languages**

A breakdown of interviews by 'language interviewed in' is provided in the Technical Report on Fieldwork (NOP 2006).

## **In-field Tracing**

Families who had moved from the issued address were traced in the field by NOP interviewers. Families who could not be successfully traced by interviewers were returned to CLS for additional tracing by the Tracing team. Details of in-field tracing activities can be found in the Technical Report on Fieldwork (NOP 2006).

## **3. Fieldwork for MCS3**

Following a competitive tender process the NatCen was appointed to carry out the fieldwork for MCS3. The fieldwork in Northern Ireland was sub-contracted by NatCen to the Central Survey Unit of NISRA (the Northern Ireland Statistics and Research Agency). The main stage of this fieldwork took place within the calendar year of 2006, starting in England and Wales in January 2006, and in Scotland and Northern Ireland in April 2008. The survey also included a follow-on survey of teachers outside England extending into 2007.

## **Briefings**

Interviewers were briefed in 3-day training sessions. These sessions were conducted jointly by researchers from NatCen and CLS. For further details see NatCen (2007).

## **Fieldwork Timetable**

The fieldwork timetable for MCS3 was driven by the requirement to interview the family during the child's first year of compulsory schooling (Reception Class in England and Wales and Primary One in Scotland and Northern Ireland). As a result, fieldwork was compressed into school years. In England and Wales, the cohort's birth dates span a single school year. However, in Scotland and Northern Ireland the birth dates are spread over more than one school year. In England, Wales and Northern Ireland, school year is normally determined by date of birth. In Scotland, school year is determined by parental preference in addition to date of birth. For this reason, school year was known with less certainty in advance in Scotland. During the first wave of fieldwork in Scotland, interviewers were asked to find out, before conducting the interview, whether the child had started school. If the child had not yet started school, the interview was deferred until the second wave of fieldwork.

**Table 9: Fieldwork timetable for MCS3 – Main Survey**

| Wave | Country             | Dates of birth                                                                           | Fieldwork      |
|------|---------------------|------------------------------------------------------------------------------------------|----------------|
| E1   | England             | 1 Sep 2000 – 28 Feb 2001                                                                 | Jan – May 2006 |
| E2   | England             | 1 Mar 2001 – 11 Jan 2002                                                                 | Apr – Jul 2006 |
| W1   | Wales               | 1 Sep 2000 – 28 Feb 2001                                                                 | Jan – May 2006 |
| W2   | Wales               | 1 Mar 2001 – 11 Jan 2002                                                                 | Apr – Jul 2006 |
| S1   | Scotland            | 1 Sep 2000- 28 Feb 2001<br>(starting school in Aug 2005)                                 | Apr – Jul 2006 |
| S2   | Scotland            | 1 Sep 2000- 28 Feb 2001<br>(starting school in Aug 2006)<br>and 1 Mar 2001 – 11 Jan 2002 | Aug – Dec 2006 |
| N1   | Northern<br>Ireland | 24 Nov 2000 – 1 July 2001                                                                | Apr – Jul 2006 |
| N2   | Northern<br>Ireland | 2 July 2001 – 11 Jan 2002                                                                | Sep – Dec 2006 |

**Table 10: Fieldwork timetable for MCS3 – Teacher Survey in Wales, Scotland and Northern Ireland**

| Teacher Wave | Country                              | Main Fieldwork Wave     | Teacher Fieldwork  |
|--------------|--------------------------------------|-------------------------|--------------------|
| T1           | Wales, Scotland and Northern Ireland | W1 & W2, S1 and N1      | Sep 2006- Jan 2007 |
| T1 – mop-up  | Wales, Scotland and Northern Ireland | W1 & W2, S1 and N1      | Jan – May 2007     |
| T2           | Wales, Scotland and Northern Ireland | W2, S1 & S2 and N1 & N2 | Mar – Jun 2007     |

## Languages

A breakdown of interviews by ‘language interviewed in’ is provided in the Technical Report on Fieldwork (NatCen 2007).

## In-field Tracing

Families who had moved from the issued address were traced in the field by NatCen interviewers. Families who could not be successfully traced by interviewers were returned to CLS for additional tracing by the Tracing Unit. Details of in-field tracing activities can be found in the Technical Report on Fieldwork (NatCen 2007).

## 4. Fieldwork for MCS4

Following a competitive tender process the NatCen was appointed to carry out the fieldwork for MCS4. This was a planned extension to their existing contract for MCS3. The fieldwork in Northern Ireland was sub-contracted by NatCen to the Central Survey Unit of NISRA (the Northern Ireland Statistics and Research Agency). The first wave of the main stage fieldwork commenced in England and Wales in January 2008 and in Scotland and Northern Ireland in April 2008. The survey also included a follow-on survey of extending into 2009.

## Briefings

Interviewers new to the study were briefed in 3-day training sessions. Interviewers who had worked on MCS3 were briefed in 2-day training sessions. Some of these

sessions were large ‘conference style’ briefings’. These sessions were conducted jointly by researchers from NatCen and CLS (see NatCen 2010).

### Fieldwork Timetable

The fieldwork timetable for MCS4 was driven by the requirement to interview the family during the child’s third year of compulsory schooling (Year 2 in England and Wales, and Primary Three in Scotland and Northern Ireland). As at MCS3, fieldwork was compressed into school years. In England and Wales, the cohort’s birth dates span a single school year. However, in Scotland and Northern Ireland the birth dates are spread over more than one school year. In England, Wales and Northern Ireland, school year is normally determined by date of birth. In Scotland, school year is determined by parental preference in addition to date of birth.

**Table 11: Fieldwork timetable for MCS4 – Main Survey**

| Wave | Country             | Dates of birth                                                                          | Fieldwork        |
|------|---------------------|-----------------------------------------------------------------------------------------|------------------|
| E1   | England             | 1 Sep 2000 – 28 Feb 2001                                                                | Jan – May 2008   |
| E2   | England             | 1 Mar 2001 – 11 Jan 2002                                                                | April – Aug 2008 |
| W1   | Wales               | 1 Sep 2000 – 28 Feb 2001                                                                | Jan – May 2008   |
| W2   | Wales               | 1 Mar 2001 – 11 Jan 2002                                                                | April – Aug 2008 |
| S1   | Scotland            | 1 Sep 2000- 28 Feb 2001<br>(started school in Aug 2005)                                 | April – Aug 2008 |
| S2   | Scotland            | 1 Sep 2000- 28 Feb 2001<br>(started school in Aug 2006)<br>and 1 Mar 2001 – 11 Jan 2002 | Aug – Dec 2008   |
| N1   | Northern<br>Ireland | 24 Nov 2000 – 1 Jul 2001                                                                | April – Aug 2008 |
| N2   | Northern<br>Ireland | 2 Jul 2001 – 11 Jan 2002                                                                | Sep – Dec 2008   |

**Table 12: Fieldwork timetable for MCS4 – Teacher Survey**

| <b>Teacher Wave</b> | <b>Country</b>                             | <b>Main Fieldwork Wave</b>                              | <b>Teacher Fieldwork</b> |
|---------------------|--------------------------------------------|---------------------------------------------------------|--------------------------|
| Wave 1              | England and Wales                          | Interviews in E1, E2, W1, W2 up to end-Apr 2008         | Jun-Nov 2008             |
| Wave 2a             | Scotland and Northern Ireland              | Interviews in S1 and N1 up to end-Apr 2008              | Jul-Dec 2008             |
| Wave 2b             | England and Wales                          | Interviews in E1, E2, W1, W2 up to end-May 2008         | Jul-Dec 2008             |
| Wave 3              | England, Wales, Scotland, Northern Ireland | Interviews in E1, E2, W1, W2, S1, N1 up to end-Aug 2008 | Oct 2008-Feb 2009        |
| Wave 4              | Scotland and Northern Ireland              | Interviews in S2 and N2 up to end-Dec 2008              | Feb-Jul 2009             |

## Languages

A breakdown of interviews by 'language interviewed in' is provided in the Technical Report on Fieldwork (NatCen 2010).

## In-field Tracing

Families who had moved from the issued address were traced in the field by NatCen interviewers. Families who could not be successfully traced by interviewers were returned to CLS for additional tracing by the Cohort Maintenance Team. Details of in-field tracing activities can be found in the Technical Report on Fieldwork (NatCen 2010).

## PART SIX: THE ACHIEVED SAMPLE

*For a more comprehensive discussion of the achieved sample, please refer to the Technical Report on Sampling (4<sup>th</sup> Edition) (Plewis 2007) and Technical Report on Response (1<sup>st</sup> Edition) (Plewis and Ketende 2006).*

### 1. The MCS1 Achieved Sample

The survey reached 18,552 families, which, after allowance for 246 sets of twins and 10 sets of triplets, amounted to 18,818 cohort children. Six families have 2 singletons in the sample, they have been assigned different serial numbers but can be identified from the variable ADDUAL00. The table below shows how these respondents are distributed over the 4 countries of the UK. Further details by stratum appear in the Technical Report on Sampling (4<sup>th</sup> Edition) (Plewis 2007).

**Table 13: MCS1 Sample Size – Clusters, Children Families, by Country**

|            | Number of sample 'wards' * | Target sample as boosted | Achieved Responses ** |                      |
|------------|----------------------------|--------------------------|-----------------------|----------------------|
|            |                            |                          | Children              | Families interviewed |
| Total UK   | 398                        | 20,646                   | 18,818                | 18,552               |
| England    | 200                        | 13,146                   | 11,695                | 11,533               |
| Wales      | 73                         | 3,000                    | 2,798                 | 2,760                |
| Scotland   | 62                         | 2,500                    | 2,370                 | 2,336                |
| N. Ireland | 63                         | 2,000                    | 1,955                 | 1,923                |

\* Counting amalgamations into 'superwards' as a single unit.

\*\* All productive contacts.

After the extension of sample birth dates in Scotland and Northern Ireland, the achieved sample of children came within 5 per cent and 2 per cent respectively of their targets of 2,500 and 2,000. With only 52 weeks of births in England and Wales, the achieved sample was 10 per cent below target. About half of this shortfall can be attributed to the drop in births which accompanied the millennium (see Technical Report on Sampling (4<sup>th</sup> Edition) (Plewis 2007). The rest was due to the lower response rate than expected in some parts of the country at some stages of the recruitment process.

### Response Rates

The overall response can be thought of as the combination of the leakages to numbers between the eligible population in the selected wards and the sample

issued to field and the success the interviewers then have at securing interviews, from the issued sample. The leakages between the eligible population and the 'in scope' population are set out in the Technical Report on Sampling (4th Edition) (Plewis 2007) as:

- Families opting out of the survey
- Families excluded by DWP
- Families excluded from the sampling frame because their postcode could not be allocated to a ward
- Undetected in-migrants.

It is arguable that the eligible population should also include families who do not claim Child Benefit; but we make the simplifying assumption that numbers of such families who are permanently resident in the UK are negligible. The Technical Report on Sampling (4th Edition) (Plewis 2007) makes two alternative assumptions about how many undetected in-movers there are. The estimates quoted in Table 14 below assume that there is an undetected in-comer for every detected out-mover, on average, in each stratum.

The alternative estimate sets undetected in-moves to zero, which raises all overall response rates (except Northern Ireland) above the target or assumed response rate set in the design of the survey and shown in the first column. This table shows that, when undetected in-migration is counted as a leakage, the overall response rate is 68 per cent for the (unweighted) UK sample, modestly below the 71 per cent expected. It is below target in every stratum except the advantaged wards of Wales, but only markedly so in Northern Ireland, with overall response rates in the combined strata of 63 per cent where 71 per cent had been somewhat optimistically set, given the lack of a tradition of such surveys in that country. Northern Ireland is also the only country where inability to assign Child Benefit claimants to a ward was a significant problem. Survey work in the ethnic areas of England was also something of an unknown quantity. A cautious target of 65 per cent was missed by 3 percentage points.

**Table 14: Response Rates by Stratum and Country for MCS1**

|         |               | <b>Expected<br/>Overall<br/>Response Rate</b> | <b>Achieved<br/>Overall<br/>Response Rate</b> | <b>In-scope<br/>Response Rate<br/>Fieldwork</b> |
|---------|---------------|-----------------------------------------------|-----------------------------------------------|-------------------------------------------------|
| England | Advantaged    | 75%                                           | 73%                                           | 86%                                             |
|         | Disadvantaged | 70%                                           | 68%                                           | 82%                                             |
|         | Ethnic        | 65%                                           | 62%                                           | 76%                                             |
|         | Total         | 70%                                           | 68%                                           | 82%                                             |

|            |               | <b>Expected<br/>Overall<br/>Response Rate</b> | <b>Achieved<br/>Overall<br/>Response Rate</b> | <b>In-scope<br/>Response Rate<br/>Fieldwork</b> |
|------------|---------------|-----------------------------------------------|-----------------------------------------------|-------------------------------------------------|
| Wales      | Advantaged    | 75%                                           | 78%                                           | 89%                                             |
|            | Disadvantaged | 70%                                           | 69%                                           | 83%                                             |
|            | Total         | 71%                                           | 72%                                           | 84%                                             |
| Scotland   | Advantaged    | 75%                                           | 73%                                           | 86%                                             |
|            | Disadvantaged | 70%                                           | 68%                                           | 83%                                             |
|            | Total         | 71%                                           | 70%                                           | 85%                                             |
| N. Ireland | Advantaged    | 75%                                           | 65%                                           | 81%                                             |
|            | Disadvantaged | 70%                                           | 61%                                           | 78%                                             |
|            | Total         | 71%                                           | 63%                                           | 79%                                             |
| UK         | All           | 71%                                           | 68%                                           | 82%                                             |

*Source: MCS Technical Report on Sampling (4<sup>th</sup> Edition) (Plewis 2007).*

Out of the cases issued to field some have been deemed ineligible because they are known or thought to have moved out of the survey area before the child reached 9 months of age. Of the remaining eligible or 'in-scope' sample, the response in fieldwork averaged 82 per cent giving at least one interview. It varied by stratum as expected, but more so. The ethnic wards as anticipated had least 76 per cent, and the advantaged areas of Wales the highest 89 per cent, with both strata in Northern Ireland being below the stratum average for Great Britain.

Complete evaluation of sources of the known characteristics in case of survey loss before and after the Child Benefit stage are included in a later edition of the Technical Report on Sampling (4<sup>th</sup> Edition) (Plewis 2007). Those lost before issue to field do not appear systematically biased. A greater propensity of families in the disadvantaged areas to be excluded by DWP is balanced by a greater propensity of the inhabitants of advantaged areas to opt out.

## **2. The MCS2 Achieved Sample**

There are two components to the MCS2 issued sample, families that were productive in MCS1 and the so-called new families. There were 18,552 productive families in the first survey of the Millennium Cohort Study. The new families were families that although eligible, did not participate in MCS1. These were identified through DWP, of whom 1,389 families were eligible to be issued for MCS2 fieldwork.

From the paragraph above, the issued sample should have been 19,941 i.e. 18,552 +1,389, but 71 families from the MCS1 productive families were not issued to the field for various reasons. Their outcomes were known and recorded before the start of the fieldwork. Therefore, the MCS2 issued sample was 19,870; 18481 were the productive families in MCS1 and the 1,389 new families.

MCS2 response is reported in three groups: 1) all families initially thought to be eligible for MCS2 survey; 2) families that were productive in MCS1; and: 3) the New Families.

All response frequencies in this report are unweighted. The outcome codes were derived as:

|                       |                                                                                                                                                                                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Productive            | All families with some data from one of 6 data collection instruments other than what was carried forward. The 6 data collection instruments were: Main Interview, Partner Interview, Proxy Partner Interview, BAS, Bracken, Height and Weight. |
| Ineligible            | Emigrant families, deaths.                                                                                                                                                                                                                      |
| Uncertain Eligibility | Families that were away temporarily and those whose eligibility was uncertain, including untraced movers.                                                                                                                                       |
| Unproductive          | Refusals, non-contacts, other non-responses, including language problems, ill/incapacitated, deleted/lost data (lost CAPI).                                                                                                                     |

### All MCS2 Families Response

There were 19,941 families originally considered eligible for MCS2 survey, 15,590 of these were productive in the survey which is 78 per cent of all MCS2 families. There were 15,808 cohort members in the 15,590 productive families.

**Table 15: MCS2 Overall response**

| Outcome code                                      |             | n      | (%)  |
|---------------------------------------------------|-------------|--------|------|
| Productive                                        |             | 15,590 | 78.0 |
| Ineligible*                                       |             | 255    | 1.3  |
| Uncertain Eligibility (including untraced movers) |             | 868    | 4.4  |
| Unproductive                                      | Refusal     | 2,002  | 10.0 |
|                                                   | Non-Contact | 1070   | 5.4  |

| Outcome code |       | n      | (%)   |
|--------------|-------|--------|-------|
|              | Other | 156    | 0.8   |
| Total        |       | 19,941 | 100.0 |

\* **Ineligible:** deaths (n=16), permanent emigrants (n=169), failed eligibility (n=70).

Table 16 below shows that a slightly higher proportion of those that were productive in MCS1 (80 per cent) took part in MCS2 compared to the overall proportion in Table 13, which also includes New Families.

**Table 16: MCS2 Overall response for families that were productive in MCS1**

| Outcome code                                      |             | n      | (%)   |
|---------------------------------------------------|-------------|--------|-------|
| Productive                                        |             | 14,898 | 80.0  |
| Ineligible*                                       |             | 167    | 0.9   |
| Uncertain Eligibility (including untraced movers) |             | 687    | 3.7   |
| Unproductive                                      | Refusals    | 17,39  | 9.4   |
|                                                   | Non-Contact | 930    | 5.0   |
|                                                   | Other       | 131    | 0.71  |
| Total                                             |             | 18,552 | 100.0 |

\* **Ineligible:** deaths (n=14), permanent emigrants (n=153).

### The New Families Response

Only about 50 per cent of the New Families were productive.

**Table 17: Overall Response for the New Families**

| Outcome code                                      |             | N     | (%)   |
|---------------------------------------------------|-------------|-------|-------|
| Productive                                        |             | 692   | 50.0  |
| Ineligible **                                     |             | 88    | 6.3   |
| Uncertain Eligibility (including untraced movers) |             | 181   | 13.0  |
| Unproductive                                      | Refusals    | 263   | 19.0  |
|                                                   | Non-Contact | 140   | 10.0  |
|                                                   | Other       | 25    | 1.8   |
| Total                                             |             | 1,389 | 100.0 |

\*\* **Ineligible**: deaths (n=2), permanent emigrants (n=16), failed eligibility (n=70).

### 3. The MCS3 Achieved Sample

There were 19,244 families potentially eligible for inclusion in the issued sample. These were 18,552 families who were productive at MCS1 and 692 'New Families' who were productive at MCS2. However, 718 families were not issued to the field due to ineligibility (death or emigration), permanent refusal and sensitive family circumstances. Their outcomes were known and recorded before the start of the fieldwork. The families not issued due to sensitive family circumstances are recorded as 'unproductive other'. Two families who were not productive at either MCS1 or MCS2 were issued in error.

Therefore, the MCS3 issued sample was 18,528 (19,244 – (718 + 2)).

This section provides MCS3 response for the 19,244 families, i.e. including the 718 families not issued and excluding the two families issued in error.

All response frequencies here are unweighted. The outcome codes in this report were again derived as for MCS2. There were 19,244 families potentially eligible for the MCS3 survey, 15,246 of these were productive in the survey, which is 79.2 per cent of all MCS3 families. There were 15,459 cohort children in the 15,246 productive families.

**Table 18: MCS3 Overall response**

| Outcome code                                      |             | n      | (%)   |
|---------------------------------------------------|-------------|--------|-------|
| Productive                                        |             | 15,246 | 79.2  |
| Ineligible **                                     |             | 300    | 1.6   |
| Uncertain Eligibility (including untraced movers) |             | 547    | 2.8   |
| Unproductive                                      | Refusals    | 2,798  | 14.5  |
|                                                   | Non-Contact | 63     | 0.3   |
|                                                   | Other       | 290    | 1.5   |
| Total                                             |             | 19,244 | 100.0 |

\*\* **Ineligible:** deaths (n=18), permanent emigrants (n=282).

#### 4. The MCS4 Achieved Sample

There were 19,244 families potentially eligible for inclusion in the issued sample. However, 2213 families were not issued to the field due to ineligibility due to death or emigration (n=362), permanent refusal (n=1,705), permanent untraced (n=136) and sensitive family circumstances (n=10). Their outcomes were known and recorded before the start of the fieldwork. The families not issued due to sensitive family circumstances are recorded as 'unproductive other'.

Therefore, the MCS4 issued sample was 17,031 (19,244 - 2213).

This section provides MCS4 response for the 19,244 families, i.e. including the 2213 families not issued. All response frequencies in this report are unweighted.

##### Outcome codes are:

Productive

All families with some data from one of five data collection instruments other than what was carried forward. The 5 data collection instruments were: Main Interview, Partner Interview, Proxy Partner Interview, Cohort Child Cognitive Assessments and Cohort Child Physical Measurements.

Ineligible

Emigrant families, deaths.

**Uncertain Eligibility**

Families that were away temporarily and those whose eligibility was uncertain, including untraced movers.

**Unproductive**

Refusals, non-contacts, other non-responses, including language problems, ill/incapacitated, deleted/lost data (lost CAPI).

*In total, 13,857 families were productive in the survey, which is 72.20 per cent of all MCS families. There were 14,043 cohort children in the 13,857 productive families.*

**Table 19: MCS4 Overall response**

| Outcome code                                      |             | n      | (%)   |
|---------------------------------------------------|-------------|--------|-------|
| Productive                                        |             | 13857  | 72.0  |
| Ineligible                                        |             | 488    | 2.5   |
| Uncertain Eligibility (including untraced movers) |             | 848    | 4.4   |
| Unproductive                                      | Refusals    | 3,516  | 18.3  |
|                                                   | Non-Contact | 149    | 0.8   |
|                                                   | Other       | 386    | 2.0   |
| Total                                             |             | 19,244 | 100.0 |

## PART SEVEN: THE DATA

### 1. Structure of the Datasets

There are two sets of 9 data files (one set of SPSS data files and one of STATA data files):

- 1) *A longitudinal file containing information on the family which is consistent over time or is the most current version of a longitudinal variable:*

- MCS Longitudinal Family Level Information.

This file contains one row for all families in the longitudinal sample: that is families who have taken part in MCS1 or MCS2 (n=19,244 (18,552+692)).

- 2) *The cross-sectional data from the Household Questionnaire, Main, Partner and Proxy Interviews:*

- MCS1 Parent Interview Data
- MCS2 Parent Interview Data
- MCS3 Parent Interview Data
- MCS4 Parent Interview Data\*.

These files contain one row for each productive family at that sweep.

\*The three bracketed income datasets have been separated out from the main and partner data at MCS4 to reduce the size of the main and partner interview. The summary information derived is deposited in the main, partner and proxy information. If you wish to explore further the data, the full data is available in these datasets. The MCS4 CAPI Questionnaire section 1.2.8 explains the way in which this data was collected.

- 3) *Cross-sectional household grid data:*

- MCS1 Household Grid
- MCS2 Household Grid
- MCS3 Household Grid
- MCS4 Household Grid.

These files contain one row for each person in the household grid in productive families at that sweep.

- 4) *Child Assessment and Measurement Files:*

- MCS2 Child Measurement Data
- MCS2 Child Assessment Data
- MCS3 Child Measurement Data
- MCS3 Child Assessment Data
- MCS4 Child Measurement Data
- MCS4 Child Assessment Data

- MCS4 Child Our Adventures (Wales) Data.

5) *Teacher Survey and School Linkage:*

- MCS3 Foundation Stage Profile Dataset
- MCS3 Teacher Survey
- MCS4 Teacher Survey.

6) *Older Siblings Data:*

- MCS2 Older Siblings Data
- MCS3 Older Siblings Data.

These files contain one row per older sibling who were reported upon by the main respondent and also those older siblings who completed the paper self-completion questionnaire.

7) *Neighbourhood Assessment Data:*

- MCS2 Neighbourhood Assessment Data June 2006.

These files contain one row for each visit to the productive families at that sweep.

8) *Geographically Linked Data including IMD and Rural Urban Indicators:*

- MCS1 Geographically Linked Data
- MCS2 Geographically Linked Data
- MCS3 Geographically Linked Data
- MCS4 Geographically Linked Data.

9) *Derived Variables:*

- MCS1 Derived Variables
- MCS2 Derived Variables
- MCS3 Derived Variables
- MCS4 Derived Variables.

These files contain one row for each productive family at that sweep.

## 2. How to Link the Datasets

Data can be linked on **mcsid**, which is a unique identifier for each family. Family-level files can be linked on this identifier only.

Data that can be linked using solely mcsid includes the information which spans sweeps such as weights and variables to carry out analysis on stratum which are held on the MCS Longitudinal Family Level Information file and the Parent Interview files.

The data also contain unique, longitudinally consistent individual identifiers for cohort members and other people in the household.

The individual identifier for cohort members is **cnum**: cohort member number (to identify separately twins and triplets) and the individual identifier for all other people in the household is **pnum**: person number. These variables appear on all the data files (except the Longitudinal Family Level Information file).

They are prefixed and suffixed differently, depending on the file.

In the Parent Interview data, they are ahcnuma0 (cohort member 1), ahcnumb0 (cohort member 2), ahcnumc0 (cohort member 3) at MCS1. At MCS2 the leading 'a' is replaced with a 'b' and all other digits remain the same. At MCS3 the leading 'b' becomes a 'c' and at MCS4 the leading digit is a 'd' with all other digits remaining the same.

In the Household Grid and Child Measurement and Child Assessment data they are ahcnum00 (MCS1), bhcnun00 (MCS2), chcnun00 (MCS3) and dhcnun00 (MCS4).

In order to provide data which can be used for a variety of different purposes, a separate file (the household grid) has been supplied to enable linkage by cohort member or other respondent or member of the household, e.g. older sibling at each sweep.

### **A cohort member dataset can be constructed by linking on**

mcsid, ahcnum00, bhcnun00, chcnun00 or dhcnun00 from the household grid with:

- mcsid and ahcnuma0, ahcnumb0, ahcnumc0 – MCS1
- mcsid and bhcnuna0, bhcnumb0, bhcnunc0 – MCS2
- mcsid and chcnuna0, chcnumb0, chcnunc0 – MCS3
- mcsid and dhcnuna0, dhcnumb0, dhcnunc0 – MCS4

**A respondent dataset can be constructed by linking on**

mcsid, ahpnum00, bhpnum00, chpum00 or dhpum00 from the household grid with:

mcsid and ampnum00 – MCS1 – main respondent  
 mcsid and bmpnum00 – MCS2 – main respondent  
 mcsid and cmpnum00 – MCS3 – main respondent  
 mcsid and dmpnum00 – MCS4 – main respondent  
 mcsid and appnum00 – MCS1 – partner respondent  
 mcsid and bppnum00 – MCS2 – partner respondent  
 mcsid and cppnum00 – MCS3 – partner respondent  
 mcsid and dppnum00 – MCS4 – partner respondent

**The older siblings datasets can be constructed by linking on:**

mcsid, bhpnum00 from the household grid with:

mcsid and bopnum00 – MCS2 – older sibling respondent  
 mcsid and chpum00 – MCS3 – older sibling respondent

**Example Stata do jobs:**

```
use 'mcs_longitudinal_family_file.dta'
sort mcsid
merge 1:1 mcsid using 'mcs1_parent_interview.dta'
sort mcsid
merge 1:1 mcsid using 'mcs2_parent_interview.dta'
sort mcsid
merge 1:1 mcsid using 'mcs3_parent_interview.dta'
sort mcsid
merge 1:1 mcsid using 'mcs4_parent_interview.dta'
sort mcsid
```

**Example SPSS syntax:**

```
GET FILE='mcs_longitudinal_family_file.sav'.
SORT CASES by mcsid.
MATCH FILES /TABLE=*
  /FILE='mcs1_parent_interview.sav'
  /FILE='mcs2_parent_interview.sav'
  /FILE='mcs3_parent_interview.sav'
  /FILE='mcs4_parent_interview.sav'
  /BY mcsid.
SORT CASES by mcsid.
SAVE OUTFILE='Family Level.sav'.
```

### 3. The Household Grid

The household grid files contain two types of information: individual identifiers and identifying characteristics (number, sex and date of birth) and cross-sectional variables collected about everyone in the household (e.g. relationships between household members).

At MCS2, the household grid was collected independently from MCS1, i.e. the MCS1 grid was not fed forward. In order to create longitudinally consistent individual identifiers, the two household grids were matched. This involved matching people using their individual identifying characteristics (name, sex and date of birth). All people present in the household at MCS1 retained their original person number<sup>1</sup> and any new entrants were given the next available person number.

At MCS2, with information only available from two sweeps, it was not always possible to determine which data were correct when information was inconsistent. With MCS3 we were in a better position to resolve these issues. Our approach has been to clean data only where it is clear that the corrections can be made with certainty. The sex variable was checked by reference to names collected at MCS2 and MCS3. Cleaning of relationships was restricted to differences in report which straddled the adult/child boundary, e.g. grandparent / grandchild, father / son. Other relationships which are possible, even where unlikely, such as step-parent / other non-relative or natural / adopted / foster, were not changed.

The household grid contains one record for each person who has ever appeared in the household for each family that participated in that sweep.

There is a variable which indicates for each person whether or not they are present at any particular sweep: ahcprs00, bhcprs00, chcprs00 and dhcprs00, for cohort members in MCS1, MCS2, MCS3 and MCS4, respectively; and ahpres00, bhpres00, chpres00 and dhpres00 for all other people in MCS1, MCS2, MCS3 and MCS4, respectively. These can be used to identify people moving in, out and back into the household by merging the three household grid files. For cases where the main interview was not conducted at MCS2 (i.e. only a partner interview was conducted) and a main interview was completed at MCS1, bhpres00 was labelled as 'Not Known'.

The other information on the household grid file (relationships and other cross-sectional information) is retained as reported at that sweep (with the exception of some limited cleaning of relationships longitudinally to attempt to correct for mis-keying).

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<sup>1</sup> Except for part-time partners who at MCS1 were all assigned a person number of 12. These people were assigned the next available person number in the household at MCS1.

## Analysis using the household grid

The individual details and cross-sectional information from the household grid which relates to cohort members, main and partner respondents, appears on the Parent Interview Data. This means that any derived variables using the sex and date of birth of cohort members, main and partner respondents and/or relationships of other household members to cohort members, main and partner respondents, can be derived solely from the Parent Interview files.

Any derived variables using the sex and date of birth of other people in the household, relationships between other people in the household and detailed analysis of change in household composition<sup>2</sup> must be done using the household grid data.

## Data collection problems relating to the household grid

At MCS1, the household grid had to be completed before carrying out any interviews and was collected for all families (including those in which a Main interview was not done). At MCS2, the household grid was collected as part of the Main interview. As a result, it was not completed in households in which there was not a main interview.

For households which were not the subject of a Main interview at MCS2 but took part in MCS1, the household grid contains the individual details for everyone who was in the household at MCS1. In order for these families to be interviewed, the interviewer would have established that the cohort member was present. So, for those households the cohort member present flag (bhcpres00) indicates that they were present. Also, in these households, we have indicated that the person who was the main respondent at MCS1 was present at MCS2 (bhpres00) (and that they were eligible for the Main interview) although the cross-sectional variables (relationships, etc) are not available in these families. In addition, if a partner or partner proxy interview has been conducted in these households, the person who completed this interview is indicated as present (bhpres00), although the cross-sectional variables (relationships, etc) are not available in these families. For all other people in these households bhpres00 indicates that we do not know whether or not they are present.

Some of the families in which the Main interview did not take place were 'New Families', i.e. those that did not take part in MCS1 because they moved into the eligible areas too late to be included in the initial survey. For such families, the individual details of the Child Benefit claimant and, if applicable, the person who completed the partner or partner proxy interview appears in the household grid and these people are coded as present (bhpres00) and the Child Benefit claimant as eligible for the Main interview.

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<sup>2</sup> There are derived variables about change in household composition relating to parents/carers.

At MCS3 and MCS4, the procedure was the same as for MCS1. The household grid was completed before carrying out any interviews and was collected for all families (including those in which a Main interview was not done).

### **Identification of Main and Partner**

At each of MCS1, 2, 3 and 4 there were three different parent interviews which could be completed with up to two different people per family. The three interviews were: Main, Partner and Partner Proxy. The selection of household members for the different interviews was done by the CAPI program, based on relationship to the cohort member and relationships between different household members. In general, any parents (including step, foster and adoptive) of cohort members and partners (including same-sex partners) of parents were eligible for interview. If there were no parents in the household, the main carer of the cohort member (and their partner) was selected for interview. In each household, there should always have been someone selected for the Main interview. A different person would have been selected for the Partner interview. If the person selected for the Partner interview was away for the fieldwork period or incapacitated, they became eligible for the Partner Proxy interview instead of the Partner interview, which was completed by the Main respondent on behalf of their partner.

At Sweep 1 there was a priority for the natural mother, if present, to do the main interview as it contained questions about pregnancy and delivery. In the few cases where mothers did the Partner interview it was due to language problems. At sweep 2 the preference was for the same person who had done the Main interview at sweep 1 to do it again if possible. If the Main respondent from sweep 1 was no longer in the cohort child's household, but at least one biological parent of the child was, then that person was selected as the new Main informant, even if he or she was not the main carer of the child. If there was no biological parent in the household then whoever was the main carer for the cohort child was selected for the main parent questionnaire. At sweeps 3 and 4 the presumption again was that the natural mother, the natural father in her absence, the previous Main informant or the main carer, in that order, would be selected as the main informant. But families could elect to follow other arrangements where, for example, the father was the main carer and the mother chose not to do either interview. Tables 20-23 summarise the different combinations of Parent interviews at each sweep.

**Table 20: MCS1 Parent interview response by sex of respondent and relationship to cohort member**

|                                                            | All       |          | Female         |       | Male           |       |
|------------------------------------------------------------|-----------|----------|----------------|-------|----------------|-------|
|                                                            | Frequency | Per cent | Natural mother | Other | Natural father | Other |
| 1. Main respondent in person (no-one eligible for partner) | 3,194     | 17.22    | 3,172          | 4     | 18             | 0     |
| 2. Main and partner respondent in person                   | 13,205    | 71.18    | 13,193         | 5     | 7              | 0     |
| 3. Main in person, partner by proxy                        | 216       | 1.16     | 215            | 0     | 1              | 0     |
| 4. Main in person, partner eligible but no response        | 1,917     | 10.33    | 1,915          | 0     | 2              | 0     |
| 5. No main interview, partner interviewed in person        | 20        | 0.11     | 20             | 0     | 0              | 0     |
| Total                                                      | 18,552    | 100      | 18,515         | 9     | 28             | 0     |

**Table 21: MCS2 Parent interview response by sex of respondent and relationship to cohort member**

|                                                            | All       |          | Female         |       | Male           |       | Not known |
|------------------------------------------------------------|-----------|----------|----------------|-------|----------------|-------|-----------|
|                                                            | Frequency | Per cent | Natural mother | Other | Natural father | Other |           |
| 1. Main respondent in person (no-one eligible for partner) | 2,655     | 17.03    | 2,574          | 19    | 61             | 1     | 0         |
| 2. Main and partner respondent in person                   | 10,418    | 66.82    | 10,281         | 28    | 108            | 1     | 0         |
| 3. Main in person, partner by proxy                        | 221       | 1.42     | 218            | 2     | 1              | 0     | 0         |
| 4. Main in person, partner eligible but no response        | 2,154     | 13.82    | 2,136          | 5     | 13             | 0     | 0         |
| 5. No main interview, partner interviewed in person        | 61        | 0.39     | 3              | 1     | 1              | 0     | 56        |
| 6. No main interview, partner interviewed by proxy         | 2         | 0.01     | 0              | 0     | 0              | 0     | 2         |
| 7. No parent interviews                                    | 79        | 0.51     | 67             | 0     | 1              | 0     | 11        |
| Total                                                      | 15,590    | 100      | 15,279         | 55    | 185            | 2     | 69        |

**Table 22: MCS3 Parent interview response by sex of respondent and relationship to cohort member**

|                                                            | All       |          | Female         |       | Male           |       |
|------------------------------------------------------------|-----------|----------|----------------|-------|----------------|-------|
|                                                            | Frequency | Per cent | Natural mother | Other | Natural father | Other |
| 1. Main respondent in person (no-one eligible for partner) | 3,021     | 19.82    | 2,930          | 19    | 72             | 0     |
| 2. Main and partner respondent in person                   | 10,475    | 68.71    | 10,193         | 25    | 255            | 2     |
| 3. Main in person, partner by proxy                        | 287       | 1.88     | 267            | 6     | 14             | 0     |

|                                                     | All       |          | Female         |       | Male           |       |
|-----------------------------------------------------|-----------|----------|----------------|-------|----------------|-------|
|                                                     | Frequency | Per cent | Natural mother | Other | Natural father | Other |
| 4. Main in person, partner eligible but no response | 1,408     | 9.24     | 1,352          | 7     | 49             | 0     |
| 5. No main interview, partner interviewed in person | 19        | 0.12     | 18             | 0     | 1              | 0     |
| 7. No parent interviews                             | 36        | 0.24     | 32             | 1     | 3              | 0     |
| Total                                               | 15,246    | 100      | 14,792         | 58    | 394            | 2     |

**Table 23: MCS4 Parent interview response by sex of Main respondent and relationship to cohort member**

|                                                                 | All       |          | Female         |       | Male           |       |
|-----------------------------------------------------------------|-----------|----------|----------------|-------|----------------|-------|
|                                                                 | Frequency | Per cent | Natural mother | Other | Natural father | Other |
| 1 Main respondent in person (no-one eligible for partner)       | 2903      | 20.9     | 2784           | 20    | 98             | 1     |
| 2 Main and partner respondent In person                         | 9161      | 66.1     | 8885           | 33    | 236            | 7     |
| 3 Main in person, partner by Proxy                              | 249       | 1.8      | 236            | 1     | 11             | 1     |
| 4 Main in person, partner eligible but no response              | 1484      | 10.7     | 1431           | 7     | 45             | 1     |
| 5 Main eligible but no interview, partner Interviewed in person | 19        | 0.1      | 18             | 0     | 0              | 1     |
| 7 Main eligible , no response (no-one eligible for partner)     | 18        | 0.1      | 17             | 0     | 1              | 0     |
| 8 Main and partner eligible, no response from either            | 23        | 0.2      | 21             | 0     | 1              | 1     |
| Total                                                           | 13857     | 100      | 13392          | 61    | 392            | 12    |

*(Sex is of Main respondent, even if they were not interviewed.)*

At MCS1, there was a Main interview in 18,532 of the 18,552 families. There was someone eligible for a Partner interview in 15,358 families and an interview was

completed in 13,225 of these cases. Proxy data were collected on 216 partners (of the 235 who were eligible); but interview data are completely missing for 1,917 two-'parent' families (adresp00). Table 20 also shows that the vast majority of the Main respondents were female. 18,524 (out of 18,815) were natural mothers. There were 28 male Main respondents, all natural fathers, 18 of whom were lone fathers. All but 61 (99.6 per cent) of the 15,358 partners identified in the families visited were natural fathers. Since the Main respondent was asked questions about pregnancy and delivery the presumption was that, wherever possible, the natural mother should be the main informant. Some of the cases where roles were reversed were because of language problems.

At MCS2, there was someone eligible for the Main interview in 15,588 of the 15,590 productive families<sup>3</sup> and an interview was completed in 15,448 cases. There was someone eligible for a Partner interview in 12,856 families and an interview was completed in 10,479 of these cases, with data by proxy in 233. There were 2,154 two-'parent' families with data missing on the partner, and 63 with data missing from the Main. There were also 79 families with some data (e.g. child assessments) but no interview data from either a Main or a Partner respondent. The Main respondents were again overwhelmingly female, but the number of them who were not natural mothers increased since MCS1 from 9 to 55. The number of male Main respondents increased from 28 at MCS1 to 187 (2 of whom were not natural fathers). Part of this change was an increase of lone-father informants (to 62), but it was mostly due to a rise in the number of two-parent families where the Main response was collected from the father (97 per cent of the partners were natural fathers).

At MCS3, a Main interview was conducted in 15,210 of the 15,246 families. There was someone eligible for a Partner interview in 12,189 families and an interview was completed in 10,475 cases, with proxy data collected in a further 287. Information was not collected on partners in 1,408 couples, and from main respondents in 19 families where the partner responded. In 36 cases there were no interviews in the dataset from any parent. The proportion of Main informants who were natural mothers again dropped, to 97 per cent (14,792). The number of female Main respondents who were not natural mothers hardly changed from MCS2 (58). But the number of Main respondents who were men changed by a significant amount. The number of natural fathers completing the main interview was 394 (more than double the 185 at the age 3 survey). Seventy-two were lone fathers and the rest were part of a couple. The switch to a male informant would have arisen in cases where the natural mother no longer lived with the child, and where the father elected to be treated as the main carer.

At MCS4, a Main interview was conducted in 13,797 of the 13,857 families. There was someone eligible for a Partner interview in 10,687 families and an interview was completed in 9,180 of these, with proxy data collected in a further 249 cases.

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<sup>3</sup> There were 2 families in which the person who should have been eligible for the Main interview actually completed the Partner interview, and there was no-one else eligible for interview.

Information was not collected from partners in 1,484 couples where the Main responded. A further 19 families had information from the partner but not from the Main respondent. In 41 cases there were no Parent interviews. The proportion of informants eligible to respond as Main, and who were natural mothers, dropped slightly from 97.0 per cent at MCS3 to 96.6 per cent (13,392). The number of natural fathers eligible to complete the Main interview at MCS4 was 392, which hardly changed from MCS3. Ninety-nine of these (compared with only 72 at MCS3) were lone fathers and the rest were part of a couple.

## Implications

One implication of these patterns is that researchers cannot automatically assume that the Main informant is the same person across sweeps, or even that she is the child's natural mother. This is increasingly unlikely to be a good approximation of the truth as time goes by.

Another implication is that, although there are some examples of parental arrangements other than one or two natural parent families, such as fostered, adoptive and step-families, there are insufficient numbers in MCS to date for separate analysis. Those interested in same-sex partnerships as a contemporary family form should be aware that the number of pairs of same-sex respondents was 4 at MCS1; 8 at MCS2, 10 at MCS3; and 12 at MCS4.

In the vast majority of cases at all sweeps, the natural mother did the Main interview and the natural father the Partner interview. There are derived variables on the Parent interview data which give details of the identity and interview status for Main and Partner respondents: **admres00**, **adpres00**, **bdmres00**, **bdpres00**, **cdmres00**, **cdpres00**, **ddmres00**, **ddpres00**.

On the Household Grid files, the Main and Partner respondents and their interview status are identified by the variables **ahelig00** and **ahresp00** (MCS1), **bhelig00** and **bhresp00** (MCS2), **chelig00** and **chresp00** (MCS3), **dhelig00** and **dhresp00** (MCS4).

At MCS1, 3 and 4, the identity of the person eligible for the Main and Partner interviews was derived from the household grid and available for all families (regardless of whether or not the individual interviews were completed). At MCS2, the identity of the individuals eligible for the Main and Partner interviews was not known if the interview was not conducted. As discussed above, where the main interview was not carried out at MCS2, we indicated that the Main respondent from MCS1 was present and eligible for the Main interview. Where the Main interview was not done at MCS2, household composition information was not collected; so unless a partner interview was done, there was no-one recorded as eligible for the Partner interview. In households in which the Main interview was done but there was no Partner interview, the person eligible for the Partner interview was derived using relationships between household members. In these families, the Partner was

assumed to be eligible for interview in person (rather than by proxy). This explains why the number eligible and responding to the Partner Proxy interview are identical.

## 4. Cohort Member Cognitive Assessments

A number of assessments have been administered to the MCS children since they were aged 3. The following assessments were administered to the MCS children at different sweeps:

### Assessments by Sweep Collected

| Assessment               | MCS Sweep |       |       |
|--------------------------|-----------|-------|-------|
|                          | MCS 2     | MCS 3 | MCS 4 |
| BAS Naming Vocabulary    | X         | X     |       |
| Bracken School Readiness | X         |       |       |
| BAS Picture Similarity   |           | X     |       |
| BAS Pattern Construction |           | X     | X     |
| BAS Word Reading         |           |       | X     |
| NFER Number Skills       |           |       | X     |

## 4.1 The British Ability Scales

The British Ability Scales (BAS) is a battery of individually administered tests of cognitive abilities and educational achievements suitable for use with children and adolescents aged from 2 years 6 months to 7 years 11 months.

### 1) BAS Naming Vocabulary (MCS 2 and MCS 3)

Following consultation with advisers and piloting, the BAS Naming Vocabulary scale was administered by interviewers to cohort members during the MCS2 data collection.

The Naming Vocabulary is a verbal scale for children aged 2 years 6 months to 7 years 11 months. It assesses the spoken vocabulary of young children. The test items consist of a booklet of coloured pictures of objects which the child is shown one at a time and asked to name. The scale measures expressive language ability, and successful performance depends on the child's previous development of a vocabulary of nouns. Picture recognition is also crucial; however, the pictures are

large and brightly coloured and are unlikely to cause problems except for children with major visual impairments or with no experience of picture books. The items require the child to recall words from long-term memory rather than to recognise or understand the meaning of words or sentences.

## Scores

Naming Vocabulary scores may reflect:

- Expressive language skills
- Vocabulary knowledge of nouns
- Ability to attach verbal labels to pictures
- General knowledge
- General language development
- Retrieval of names from long-term memory
- Level of language stimulation.

Low scores may reflect reluctance to speak.

**The datasets provide the following scores:**

| Variable | Description                                    |
|----------|------------------------------------------------|
| bcnsco00 | S2 COG: Total score for Naming Vocabulary test |
| bdnvabil | S2 COG: Naming Vocabulary ability score        |
| bdnvtscr | S2 COG: Naming Vocabulary T-score              |
| ccnsco00 | S3 COG: Total score for Naming Vocabulary test |
| cdnvabil | S3 COG: Naming Vocabulary ability score        |
| cdnvtscr | S3 COG: Naming Vocabulary T-score              |

## 2) BAS Picture Similarity (MCS3)

Children are shown a row of 4 pictures on a page and asked to place a card with a fifth picture under the picture most similar to it. This assessment measures children's problem solving abilities.

**The dataset provides the following scores:**

| Variable | Description                                |
|----------|--------------------------------------------|
| ccpsco00 | S3 COG: Picture Similarity Total raw score |
| cdpsabil | S3 COG: Picture Similarity ability score   |
| cdpstscr | S3 COG: Picture Similarity T-score         |

**3) BAS Pattern Construction (MCS3 and MCS4)**

The child constructs a design by putting together flat squares or solid cubes with black and yellow patterns on each side. The child's score is based on accuracy and speed. This assessment tests spatial awareness but can also be used to observe dexterity and coordination, as well as traits like perseverance and determination.

**The dataset provides the following scores:**

| Variable | Description                                   |
|----------|-----------------------------------------------|
| cccsc00  | S3 COG: Total Score for Pattern Construction  |
| cdpcabil | S3 COG: Pattern Construction ability score    |
| Cdpctscr | S3 COG: Pattern Construction T-score          |
| dctots00 | S4 CM Pattern Construction Total Raw Score    |
| dcpcab00 | S4 CM Pattern Construction Ability Scores     |
| dcpcts00 | S4 CM Pattern Construction age-based T-Scores |

**4) BAS Word Reading (MCS 4)**

Word Reading is an assessment from the British Ability Scales: Second Edition (BAS 2) which assesses children's English reading ability.

The child reads aloud a series of words presented on a card. The assessment consists of 90 words in total. The words are organised into 9 blocks of 10 words in ascending order of difficulty. The child is asked to read each word in a block out loud to the interviewer. The number of blocks of words the child is asked to attempt to read is dependent on the child's performance during the assessment. This assessment is designed to be used with children aged from 5 years to 17 years and 11 months. All of the children in MCS4 started at the first item, as this was the starting point for children of their age.

A child's progression through the assessment is dependent on the number of words they read correctly. If a child makes 8 errors in a block of 10 words, then the assessment stops.

**The dataset provides the following scores:**

| Variable | Description                             |
|----------|-----------------------------------------|
| dcwrsc00 | S4 CM Total score for Word Reading test |
| dcwrab00 | S4 CM Word Reading Ability Score        |
| dcwrsd00 | S4 CM Word Reading Standard Score       |

In Wales a different test was carried out (see Section 4.3 below).

**Scores for the BAS assessments**

There are three types of score provided for each scale of the BAS: raw score, ability score and T-scores or standardised scores. Each type has its uses and limitations.

*Raw Scores*

Raw scores are simply the number of items the cohort member child answered correctly. They do not take into account the stop and start points of the items administered; for this reason, the raw scores have little meaning and should not be used.

*Ability Scores*

The ability scores are a transformation of the raw scores that take into account the specific item set administered. They are not adjusted for anything else, so are the scores to consult for unadjusted cognitive scores.

There are some issues to keep in mind when using ability scores. The first is that it not a truly continuous scale. The table below shows the correspondence between some example raw scores and ability scores. As can be seen from this table, there are ability scores that cannot be obtained.

Correspondence of raw scores to ability scores for an example BAS scale:

| Raw Score | Ability Score |
|-----------|---------------|
| 4         | 26            |
| 5         | 30            |
| 6         | 33            |
| 7         | 37            |

|   |    |
|---|----|
| 8 | 40 |
| 9 | 43 |

For convenience, the ability scores for each scale start with a value of 10, which reflects a raw score of 0 on the easiest possible set of items in a scale. The upper limit of ability scores varies from scale to scale. Because the ability scale uses an arbitrary numbering system, comparing ability scores from different scales is not meaningful, just as comparing raw scores from different scales is not meaningful.

The other issue is that the ability scores are not adjusted for age. Children of a large range of ages take the same BAS tests, and the general trend is that older children score higher. When using ability scores, one should control for child age. The issue of age and the BAS scales is discussed in further detail in the section below on BAS Scales and Age.

### *T-Scores and Standardised Scores*

Also available for all scales are T-scores or a standardised score. These scores are adjusted for the cohort member child's age group and for the mean scores of the BAS norming group. They are computed using the BAS manual's conversion tables. For each 3-month age group, there is a table showing the conversion of ability scores to T-scores or standardised scores. The T-scores have a mean of 50 and standard deviation of 10 within the norming sample of a given age group. A cohort child who has an ability score that is the same as the mean for the norming group in his or her age group will have a T-score of 50. A child with a T-score of 60 had an ability score that was one standard deviation above the norming sample mean for his or her age group.

All of the scales used with the MCS sample in sweeps 2 through 4 have T-scores, with the one exception of Word Reading at MCS 4. That scale has a standardised score rather than a T-score. The only difference between the standardised score and the T-scores is that the former does not have a mean of 50 and standard deviation of 10. It is otherwise computed the same as the T-score, adjusting for age group and norming sample mean and standard deviation.

There are pros and cons to using T-scores or standardised scores. While these scores take into account child age, they are based on 3-month age groupings of the norming sample. They don't take into account the score variation with each group of 3 months. They also are based on the relationship between age and score in the norming sample rather than within the MCS sample. Using the age of the MCS sample one is using as a control will be a more accurate adjustment for age than using the T-scores (see the section on age equivalence below for more information). However, if one is looking at univariate relationships and cannot control for MCS child age, it could be beneficial to use the T-scores or standardised scores, especially in cases in which the variables of interest may be related to child age.

As the T-scores and standardised scores remove the mean and standard deviation of the norming sample from each score, they may hide differences in variance at

different ages. If one is interested in how variance in BAS scores differs across age or sweep, one may want to avoid using the T-scores or standardised scores so that the actual variance in the sample is clear.

As was the case for the ability scores, the T-scores and standardised scores are not truly continuous.

Below is a list of variables by MCS sweep for the different score types:

### Score variables for the BAS Assessments by Sweep

| Assessment           | Sweep | Raw Score | Ability Score | T-Score  |
|----------------------|-------|-----------|---------------|----------|
| Naming Vocabulary    | MCS2  | bdbasr00  | bdbasa00      | bdbast00 |
|                      | MCS3  | ccnsco00  | cdnvabil      | cdnvtscr |
| Picture Similarity   | MCS3  | ccpsco00  | cdpsabil      | cdpstscr |
| Pattern Construction | MCS3  | cccsc00   | cdpcabil      | cdpctscr |
|                      | MCS4  | dctots00  | dcpcab00      | bcpcts00 |
| Word Reading         | MCS4  | dcwrsc00  | dcwrab00      | dcwrsc00 |

### Further information

For more information about the development, administration, scoring and interpretation of the BAS scores see:

Elliott, C.D., Smith, P, and McCulloch, K (1996). *British Ability Scales Second Edition (BAS II): Administration and Scoring Manual*. London: NFER-Nelson.

Elliott, C.D., Smith, P, and McCulloch, K (1997). *British Ability Scales Second Edition (BAS II): Technical Manual*. London: NFER-Nelson.

## 4.2 The Bracken School Readiness Score (MCS2)

The Bracken Basic Concept Scale – Revised (BBCS-R) is used to assess the basic concept development in children in the age range of 2 years 6 months to 7 years 11 months. BBCS-R measures the comprehension of 308 functionally relevant educational concepts in 11 subtests or concept categories. Following consultation with advisers and piloting, only subtests 1-6 were administered by interviewers to the members of the cohort during the MCS2 data collection.

The sub-tests administered together form the Bracken School Readiness Assessment (BSRA) which evaluates 88 concepts relating to:

1. **Colours:** represents both primary colours and basic colour terms.
2. **Letters:** measures knowledge of both upper- and lower-case letters.
3. **Numbers/Counting:** measures recognition of single- and double-digit numbers, and samples the ability to assign a number value to a set of objects.
4. **Sizes:** includes concepts that describe one, two, and three dimensions.
5. **Comparisons:** measures ability to match and/or differentiate objects based on one or more of their salient characteristics.
6. **Shapes:** includes one-, two-, and three-dimensional shapes. The one-dimensional category includes linear shapes; two-dimensional shapes are represented by concepts such as the circle, square, and triangle; and three-dimensional shapes include concepts such as the cube and pyramid.

The readiness concepts assessed in these sub-tests are argued to be directly related to early childhood education and to predict readiness for more formal education. The test is individually administered and suitable for children aged 2 years 6 months to 7 years 11 months.

## Scores

The dataset provides the following scores:

*Raw Scores:* The total number of correct answers for each of the six BRSA sub-tests.

| Variable | Description                               |
|----------|-------------------------------------------|
| bdcosc00 | S2 DV Bracken: Colours Raw Score          |
| bdlesc00 | S2 DV Bracken: Letters Raw Score          |
| bdnosc00 | S2 DV Bracken: Numbers Raw Score          |
| bdszsc00 | S2 DV Bracken: Sizes Raw Score            |
| bdcmsc00 | S2 DV Bracken: Comparisons Raw Score      |
| bdshsc00 | S2 DV Bracken: Shapes Raw Score           |
| bdbsrc00 | S2 DV Bracken: School Readiness Composite |

*Percentage mastery:* The raw score as a percentage of the maximum possible score for each sub-test.

| Variable | Description                          |
|----------|--------------------------------------|
| bdcmas00 | S2 DV Bracken: Colours % mastery     |
| bdlmas00 | S2 DV Bracken: Letters % mastery     |
| bdnmas00 | S2 DV Bracken: Numbers % mastery     |
| bdsmas00 | S2 DV Bracken: Size % mastery        |
| bdomas00 | S2 DV Bracken: Comparisons % mastery |
| bdhmas00 | S2 DV Bracken: Shapes % mastery      |

*School Readiness Composite Percentage mastery:* The total number of correct answers in all 6 sub-tests as a percentage of the maximum possible score (bdsrcom00).

### Age-adjusted scores

The following variables are derived from bdsrcom00 (School Readiness Composite) which is the total number of correct answers adjusted for age.

*School Readiness Composite Standard Score:* The total number of correct answers on all six sub-tests (bdsrcom00).

*Normed scores:* Derived from standard tables in the BSRA manual and defined with reference to the standardisation sample used in developing the assessments. The standardisation sample was composed of 1,100 children aged between 2 years 6 months and 8 years 0 months representative of the general US population and was stratified by age, gender, race/ethnicity and parental education:

- *Standard Scores* with mean of 15 and standard deviation of 15.
- *Percentile ranks* representing the percentage of children in the normative sample who are ranked at or below the child's score (bdsrcom00).
- *Descriptive Classification* of the normed scores into five categories: Very advanced; Advanced; Average; Delayed; and Very delayed (bdsrcom00).

### Further information

For more information about the development, administration, scoring and interpretation of the BSRA see:

Bracken, B.A. (2002). *Bracken School Readiness Assessment: Administration Manual*. San Antonio, Texas: Psychological Corporation.

### 4.3 NFER Number Skills (MCS4)

This test was adapted from the NFER Progress in Maths test which is aimed for 7-year-olds and was originally developed and nationally UK standardised in 2004. The whole test has a maximum raw score of 28. The national mean raw score in 2004 was 19.3 with a standard deviation of 5.3. The scores were nationally age standardised to a mean of 100 and SD of 15.

The edition of this test used in the MCS is an adaptive version of the test created by Cres Fernandes of NFER. All children have to complete an initial test and based on their score they are routed to easier, medium or harder sections. The sections were devised to save administration time, as it means each child completes around half the original number of questions.

An item response scaling method (Rasch) was used to scale the results of the easy, medium and hard subtest scores to the equivalent original raw scores. The variable **maths7scale** can be considered to be the estimated raw score based on the original test. The variable **maths7sas** is the standardised age adjusted score based on the national standardisation lookup tables in 2004.

The dataset provides the following scores:

| Variable | Description                                          |
|----------|------------------------------------------------------|
| mtotscor | S4 CM Number Skills Total Raw Score                  |
| maths7sc | S4 CM Estimated raw score based on original PM7 test |
| maths7sa | S4 CM Standardise age-adjusted score                 |

### 4.4 Our Adventures (MCS4) – Wales Only

Our Adventures is part of the All Wales Reading Test, which was developed in Wales to assess the reading skills of children in Welsh schools. The test is available in Welsh and English.

In MCS4, parents of children in Wales were given the option of having their child's reading skills assessed in either Welsh or English. The Welsh version of Our Adventures was used for children whose parents opted for the Welsh medium to be used, and the Word Reading assessment was used for children whose parents opted for the English medium to be used.

It was decided to use the Welsh medium All Wales Reading Test, rather than a Welsh translation of the Word Reading assessment because the Word Reading assessment is designed only to assess English reading ability and if translated the results are not valid.

The Our Adventures assessment is a paper booklet that shows a story in pictures and words; underneath each picture is a sentence that has one missing word, and a list of words that can complete the sentence. The child has to circle the word that best completes the sentence. There are a total of 59 items, and the assessment has a time limit of 30 minutes. The assessment continues until the time limit has been reached, or the child completes the last item.

This assessment is designed to be used with children from age 6 years 10 months to 9 years 9 months.

| <b>Variable</b> | <b>Description</b>      |
|-----------------|-------------------------|
| DCCSEX00        | S4 CM Cohort member Sex |
| DCOQ0100        | S4 CM Picture 1         |
| DCOQ0200        | S4 CM Picture 2         |
| DCOQ0300        | S4 CM Picture 3         |
| DCOQ0400        | S4 CM Picture 4         |
| DCOQ0500        | S4 CM Picture 5         |
| DCOQ0600        | S4 CM Picture 6         |
| DCOQ0700        | S4 CM Picture 7         |
| DCOQ0800        | S4 CM Picture 8         |
| DCOQ0900        | S4 CM Picture 9         |
| DCOQ1000        | S4 CM Picture 10        |
| DCOQ1100        | S4 CM Picture 11        |
| DCOQ1200        | S4 CM Picture 12        |
| DCOQ1300        | S4 CM Picture 13        |
| DCOQ1400        | S4 CM Picture 14        |
| DCOQ1500        | S4 CM Picture 15        |
| DCOQ1600        | S4 CM Picture 16        |
| DCOQ1700        | S4 CM Picture 17        |
| DCOQ1800        | S4 CM Picture 18        |
| DCOQ1900        | S4 CM Picture 19        |
| DCOQ2000        | S4 CM Picture 20        |
| DCOQ2100        | S4 CM Picture 21        |
| DCOQ2200        | S4 CM Picture 22        |
| DCOQ2300        | S4 CM Picture 23        |
| DCOQ2400        | S4 CM Picture 24        |
| DCOQ2500        | S4 CM Picture 25        |
| DCOQ2600        | S4 CM Picture 26        |
| DCOQ2700        | S4 CM Picture 27        |

| Variable | Description      |
|----------|------------------|
| DCOQ2800 | S4 CM Picture 28 |
| DCOQ2900 | S4 CM Picture 29 |
| DCOQ3000 | S4 CM Picture 30 |
| DCOQ3100 | S4 CM Picture 31 |
| DCOQ3200 | S4 CM Picture 32 |
| DCOQ3300 | S4 CM Picture 33 |
| DCOQ3400 | S4 CM Picture 34 |
| DCOQ3500 | S4 CM Picture 35 |
| DCOQ3600 | S4 CM Picture 36 |
| DCOQ3700 | S4 CM Picture 37 |
| DCOQ3800 | S4 CM Picture 38 |
| DCOQ3900 | S4 CM Picture 39 |
| DCOQ4000 | S4 CM Picture 40 |
| DCOQ4100 | S4 CM Picture 41 |
| DCOQ4200 | S4 CM Picture 42 |
| DCOQ4300 | S4 CM Picture 43 |
| DCOQ4400 | S4 CM Picture 44 |
| DCOQ4500 | S4 CM Picture 45 |
| DCOQ4600 | S4 CM Picture 46 |
| DCOQ4700 | S4 CM Picture 47 |
| DCOQ4800 | S4 CM Picture 48 |
| DCOQ4900 | S4 CM Picture 49 |
| DCOQ5000 | S4 CM Picture 50 |
| DCOQ5100 | S4 CM Picture 51 |
| DCOQ5200 | S4 CM Picture 52 |
| DCOQ5300 | S4 CM Picture 53 |
| DCOQ5400 | S4 CM Picture 54 |
| DCOQ5500 | S4 CM Picture 55 |
| DCOQ5600 | S4 CM Picture 56 |
| DCOQ5700 | S4 CM Picture 57 |
| DCOQ5800 | S4 CM Picture 58 |

The dataset provides the following scores:

### *Age Equivalents*

Age equivalents are not provided in the datasets, but users may wish to compute their own. Tables 24 to 30 below give conversions from ability scores to age equivalents for the assessments. **It is very important that users read and understand the limitations of these estimated age equivalents before using them.**

The tables contain the median ability score for the 3-month age groups that are around the ages of the cohort members at MCS2, MCS3, and MCS4. These

numbers are taken from the BAS II manual and the Bracken manual which are unfortunately not available to the public.

The gap between each age group and the previous age group are computed (Gap) and the average of these gaps is taken to get the average ability-score difference for children 3-months apart in age. This is then divided by 3 to get the average ability score difference for a 1-month age difference.

As the method uses differences at the median, averaged across a range of ages, the resulting “differences in age equivalents” are very much rough estimates. Any use of these should be accompanied by an explanation that they are rough estimates and should be interpreted carefully.

This method of using average change per month over a range of ages relies on assumptions about the data that are not necessarily true. The main assumption is that the change in score that is seen with a change of 1 month in age is consistent across a range of ages. However, examination of the MCS data shows that this is not the case. Older children tend to score higher on the BAS scales; but the increase in score by increase in age is not consistent across the age range. This is also evident in the tables provided: for example, 3-month score gaps range from 3 to 5 on Naming Vocabulary at MCS2 and 5 to 8.5 on Word Reading at MCS4.

The average gaps are also based on the differences in median scores at different ages. The gaps by age at other points of the distribution (e.g. a standard deviation below or above the median) may not necessarily be the same as the gaps at the median.

In addition, other contextual variables such as school experience may influence scores more than age alone. We looked at the BAS scores across age and sweep. The older children at sweep 2 were close in age to the younger children at sweep 3. We found that the increase in score between the two sweeps was larger than one would expect based on increase in age alone. That is, the younger children at sweep 3 scored much higher than the older children at sweep 2, even though the age difference between them would have predicted a much smaller difference. This suggests that other factors may have influenced their scores. Though people may more readily understand scores reported as age equivalents, and score is indeed related to age, it is important to remember that other factors are also strongly related to scores and that there is great variation in scores for a given age group.

Tables 24-30 provide the age equivalence tables for the various assessments.

**Table 24: MCS2 Bracken School Readiness**

| Ability score gap | Age equiv (months) | Age group | Median ability score | Gap | Average 3-month gap | Average 1-month gap |
|-------------------|--------------------|-----------|----------------------|-----|---------------------|---------------------|
| 1                 | 1.04               | 2y6m      | 14                   |     |                     |                     |
| 2                 | 2.08               | 2y9m      | 16.5                 | 2.5 | 3.125               | 1.04                |
| 3                 | 3.13               | 3y0m      | 20                   | 3.5 |                     |                     |
| 4                 | 4.17               | 3y3m      | 24                   | 4   |                     |                     |
| 5                 | 5.21               | 3y6m      | 26.5                 | 2.5 |                     |                     |
| 6                 | 6.25               |           |                      |     |                     |                     |
| 7                 | 7.29               |           |                      |     |                     |                     |
| 8                 | 8.33               |           |                      |     |                     |                     |
| 9                 | 9.38               |           |                      |     |                     |                     |
| 10                | 10.42              |           |                      |     |                     |                     |
| 11                | 11.46              |           |                      |     |                     |                     |
| 12                | 12.50              |           |                      |     |                     |                     |
| 13                | 13.54              |           |                      |     |                     |                     |
| 14                | 14.58              |           |                      |     |                     |                     |
| 15                | 15.63              |           |                      |     |                     |                     |
| 16                | 16.67              |           |                      |     |                     |                     |

**Table 25: MCS2 BAS Naming Vocabulary**

| Ability score gap | Age equiv (months) | Age group | Median ability score | Gap | Average 3-month gap | Average 1-month gap |
|-------------------|--------------------|-----------|----------------------|-----|---------------------|---------------------|
| 1                 | 1.25               | 2y6m      | 68                   |     |                     |                     |
| 2                 | 2.5                | 2y9m      | 71                   | 3   | 3.75                | 1.25                |
| 3                 | 3.75               | 3y0m      | 76                   | 5   |                     |                     |
| 4                 | 5                  | 3y3m      | 80                   | 4   |                     |                     |

| Ability score gap | Age equiv (months) | Age group | Median ability score | Gap | Average 3-month gap | Average 1-month gap |
|-------------------|--------------------|-----------|----------------------|-----|---------------------|---------------------|
| 5                 | 6.25               | 3y6m      | 83                   | 3   |                     |                     |
| 6                 | 7.5                |           |                      |     |                     |                     |
| 7                 | 8.75               |           |                      |     |                     |                     |
| 8                 | 10                 |           |                      |     |                     |                     |
| 9                 | 11.25              |           |                      |     |                     |                     |
| 10                | 12.5               |           |                      |     |                     |                     |
| 11                | 13.75              |           |                      |     |                     |                     |
| 12                | 15                 |           |                      |     |                     |                     |
| 13                | 16.25              |           |                      |     |                     |                     |
| 14                | 17.5               |           |                      |     |                     |                     |
| 15                | 18.75              |           |                      |     |                     |                     |
| 16                | 20                 |           |                      |     |                     |                     |
| 17                | 21.25              |           |                      |     |                     |                     |
| 18                | 22.5               |           |                      |     |                     |                     |
| 19                | 23.75              |           |                      |     |                     |                     |
| 20                | 25                 |           |                      |     |                     |                     |
| 21                | 26.25              |           |                      |     |                     |                     |
| 22                | 27.5               |           |                      |     |                     |                     |
| 23                | 28.75              |           |                      |     |                     |                     |
| 24                | 30                 |           |                      |     |                     |                     |
| 25                | 31.25              |           |                      |     |                     |                     |

**Table 26: MCS3 BAS Picture Similarity**

| Ability score gap | Age equiv (months) | Age equiv (months) | Ability score gap | Age group | Median ability score | Gap | Average 3-month gap | Average 1-month gap |
|-------------------|--------------------|--------------------|-------------------|-----------|----------------------|-----|---------------------|---------------------|
| 1                 | 1.61               | 1                  | 0.62              |           |                      |     |                     |                     |
| 2                 | 3.23               | 2                  | 1.24              | 4:3       | 70                   |     |                     |                     |
| 3                 | 4.84               | 3                  | 1.86              | 4:6       | 73                   | 3   |                     |                     |
| 4                 | 6.45               | 4                  | 2.48              | 4:9       | 75                   | 2   |                     |                     |
| 5                 | 8.06               | 5                  | 3.10              | 5:0       | 77                   | 2   |                     |                     |
| 6                 | 9.68               | 6                  | 3.71              | 5:3       | 79                   | 2   | 1.86                | 0.62                |
| 7                 | 11.29              | 7                  | 4.33              | 5:6       | 81                   | 2   |                     |                     |
| 8                 | 12.90              | 8                  | 4.95              | 5:9       | 82.5                 | 1.5 |                     |                     |
| 9                 | 14.52              | 9                  | 5.57              | 6:0       | 83                   | 0.5 |                     |                     |
| 10                | 16.13              | 10                 | 6.19              |           |                      |     |                     |                     |
| 11                | 17.74              | 11                 | 6.81              |           |                      |     |                     |                     |
| 12                | 19.35              | 12                 | 7.43              |           |                      |     |                     |                     |
| 13                | 20.97              | 13                 | 8.05              |           |                      |     |                     |                     |
| 14                | 22.58              | 14                 | 8.67              |           |                      |     |                     |                     |
| 15                | 24.19              | 15                 | 9.29              |           |                      |     |                     |                     |
| 16                | 25.81              | 16                 | 9.90              |           |                      |     |                     |                     |
| 17                | 27.42              | 17                 | 10.52             |           |                      |     |                     |                     |
| 18                | 29.03              | 18                 | 11.14             |           |                      |     |                     |                     |
| 19                | 30.65              | 19                 | 11.76             |           |                      |     |                     |                     |
| 20                | 32.26              | 20                 | 12.38             |           |                      |     |                     |                     |
| 21                | 33.87              | 21                 | 13.00             |           |                      |     |                     |                     |
| 22                | 35.48              | 22                 | 13.62             |           |                      |     |                     |                     |
| 23                | 37.10              | 23                 | 14.24             |           |                      |     |                     |                     |
| 24                | 38.71              | 24                 | 14.86             |           |                      |     |                     |                     |

| Ability score gap | Age equiv (months) | Age equiv (months) | Ability score gap | Age group | Median ability score | Gap | Average 3-month gap | Average 1-month gap |
|-------------------|--------------------|--------------------|-------------------|-----------|----------------------|-----|---------------------|---------------------|
| 25                | 40.32              | 25                 | 15.48             |           |                      |     |                     |                     |

**Table 27: MCS3 BAS Naming Vocabulary**

| Ability score gap | Age equiv (months) | Age equiv (months) | Ability score gap | Age group | Median ability score | Gap | Average 3-month gap | Average 1-month gap |
|-------------------|--------------------|--------------------|-------------------|-----------|----------------------|-----|---------------------|---------------------|
| 1                 | 1.20               | 1                  | 0.83              |           |                      |     |                     |                     |
| 2                 | 2.40               | 2                  | 1.67              | 4:3       | 92.5                 |     |                     |                     |
| 3                 | 3.60               | 3                  | 2.50              | 4:6       | 95.5                 | 3   |                     |                     |
| 4                 | 4.80               | 4                  | 3.33              | 4:9       | 99                   | 3.5 |                     |                     |
| 5                 | 6.00               | 5                  | 4.17              | 5:0       | 103                  | 4   |                     |                     |
| 6                 | 7.20               | 6                  | 5.00              | 5:3       | 105.5                | 2.5 | 2.50                | 0.83                |
| 7                 | 8.40               | 7                  | 5.83              | 5:6       | 107                  | 1.5 |                     |                     |
| 8                 | 9.60               | 8                  | 6.67              | 5:9       | 108.5                | 1.5 |                     |                     |
| 9                 | 10.80              | 9                  | 7.50              | 6:0       | 110                  | 1.5 |                     |                     |
| 10                | 12.00              | 10                 | 8.33              |           |                      |     |                     |                     |
| 11                | 13.20              | 11                 | 9.17              |           |                      |     |                     |                     |
| 12                | 14.40              | 12                 | 10.00             |           |                      |     |                     |                     |
| 13                | 15.60              | 13                 | 10.83             |           |                      |     |                     |                     |
| 14                | 16.80              | 14                 | 11.67             |           |                      |     |                     |                     |
| 15                | 18.00              | 15                 | 12.50             |           |                      |     |                     |                     |
| 16                | 19.20              | 16                 | 13.33             |           |                      |     |                     |                     |
| 17                | 20.40              | 17                 | 14.17             |           |                      |     |                     |                     |
| 18                | 21.60              | 18                 | 15.00             |           |                      |     |                     |                     |
| 19                | 22.80              | 19                 | 15.83             |           |                      |     |                     |                     |
| 20                | 24.00              | 20                 | 16.67             |           |                      |     |                     |                     |
| 21                | 25.20              | 21                 | 17.50             |           |                      |     |                     |                     |

|    |       |    |       |  |  |  |  |  |
|----|-------|----|-------|--|--|--|--|--|
| 22 | 26.40 | 22 | 18.33 |  |  |  |  |  |
| 23 | 27.60 | 23 | 19.17 |  |  |  |  |  |
| 24 | 28.80 | 24 | 20.00 |  |  |  |  |  |
| 25 | 30.00 | 25 | 20.83 |  |  |  |  |  |

**Table 28: MCS3 BAS Pattern Construction**

| Ability score gap | Age equiv (months) | Age equiv (months) | Ability score gap | Age group | Median ability score | Gap | Average 3-month gap | Average 1-month gap |
|-------------------|--------------------|--------------------|-------------------|-----------|----------------------|-----|---------------------|---------------------|
| 1                 | 0.55               | 1                  | 1.81              |           |                      |     |                     |                     |
| 2                 | 1.11               | 2                  | 3.62              | 4:3       | 68.5                 |     |                     |                     |
| 3                 | 1.66               | 3                  | 5.43              | 4:6       | 74.5                 | 6   |                     |                     |
| 4                 | 2.21               | 4                  | 7.24              | 4:9       | 81.5                 | 7   |                     |                     |
| 5                 | 2.76               | 5                  | 9.05              | 5:0       | 85.5                 | 4   |                     |                     |
| 6                 | 3.32               | 6                  | 10.86             | 5:3       | 92.5                 | 7   | 5.43                | 1.81                |
| 7                 | 3.87               | 7                  | 12.67             | 5:6       | 97.5                 | 5   |                     |                     |
| 8                 | 4.42               | 8                  | 14.48             | 5:9       | 103.5                | 6   |                     |                     |
| 9                 | 4.97               | 9                  | 16.29             | 6:0       | 106.5                | 3   |                     |                     |
| 10                | 5.53               | 10                 | 18.10             |           |                      |     |                     |                     |
| 11                | 6.08               | 11                 | 19.90             |           |                      |     |                     |                     |
| 12                | 6.63               | 12                 | 21.71             |           |                      |     |                     |                     |
| 13                | 7.18               | 13                 | 23.52             |           |                      |     |                     |                     |
| 14                | 7.74               | 14                 | 25.33             |           |                      |     |                     |                     |
| 15                | 8.29               | 15                 | 27.14             |           |                      |     |                     |                     |
| 16                | 8.84               | 16                 | 28.95             |           |                      |     |                     |                     |
| 17                | 9.39               | 17                 | 30.76             |           |                      |     |                     |                     |
| 18                | 9.95               | 18                 | 32.57             |           |                      |     |                     |                     |
| 19                | 10.50              | 19                 | 34.38             |           |                      |     |                     |                     |
| 20                | 11.05              | 20                 | 36.19             |           |                      |     |                     |                     |

|    |       |    |       |  |  |  |  |  |
|----|-------|----|-------|--|--|--|--|--|
| 21 | 11.61 | 21 | 38.00 |  |  |  |  |  |
| 22 | 12.16 | 22 | 39.81 |  |  |  |  |  |
| 23 | 12.71 | 23 | 41.62 |  |  |  |  |  |
| 24 | 13.26 | 24 | 43.43 |  |  |  |  |  |
| 25 | 13.82 | 25 | 45.24 |  |  |  |  |  |

**Table 29: MCS4 Word reading**

| Ability score gap | Age equiv (mths) | Age equiv (mths) | Ability score gap | Age Group | Median Ability score | Gap | Average 3-month gap | Average 1-month gap |
|-------------------|------------------|------------------|-------------------|-----------|----------------------|-----|---------------------|---------------------|
| 1                 | 0.51             | 1                | 1.98              | 6y3m      | 71.5                 |     |                     |                     |
| 2                 | 1.01             | 2                | 3.95              | 6y6m      | 77.0                 | 5.5 | 5.93                | 1.98                |
| 3                 | 1.52             | 3                | 5.93              | 6y9m      | 82.5                 | 5.5 |                     |                     |
| 4                 | 2.02             | 4                | 7.90              | 7y0m      | 88.0                 | 5.5 |                     |                     |
| 5                 | 2.53             | 5                | 9.88              | 7y3m      | 93.5                 | 5.5 |                     |                     |
| 6                 | 3.04             | 6                | 11.86             | 7y6m      | 99.5                 | 6   |                     |                     |
| 7                 | 3.54             | 7                | 13.83             | 7y9m      | 104.5                | 5   |                     |                     |
| 8                 | 4.05             | 8                | 15.81             | 8y0m      | 113.0                | 8.5 |                     |                     |
| 9                 | 4.55             | 9                | 17.79             |           |                      |     |                     |                     |
| 10                | 5.06             | 10               | 19.76             |           |                      |     |                     |                     |
| 11                | 5.57             | 11               | 21.74             |           |                      |     |                     |                     |
| 12                | 6.07             | 12               | 23.71             |           |                      |     |                     |                     |
| 13                | 6.58             | 13               | 25.69             |           |                      |     |                     |                     |
| 14                | 7.08             | 14               | 27.67             |           |                      |     |                     |                     |
| 15                | 7.59             | 15               | 29.64             |           |                      |     |                     |                     |
| 16                | 8.10             | 16               | 31.62             |           |                      |     |                     |                     |
| 17                | 8.60             | 17               | 33.60             |           |                      |     |                     |                     |

| Ability score gap | Age equiv (mths) | Age equiv (mths) | Ability score gap | Age Group | Median Ability score | Gap | Average 3-month gap | Average 1-month gap |
|-------------------|------------------|------------------|-------------------|-----------|----------------------|-----|---------------------|---------------------|
| 18                | 9.11             | 18               | 35.57             |           |                      |     |                     |                     |
| 19                | 9.61             | 19               | 37.55             |           |                      |     |                     |                     |
| 20                | 10.12            | 20               | 39.52             |           |                      |     |                     |                     |
| 21                | 10.63            | 21               | 41.50             |           |                      |     |                     |                     |
| 22                | 11.13            | 22               | 43.48             |           |                      |     |                     |                     |
| 23                | 11.64            | 23               | 45.45             |           |                      |     |                     |                     |
| 24                | 12.14            | 24               | 47.43             |           |                      |     |                     |                     |
| 25                | 12.65            | 25               | 49.40             |           |                      |     |                     |                     |

**Table 30: MCS4 Pattern Construction**

| Ability score gap | Age equiv (mths) | Age equiv (mths) | Ability score gap | Age Group | Median Ability score | Gap | Average 3-month gap | Average 1-month gap |
|-------------------|------------------|------------------|-------------------|-----------|----------------------|-----|---------------------|---------------------|
| 1                 | 2.33             | 1                | 0.43              | 6y3m      | 110.0                |     |                     |                     |
| 2                 | 4.67             | 2                | 0.86              | 6y6m      | 110.0                | 0   | 1.29                | 0.43                |
| 3                 | 7.00             | 3                | 1.29              | 6y9m      | 110.0                | 0   |                     |                     |
| 4                 | 9.33             | 4                | 1.71              | 7y0m      | 111.0                | 1   |                     |                     |
| 5                 | 11.67            | 5                | 2.14              | 7y3m      | 113.0                | 2   |                     |                     |
| 6                 | 14.00            | 6                | 2.57              | 7y6m      | 114.5                | 1.5 |                     |                     |
| 7                 | 16.33            | 7                | 3.00              | 7y9m      | 116.0                | 1.5 |                     |                     |
| 8                 | 18.67            | 8                | 3.43              | 8y0m      | 119.0                | 3   |                     |                     |
| 9                 | 21.00            | 9                | 3.86              |           |                      |     |                     |                     |
| 10                | 23.33            | 10               | 4.29              |           |                      |     |                     |                     |
| 11                | 25.67            | 11               | 4.71              |           |                      |     |                     |                     |
| 12                | 28.00            | 12               | 5.14              |           |                      |     |                     |                     |

| Ability score gap | Age equiv (mths) | Age equiv (mths) | Ability score gap | Age Group | Median Ability score | Gap | Average 3-month gap | Average 1-month gap |
|-------------------|------------------|------------------|-------------------|-----------|----------------------|-----|---------------------|---------------------|
| 13                | 30.33            | 13               | 5.57              |           |                      |     |                     |                     |
| 14                | 32.67            | 14               | 6.00              |           |                      |     |                     |                     |
| 15                | 35.00            | 15               | 6.43              |           |                      |     |                     |                     |
| 16                | 37.33            | 16               | 6.86              |           |                      |     |                     |                     |
| 17                | 39.67            | 17               | 7.29              |           |                      |     |                     |                     |
| 18                | 42.00            | 18               | 7.71              |           |                      |     |                     |                     |
| 19                | 44.33            | 19               | 8.14              |           |                      |     |                     |                     |
| 20                | 46.67            | 20               | 8.57              |           |                      |     |                     |                     |
| 21                | 49.00            | 21               | 9.00              |           |                      |     |                     |                     |
| 22                | 51.33            | 22               | 9.43              |           |                      |     |                     |                     |
| 23                | 53.67            | 23               | 9.86              |           |                      |     |                     |                     |
| 24                | 56.00            | 24               | 10.29             |           |                      |     |                     |                     |
| 25                | 58.33            | 25               | 10.71             |           |                      |     |                     |                     |

There is no equivalent table for the Number Skills.

## General influences on test scores

It is important to note that the child's performance may have been affected by influences extraneous to those that the assessment is intended to measure. The conditions listed below can lead either to a higher or lower score than would normally be obtained.

- |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Non-standard administration of the scale</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>The specification of CAPI program and the training of interviewers was designed to ensure standard administration of the assessment.</p>                                                                                              |
| <ul style="list-style-type: none"> <li>• Non-standard scoring</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Scoring algorithms used ensure standard scoring in all cases.</p>                                                                                                                                                                     |
| <ul style="list-style-type: none"> <li>• Administration disrupted by noise or other interruptions</li> <li>• Difficulty in establishing rapport with the child</li> <li>• Child has difficulty in concentrating on the tasks or is easily distracted</li> <li>• Child is excessively anxious to the extent that concentration/flexibility of thought seems impaired</li> <li>• Child is reluctant to respond and/or refuses to persevere on more difficult items</li> <li>• Child has permanent/temporary sensory impairment (particularly vision/hearing) or motor impairment</li> <li>• Child is on medication of a type that could affect performance</li> <li>• Child is overtired or ill</li> </ul> | <p>The training of interviewers was designed to ensure that risks were minimised.</p> <p>Details of any interruptions, distractions, behaviours, health circumstances, etc., were recorded in CAPI by interviewers. See table below.</p> |

## Distractions that may impact assessment

Below is a list of variables that include information in distractions that may impact on the assessments:

**Table 31: Assessment distractions**

| Description                                                   | Variable name at MCS 2* |
|---------------------------------------------------------------|-------------------------|
| S2 COG Anyone else present during Assessment?                 | bcapre00                |
| S2 COG Was Mother present?                                    | bcmpr00                 |
| S2 COG Was father present?                                    | bcfpre00                |
| S2 COG Were other adults present?                             | bcopre00                |
| S2 COG Were other children present?                           | bccpre00                |
| S2 COG None of the above.                                     | bcnoab00                |
| S2 COG Number of other adults.                                | bcnoad00                |
| S2 COG Number of other children.                              | bcnoch00                |
| S2 COG Effect on child's performance of mother present.       | bcchpm00                |
| S2 COG Effect on child's performance of father present.       | bcchpf00                |
| S2 COG Effect on child's performance of adults present.       | bcchpo00                |
| S2 COG Effect on child's performance of other children there. | bcchpc00                |
| S2 COG What was child's energy level during BAS Assessment    | bcener00                |

\* Equivalent data are not currently available for MCS3 and 4.

## 5. Cohort Member Behavioural Development

At MCS2, 3 and 4 the Strengths and Difficulties Questionnaire (SDQ) is part of the **main self-completion instrument**.

The SDQ is a behavioural screening questionnaire for 3- to 16-year-olds. It measures **25 items on psychological attributes**.

The Main respondent is asked to comment on the following statements with: Not true, Somewhat true or Certainly true.

### i) Emotion Symptoms Scale

1. Complains of headaches/stomach aches/sickness

2. Often seems worried
3. Often unhappy
4. Nervous or clingy in new situations
5. Many fears, easily scared.

*ii) Conduct problems*

1. Often has temper tantrums
2. Generally obedient\*
3. Fights with or bullies other children
4. Can be spiteful to others
5. Often argumentative with adults.

*ii). Hyperactivity Scale*

1. Restless, overactive, cannot stay still for long
2. Constantly fidgeting
3. Easily distracted
4. Can stop and think before acting\*
5. Sees tasks through to the end\*.

*iv) Peer Problems*

1. Tends to play alone
2. Has at least one good friend\*
3. Generally liked by other children\*
4. Picked on or bullied by other children
5. Gets on better with adults.

*v) Pro-social Scale*

1. Considerate of others' feelings
2. Shares readily with others
3. Helpful if someone is hurt, upset or ill
4. Kind to younger children
5. Often volunteers to help others.

*\* Denotes items that are reversed – when generating sub scales on behaviour problems.*

Each of the 5 sub-scales can be used alone or together to create:

- 1-4 when taken together generates a **total difficulties score**.
- 1 and 4 create an **internalising problem score**.
- 2 and 3 create an **externalising conduct score**.
- 5 alone measures **pro-social behaviour**.

## Further information

For more information about the scoring and interpretation of the Strengths and Difficulties Instrument see:

Goodman, R. (1997). 'The Strengths and Difficulties Questionnaire: A Research Note.' *Journal of Child Psychology and Psychiatry*. 38: 581-586.

\_\_\_\_ (2001), 'Psychometric properties of the Strengths and Difficulties Questionnaire (SDQ).' *Journal of the American Academy of Child and Adolescent Psychiatry*. 40: 1337-1345.

Goodman, R., Meltzer, H. and Bailey, V. (1998). 'The Strengths and Difficulties Questionnaire: A pilot study on the validity of the self-report version.' *European Child and Adolescent Psychiatry*. 7: 125-130.

## 6. Cohort Member Physical Measurement

### Height

The original height variables – **byhtcm00** and **byhtmm00** (MCS2); **cyhtcm00** and **cyhtmm00** (MCS3); and **dhtcm00** (MCS4) – have not been edited.

Copies of the variables were made – **bdhcmc00** and **bdhmmc00** (MCS2); **cdhcmc00** and **cdhmmc00** (MCS3); and **dchtdv00** (MCS4) – and appropriate changes were made to them as follows:

- Where interviewer notes gave clear warnings that the height values entered were incorrect, the values were removed from **bdhcmc00**, **bdhmmc00**, **cdhcmc00**, **cdhmmc00** and **dchtdv00**.
- Where the interviewer notes gave a value to replace an incorrect entry, these were changed in **bdhcmc00**, **bdhmmc00**, **cdhcmc00**, **cdhmmc00** and **dchtdv00**.
- The variables **bdhtam00** and **cdhtam00** are flags to show if any changes were made. There were very few interviewer comments at MCS4 relating to measurements.

A variable was included to categorise the “outcome” of each height measurement based on the amended variables. The original variables **byhtri0a** and **cyhtri0a** record whether or not the measurement was “successful.” However, many cases where this was flagged as “unsuccessful” still have a height measurement recorded. Researchers may wish to use the variables **bdhtoc00** and **cdhtoc00** to exclude these cases (as well as cases containing historic data) from analyses.

At MCS4, the variable **dchtis00** indicates whether “measurement circumstances” (**dchtrz0a** to **dchtrz0d**) and/or “other information” (**dchtex0a** and **dchtex0b**) was given in relation to the height measurement, and flags up the highest and lowest 100 or so values where no other circumstances are mentioned.

## Weight

The original weight variables – **bywtcm00** and **bywtgm00** (MCS2); **cywtcm00**, **cywtgm00** (MCS3); and **dcwtcm00** (MCS4) – were not edited.

Copies of the variables were made – **bdwtkc00** and **bdwtgc00** (MCS2); **cdwtkc00** and **cdwtgc00** (MCS3); and **dcwtdv00** (MCS4) – and appropriate changes were made to them as follows:

- Where interviewer notes gave clear warnings that the weight values entered were incorrect, the values were removed from **bdwtkc00**, **bdwtgc00**, **cdwtkc00**, **cdwtgc00** and **dcwtdv00**.
- Where the interviewer notes gave a value to replace an incorrect entry, these were changed in **bdwtkc00**, **bdwtgc00**, **cdwtkc00** and **cdwtgc00**. There were very few interviewer comments at MCS4 relating to measurements.
- The variables **bdwtam00** (MCS2) and **cdwtam00** (MCS3) are flags to show if any changes were made.

A variable was included to categorise the “outcome” of each weight measurement, based on the amended variables. Researchers may wish to use the variables **bdwtoc00** (MCS2) and **cdwtoc00** (MCS3) to exclude cases containing historic data from analyses.

At MCS4, the variable **dcwtis00** indicated whether “measurement circumstances” (**dcwtrz0a** to **dcwtrz0d**) and/or “other information” (**dcwtex0a** and **dcwtex0b**) was given in relation to the weight measurement, and flags up the highest and lowest 100 or so values where no other circumstances are mentioned.

## Height, weight and BMI

The amended height and weight variables were used to calculate BMI.

The formula to compute BMI is weight (in kilos) divided by height squared (height measured in metres). This is computed for cases where we have a valid value given for both height and weight, and will be missing if either or both measurement is missing.

## Outliers

All height and weight observations have been included in the data, even where they might be considered outliers. All observations have been used to calculate the BMI measure. We leave it to individual researchers to take decisions on whether they consider any of the measurements to be outliers and what they do with such observations. Users should be warned that the dataset contains a few values that other users have considered implausible.

## 7. Income data

The MCS has collected income in a number of different ways over the different sweeps. At sweeps 1-4 income data were collected in a single banded question in addition a set of detailed questions which collected information on a range of different measures detailed in Table 32.

**Table 32: Income data collection across the cohort studies**

| Income                                  | Collection at MCS 1 |         | Collection at MCS 2 |         | Collection at MCS 3 |         | Collection at MCS 4 |         |
|-----------------------------------------|---------------------|---------|---------------------|---------|---------------------|---------|---------------------|---------|
|                                         | Main                | Partner | Main                | Partner | Main                | Partner | Main                | Partner |
| Gross Earnings                          | ✓                   | ✓       | ✓                   | ✓       | ✓                   | ✓       | ✓                   | ✓       |
| Net Earnings                            | ✓                   | ✓       | ✓                   | ✓       | ✓                   | ✓       | ✓                   | ✓       |
| Usual net Earnings                      |                     |         |                     |         | ✓                   | ✓       | ✓                   | ✓       |
| Earnings from second job                | ✓                   | ✓       | ✓                   | ✓       | ✓                   | ✓       | ✓                   | ✓       |
| Irregular earnings from occasional work | ✓                   | ✓       | ✓                   | ✓       | ✓                   | ✓       | ✓                   | ✓       |
| Earnings from Self-employment           | ✓                   | ✓       | ✓                   | ✓       | ✓                   | ✓       | ✓                   | ✓       |
| Child benefit                           | -                   |         | -                   |         | ✓                   | ✓       | ✓                   | ✓       |
| Guardian's Allowance                    | -                   |         | -                   |         | ✓                   |         | ✓                   | ✓       |
| Carer's allowance                       | -                   |         | -                   |         | ✓                   | ✓       | ✓                   | ✓       |
| State pension                           | -                   |         | -                   |         | ✓                   | ✓       | ✓                   | ✓       |
| Widow's pension                         | -                   |         | -                   |         | ✓                   | ✓       | ✓                   | ✓       |
| War disablement allowance               | -                   |         | -                   |         | ✓                   | ✓       | ✓                   | ✓       |
| Severe Disablement Allowance            | -                   |         | -                   |         | ✓                   | ✓       | ✓                   | ✓       |
| Disability Allowance                    | -                   |         | -                   |         | ✓                   | ✓       | ✓                   | ✓       |
| Job seekers allowance                   | -                   |         | -                   |         | ✓                   | ✓       | ✓                   | ✓       |

| Income                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Collection at MCS 1 |  | Collection at MCS 2 |  | Collection at MCS 3 |   | Collection at MCS 4 |   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--|---------------------|--|---------------------|---|---------------------|---|
| Pension credit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | -                   |  | -                   |  | ✓                   | ✓ | ✓                   | ✓ |
| Income support                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | -                   |  | -                   |  | ✓                   | ✓ | ✓                   | ✓ |
| Incapacity benefit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -                   |  | -                   |  | ✓                   | ✓ | ✓                   | ✓ |
| Working tax credit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ✓                   |  | ✓                   |  | ✓                   | ✓ | ✓                   | ✓ |
| Child tax credit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                     |  | ✓                   |  | ✓                   | ✓ | ✓                   | ✓ |
| Child care tax credit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | ✓                   |  | ✓                   |  |                     |   |                     |   |
| Statutory sick pay                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -                   |  | -                   |  | ✓                   | ✓ | ✓                   | ✓ |
| Grant from the social fund for maternity expenses                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | -                   |  | -                   |  | ✓                   | ✓ | ✓                   | ✓ |
| Other social fund grant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | -                   |  | -                   |  | ✓                   | ✓ | ✓                   | ✓ |
| Maternity Allowance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | -                   |  | -                   |  | ✓                   |   | ✓                   |   |
| Statutory Maternity Allowance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                     |  | -                   |  | ✓                   |   | ✓                   |   |
| Child Maintenance Payment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | *                   |  | *                   |  | ✓                   |   | ✓                   |   |
| Other Regular Payments**                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ✓                   |  | ✓                   |  | ✓                   |   | ✓                   |   |
| ✓ = Collected in full with respondents reporting amount.<br>- = Partially collected, no amount given.<br>* = Collected as an option of other regular payments.<br>** = Includes: education grants/student shops or work; training/government training scheme; employers maternity/paternity pay; maintenance allowance or other regular; regular cash help from parents; regular cash help from other relatives; rent from boarders, lodgers or sub-tenants; other income from organisations; pension from a former employer; income from investments; allowance for a foster child. Monthly amount given as a total from these sources. |                     |  |                     |  |                     |   |                     |   |

## Methods used to generate/impute income data

As in most studies, there is an amount of item non-response with the income data (see below for more details). For this reason, the banded data were used to impute missing data.

## Banded data

Respondents were shown a card with weekly, monthly and annual bands of total take-home income from all these sources and earnings after tax and other deductions. These 'sources' implicitly included state benefits, which had been the subject of more detailed previous questions. Note that, unlike other state benefits, there was no attempt to ascertain the amounts of housing benefit and council tax benefit received as separate components, so they may well have been omitted from

estimates of total net income as reported. Bands of different sizes were used for lone and 'couple' families.

### Missing income data (item non-response)

Analysis of the collected data shown in Table 33 indicates that more than 1,500 of MCS families, at each sweep, do not provide banded income data either by saying they didn't know their family income or refusing.

**Table 33: Completeness of MCS banded household net income data (number of families)**

|                                  | MCS1   | MCS2*  | MCS3   | MCS4   |
|----------------------------------|--------|--------|--------|--------|
| Missing income data (refusal)    | 482    | 439    | 673    | 510    |
| Missing income data (don't know) | 1,092  | 1,875  | 956    | 1,069  |
| Observed number of families      | 18,552 | 15,590 | 15,246 | 13,857 |

\* There were 144 families at MCS2 where there was no response to the banded income question.

### Imputation of missing and continuous income from banded data

We imputed income for the cases where it was missing using interval regression (Stewart 1983). This method allowed us also to impute a continuous value within a band where income band was available, rather than assuming that all cases in a band had the same midpoint income. This was achieved using Stata's INTREG command (StataCorp 2007; Conroy 2005). INTREG fits a model of  $y=[\text{dependent variable 1, dependent variable 2}]$  on *independent variables* where in our case, *dependent variable 1* was the log lower income band and *dependent variable 2* was log upper band. The bounds of the lowest and topmost bands are respectively considered to be left and right censored. The predictors at each sweep were variables in Table 34, as measured at that sweep.

**Table 34: Variables used to impute income using interval regression**

| Variable                           | Categories                              |
|------------------------------------|-----------------------------------------|
| Main respondent's age at interview | 20-29<br>30-34<br>35-49<br>40-44<br>45+ |
| Housing tenure                     | Own<br>Private renting                  |

| Variable                                                               | Categories                                                                                                                                                                                                                            |
|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                        | Renting from Local Authority or Housing Association<br>Other                                                                                                                                                                          |
| DV combined labour market status of main and partner respondents       | Both in work/on leave<br>Main in work/on leave, partner not in work/on leave<br>Partner in work/on leave, main not in work/on leave<br>Both not in work/on leave<br>Lone parent in work/on leave,<br>Lone parent not in work/on leave |
| Point type                                                             | Advantaged<br>Disadvantaged<br>Ethnic                                                                                                                                                                                                 |
| DV interview government office region                                  | North East<br>North West<br>Yorkshire and the Humber<br>East Midlands<br>West Midlands<br>East of England<br>London<br>South East<br>South West<br>Wales<br>Scotland<br>Northern Ireland                                              |
| Receipt of state benefit?                                              | No<br>Yes                                                                                                                                                                                                                             |
| Main respondent's ethnic group – 6 category census classification (UK) | White<br>Mixed<br>Indian<br>Pakistani and Bangladeshi<br>Black or Black British<br>Other ethnic group (inc. Chinese and other Asian)                                                                                                  |
| DV combined education highest NVQ                                      | NVQ level 1<br>NVQ level 2<br>NVQ level 3<br>NVQ level 4<br>NVQ level 5<br>Overseas qual only<br>None of these                                                                                                                        |
| Main type of accommodation                                             | A house or bungalow<br>A flat or maisonette<br>A studio flat                                                                                                                                                                          |
| Number of children                                                     | 1                                                                                                                                                                                                                                     |

| Variable                                  | Categories                             |
|-------------------------------------------|----------------------------------------|
| including cohort child                    | 2<br>3<br>4+                           |
| DV summary of parents/carers in household | Two parents/carers<br>One parent/carer |

## Equivalisation

We used modified OECD scales for equivalisation, rather than McClements (HBAI Team 2005) which were used in earlier HBAI tables and in some MCS reports. Each scale sets the family's needs relative to those of a couple with no children whose scale is set equal to 1. In the modified OECD scale, a family of one parent and one child under 14 has a scale of 0.87; one parent and two such children 1.07; and so on. A comparison of the two scales is shown in Table 35.

**Table 35: OECD household equivalence scales**

| Equivalence scales before housing cost                               | OECD scale |
|----------------------------------------------------------------------|------------|
| First adult (Main respondent)                                        | 0.67       |
| Spouse                                                               | 0.33       |
| Dependent child age between 14<=18 years old (16<=18 for McClements) | 0.33       |
| Child aged under 14 years (<16 for McClements)                       | 0.20       |

\* The user guides to initial findings of MCS 1 and MCS 2 used a simplified version of this scale where all children under 16 years were given a score of 0.23

## References

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## 8. Variable Names and Labels

### Original variable names

The MCS1 data were initially deposited with variable names that had been automatically allocated by the CAPI program (Blaise 4). Within Blaise, each question has a variable name, made up of a maximum of 8 characters, and this is used to determine the variable name on the dataset. The original allocation of variable names within the CAPI program was based on the use of mnemonics with the first character (A-L) indicating the CAPI 'module' (or 'section') within the instrument. The exception is the proxy modules for which variables begin with 'px' or 'prox'. The modules and associated characters are summarised below:

- Module A: Non-resident parents
- Module B: Father's involvement with baby
- Module C: Pregnancy, labour and delivery
- Module D: Baby's health and development
- Module E: Childcare
- Module F: Grandparents and friends
- Module G: Parents' health
- Module H: Self-completion
- Module J: Employment and education
- Module K: Housing and local area
- Module L: Interests and time with baby.

Where the question was repeated (e.g. the same question asked for each person in the household, baby, illness, hospital admission, etc, reported), Blaise automatically allocated a number suffix (e.g. name, name2, name3, name4). Unfortunately, where the variable name in the Blaise program was originally more than 6 characters long, Blaise truncated the name to allow for the suffix. As a result, there was not always a simple match between the Blaise program documentation and the data. For example, the suffix 2 in the case of some variables indicated the second child within a set of multiple births and in others it represents the partner's answer to a question also put to the main respondent.

### Revised Variable names

With the MCS2 deposit the MCS1 variables have been renamed in a longitudinally consistent way, which will be maintained as further sweeps of the survey accumulate. The explanation of relationship between question names (in the questionnaire) and variable names (in the data) is given below:

Each question name in the instrumentation is made up of 4 letters.

Each variable name in the data is 8 characters long – made up of the 4-letter question name (e.g. ETHE), 2 single-letter prefixes and 2 single-character suffixes as follows:

[Prefix1] [Prefix2] [Question name] [Suffix1] [Suffix2];

where:

Prefix1: Indicates the sweep; a= MCS1; b=MCS2; c=MCS3; and so on.

Prefix2: Identifies the instrument/respondent thus:

- m = Main respondent
- p = Partner respondent
- x = Proxy interview
- n = Question exclusive to 'new families'
- h = Household module completed by Main or Partner respondent
- d = Derived
- y = Physical measurements
- c = Cognitive assessments
- a = Survey administrative data.

Question name – the 4-letter question name in the instrumentation.

Suffix1: Identifies the iteration, i.e. where the same question is repeated for different events/individuals, 0=no iteration; a=iteration 1; b=iteration 2; c=iteration 3; and so on.

Suffix2: Identifies a multi-coded variable – i.e., where a single question produces more than one answer, 0=no multi-code; a=answer 1; b=answer 2; c= answer 3; and so on.

Hence, the variable names on the dataset have the following form:

[Sweep] [Instrument] [Question name] [Iteration] [Multi-coding]

E.g. bmfcin00 holds information from MCS2 (b), given by the main respondent (m), in response to a question named 'fcin', which was not repeated (0) or multi-coded (0).

N.B.: It is important to remember that, due to the iterations and multi-coded responses, questions can be associated with more than one variable. These different variables are distinguished by the suffixes. If questions are not repeated or multi-coded the suffixes will be represented by zeros; otherwise letters of the alphabet are used, as shown below.

It also follows that questions can be classified into:

- a) not repeated, single-coded
- b) not repeated, multi-coded
- c) repeated for child, not event, single-coded
- d) repeated for child, not event, multi-coded
- e) repeated for event, not child, single-coded
- f) repeated for child, not event, multi-coded
- g) repeated for child and event, single-coded
- h) repeated for child and event, multi-coded.

Further guidance is given below.

Please note, the prefixes identifying sweep and instrument are not included in the variable names given below.

**Table 36: Conventions for Suffixes in Variable Names**

| <b>Question Type</b>                        | <b>Suffix 1 – indicating iterations</b>                                                                                                                                                                                                                                                                                                                | <b>Suffix 2 – indicating multiple coding</b>          |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Not repeated, single-coded                  | 0                                                                                                                                                                                                                                                                                                                                                      | 0                                                     |
| Not repeated, multi-coded                   | 0                                                                                                                                                                                                                                                                                                                                                      | 'a'-'z' depending on the number of possible responses |
| Repeated for child, not event, single-coded | 'a' for the first child, 'b' for the second and 'c' for the third. The maximum number of cohort children is 3.                                                                                                                                                                                                                                         | 0                                                     |
| Repeated for child not event, multi-coded   | 'a' for the first child, 'b' for the second and 'c' for the third. The maximum number of cohort children is 3.                                                                                                                                                                                                                                         | 'a'-'z' depending on the number of possible responses |
| Repeated for event, not child, single-coded | Starting with 'a' for the first event and using subsequent letters of the alphabet for successive events.                                                                                                                                                                                                                                              | 0                                                     |
| Repeated for event not child, multi-coded   | Starting with 'a' for the first event and using subsequent letters of the alphabet for successive events.                                                                                                                                                                                                                                              | 'a'-'z' depending on the number of possible responses |
| Repeated for event and child, single-coded  | Starting with 'a' for the first event for child 1 and using subsequent letters of the alphabet for successive events. The first event for child 2 will be indicated by the next letter of the alphabet after that used for the last event for child 1, and so on. In this situation the letters will not indicate which child the variable relates to. | 0                                                     |
| Repeated for event and child, multi-coded   | Starting with 'a' for the first event for child 1 and using subsequent letters of the alphabet for successive events. The first event for child 2 will be indicated by the next letter of the alphabet after that used for the last event for child 1, and so on. In this situation the letters will not indicate which child the variable relates to. | 'a'-'z' depending on the number of possible responses |

Specific examples taken from the questionnaire are shown below.

1. *Not repeated, single-coded variable name with suffixes*

|                                                                                                                                                                                                                                                                                                                                |                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <p>FCIN</p> <p>Now a few questions about yourself. Can I just check: what is your current legal marital status?</p> <p>PRIORITY CODE:</p> <p>1. Legally separated</p> <p>2. Married, 1st and only marriage</p> <p>3. Remarried, 2nd or later marriage</p> <p>4. Single, never married</p> <p>5. Divorced</p> <p>6. Widowed</p> | <p>One response coded</p> <p>FCIN00</p> <p>FCIN00</p> <p>FCIN00</p> <p>FCIN00</p> <p>FCIN00</p> <p>FCIN00</p> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|

2. *Not repeated, multi-coded*

|                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <p>SMUS</p> <p>Can I check, do you use tobacco products such as cigarettes, cigars, a pipe or chewing tobacco at all nowadays? IF YES, PROBE: What do you smoke or use?</p> <p>CODE ALL THAT APPLY:</p> <p>1. No, does not smoke</p> <p>2. Yes, cigarettes</p> <p>3. Yes, roll-ups</p> <p>4 Yes, cigars</p> <p>5. Yes, a pipe</p> <p>6. Yes, chewing tobacco</p> <p>95. Yes, other tobacco product</p> | <p>Up to five responses coded</p> <p>SMUS0a</p> <p>SMUS0b</p> <p>SMUS0c</p> <p>SMUS0d</p> <p>SMUS0e</p> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|

3. *Repeated for child not event, single-coded*

|                                                                                                                                                           |                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <p>OTMA (child1)</p> <p>Do you make regular maintenance payments for this child?</p> <p>1. Yes</p> <p>2. No</p> <p>OTMA (child2)</p> <p>OTMA (child3)</p> | <p>One response coded for each child</p> <p>OTMAa0, OTMAa0</p> <p>OTMAb0, OTMAb0</p> <p>OTMAc0, OTMAc0</p> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|

4. *Repeated for child not event, multi-coded*

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><i>WRBI</i> (child 1)</p> <p>Was there anything wrong with <i>Jack</i> at the time of birth or at any time during the first week?</p> <p>CODE ALL THAT APPLY</p> <p>1. No problems [exclusive code]</p> <p>2. Delay in breathing at birth</p> <p>3. Breathing difficulty or distress in first week</p> <p>4. Jaundice requiring hospital treatment</p> <p>5. Infection or suspected infection</p> <p>94. Other problems in first week</p> <p>95. Other problems at birth</p> | <p>Up to six responses coded for each child</p> <p><i>WRBI</i>aa</p> <p><i>WRBI</i>ab</p> <p><i>WRBI</i>ac</p> <p><i>WRBI</i>ad</p> <p><i>WRBI</i>ae</p> <p><i>WRBI</i>af</p> |
| <p><i>WRBI</i> (child 2)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p><i>WRBI</i>ba</p> <p><i>WRBI</i>bb</p> <p><i>WRBI</i>bc</p> <p><i>WRBI</i>bd</p> <p><i>WRBI</i>be</p> <p><i>WRBI</i>bf</p>                                                 |
| <p><i>WRBI</i> (child 3)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p><i>WRBI</i>cb</p> <p><i>WRBI</i>cc</p> <p><i>WRBI</i>cd</p> <p><i>WRBI</i>ce</p> <p><i>WRBI</i>cf</p> <p><i>WRBI</i>ca</p>                                                 |

5. *Repeated for event not child, single-coded*

|                                                                                                                                               |                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| ADHO (problem 1)                                                                                                                              | One response coded for each event |
| You mentioned that you had <i>bleeding or threatened miscarriage in early pregnancy</i> . Were you admitted to a hospital for this condition? |                                   |
| 1. Yes                                                                                                                                        | ADHOa0<br>ADHOa0                  |
| 2. No                                                                                                                                         | ADHOb0<br>ADHOb0                  |
| ADHO (problem 2)                                                                                                                              | ADHOc0<br>ADHOc0                  |
| ADHO (problem 3)                                                                                                                              | ADHOd0<br>ADHOd0                  |
| ADHO (problem 4)                                                                                                                              | ADHOe0<br>ADHOe0                  |
| ADHO (problem 5)                                                                                                                              | ADHOf0<br>ADHOf0                  |
| ADHO (problem 6)                                                                                                                              | ADHOG0<br>ADHOG0                  |
| ADHO (problem 7)                                                                                                                              | ADHOh0<br>ADHOh0                  |
| ADHO (problem 8)                                                                                                                              | ADHOi0<br>ADHOi0                  |
| ADHO (problem 9)                                                                                                                              | ADHOj0<br>ADHOj0                  |
| ADHO (problem 10)                                                                                                                             |                                   |

6. *Repeated for event not child, multi-coded*

No examples in MCS1.

7. *Repeated for event and child, single-coded*

|                                                                                                                                                       |                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| ACCH (accident 1, child 1)<br>Did [babyn[loop]] go to hospital? IF YES: Was this just to casualty or was [bnheshe[loop]] admitted to a hospital ward? | One response coded for each event |
| 1. No, did not go to hospital                                                                                                                         | ACCHa0<br>ACCHa0<br>ACCHa0        |
| 2. Yes, went to Casualty/Accident and Emergency                                                                                                       | ACCHb0<br>ACCHb0<br>ACCHb0        |
| 3. Yes, was admitted to a hospital ward                                                                                                               | ACCHc0<br>ACCHc0<br>ACCHc0        |
| ACCH (accident 2, child 1)                                                                                                                            | ACCHd0<br>ACCHd0<br>ACCHd0        |
| ACCH (accident 3, child 1)                                                                                                                            | ACCHe0<br>ACCHe0<br>ACCHe0        |
| ACCH (accident 4, child 1)                                                                                                                            | ACCHf0<br>ACCHf0<br>ACCHf0        |
| ACCH (accident 1, child 2)                                                                                                                            | ACCHg0<br>ACCHg0<br>ACCHg0        |
| ACCH (accident 2, child 2)                                                                                                                            | ACCHh0<br>ACCHh0<br>ACCHh0        |
| ACCH (accident 3, child 2)                                                                                                                            | ACCHi0<br>ACCHi0<br>ACCHi0        |
| ACCH (accident 4, child 2)                                                                                                                            | ACCHj0<br>ACCHj0<br>ACCHj0        |
| ACCH (accident 1, child 3)                                                                                                                            | ACCHk0                            |

|                            |                            |
|----------------------------|----------------------------|
| ACCH (accident 2, child 3) | ACCHk0<br>ACCHk0           |
| ACCH (accident 3, child 3) | ACCHI0<br>ACCHI0<br>ACCHI0 |
| ACCH (accident 4, child 3) |                            |

8. *Repeated for event and child, multi-coded*

|                                                                                                      |                                           |
|------------------------------------------------------------------------------------------------------|-------------------------------------------|
| ACWT (accident 1, child 1)                                                                           | Up to four responses coded for each event |
| Thinking about the most severe (or only) accident or injury, what sort of accident or injury was it? |                                           |
| 1. Loss of consciousness/ knocked out                                                                | ACWTaa<br>ACWTab<br>ACWTac<br>ACWTad      |
| 2. Bang on the head/ injury to head without being knocked out                                        | ACWTba<br>ACWTbb<br>ACWTbc<br>ACWTbd      |
| 3. Broken bone or fracture                                                                           | ACWTca<br>ACWTcb<br>ACWTcc<br>ACWTcd      |
| 4. Near drowning                                                                                     | ACWTda<br>ACWTdb<br>ACWTdc<br>ACWTdd      |
| 5. Swallowed household cleaner/other poison/pills                                                    | ACWTea<br>ACWTeб<br>ACWTec<br>ACWTed      |
| 6. Swallowed object                                                                                  | ACWTfa<br>ACWTfb                          |

|                                 |                                      |
|---------------------------------|--------------------------------------|
|                                 | ACWTfc<br>ACWTfd                     |
| 7. Cut needing stitches or glue | ACWTga<br>ACWTgb<br>ACWTgc<br>ACWTgd |
| 8. Injury to mouth or tooth     | ACWTha<br>ACWThb<br>ACWThc<br>ACWThd |
| 9. Burn or scald                | ACWTia<br>ACWTib<br>ACWTic<br>ACWTid |
| 95. Other (please specify)      |                                      |
| ACWT (accident 2, child 1)      |                                      |
| ACWT (accident 3, child 1)      |                                      |
| ACWT (accident 4, child 1)      |                                      |
| ACWT (accident 1, child 2)      |                                      |
| ACWT (accident 2, child 2)      |                                      |
| ACWT (accident 3, child 2)      |                                      |
| ACWT (accident 4, child 2)      |                                      |
| ACWT (accident 1, child 3)      |                                      |
| ACWT (accident 2, child 3)      | ACWTja<br>ACWTjb<br>ACWTjc<br>ACWTjd |
| ACWT (accident 3, child 3)      | ACWTka<br>ACWTkb<br>ACWTkc<br>ACWTkd |
| ACWT (accident 4, child 3)      | ACWTla                               |

|  |                            |
|--|----------------------------|
|  | ACWTIb<br>ACWTIc<br>ACWTId |
|--|----------------------------|

### Original variable labels

The variable labels included on the dataset were also initially derived from the CAPI program. In exporting the SPSS dataset labels based on the wording of questions were automatically allocated. Subsequently, these have been individually reviewed and, where necessary, modified in an effort to ensure that labels are comprehensible and accurate. In particular, prefixes have been assigned to the variable labels which indicate whether the question is from the Household interview (HHGrid), Main interview (Main), Proxy interview (Proxy), Partner interview (Part) or whether the variable is a survey administration variable (Admin).

Again, particular problems occurred where a question was repeated (e.g. the same question was asked for each birth, relationship, job, qualification, etc., reported). When initially created, the CAPI-generated dataset had identical labels for each repeat of the question. In revising these labels, efforts were made to indicate which variables related to which household member, baby, illness, etc.

An example based on the repeated question is: *“Thinking of the baby who does not live with you now, did you have a boy or a girl? Child 1”, “Child 2”, etc.* Identify the first, second, etc., absent child reported.

**Table 37: Original variable labels for repeated questions**

“Thinking of the baby who does not live with you now, did you have a boy or a girl?”

| Respondent             | Child | Variable name | Label                       |
|------------------------|-------|---------------|-----------------------------|
| <b>Main respondent</b> | 1     | Hboygr1       | Main: child’s sex (child 1) |
|                        | 2     | Hboygr2       | Main: child’s sex (child 2) |
|                        | 3     | Hboygr3       | Main: child’s sex (child 3) |
|                        | ...   | etc. up to    |                             |
|                        | 9     | Hboygr9       | Main: child’s sex (child 9) |
| <b>Partner</b>         | 1     | Hboygr11      | Part: child’s sex (child 1) |
|                        | 2     | Hboygr12      | Part: child’s sex (child 2) |
|                        | 3     | Hboygr13      | Part: child’s sex (child 3) |

|  |     |           |                              |
|--|-----|-----------|------------------------------|
|  | ... | etc up to |                              |
|  | 10  | Hboygr20  | Part: child's sex (child 10) |

### Revised variable labels

Variables have been labelled in a consistent manner to aid navigation within the datasets. Labels have abbreviated descriptions to indicate, sweep, instrument and position in loops.

**Table 38: Abbreviations used in Variable Labels**

| Abbreviation | Description                                                                                                                                                                          |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| S1           | Sweep 1                                                                                                                                                                              |
| S2           | Sweep 2                                                                                                                                                                              |
| S3           | Sweep 3                                                                                                                                                                              |
| HHQ          | Household Module                                                                                                                                                                     |
| MAIN         | Main Interview                                                                                                                                                                       |
| PART         | Partner Interview                                                                                                                                                                    |
| PROX         | Proxy Interview                                                                                                                                                                      |
| DV           | Derived Variable                                                                                                                                                                     |
| ADMIN        | Administrative Data                                                                                                                                                                  |
| OS           | Older Siblings                                                                                                                                                                       |
| OSSC         | Older Sibling Self Completion                                                                                                                                                        |
| COG          | Cognitive Assessments, e.g. BAS and Bracken                                                                                                                                          |
| PHYS         | Physical Measurements, e.g. height and weight                                                                                                                                        |
| NA           | Neighbourhood Assessment                                                                                                                                                             |
| MC           | These appear at the end of labels and indicate a multi-coded question, e.g. MC1, MC2, MCS3                                                                                           |
| R            | These appear at the end of labels and indicate an event loop such as pregnancy R1, R2, R3, where R1 means first pregnancy, R2 means second pregnancy, R3 means third pregnancy, etc. |

## **Value labels**

The value labels are also similarly derived from the CAPI program and have similarly been reviewed and, where necessary, modified in an effort to ensure that labels are comprehensible and accurate.

## **9. Feed Forward Data**

Some information was fed forward from earlier sweeps. These variables are available on the Parent Interview Data. The feed forward data were associated with the Main respondent and the Partner respondent from the previous sweep. It was fed forward into the MCS2, MCS3 or MCS4 interview if the interviewer indicated that the Main respondent was the same as at the previous sweep (bhsmar00, chsmar00, dmmdsam00) or that the Partner respondent was the same as at the previous sweep (bhspar00, chspar00, dpdsam00). The name of the Main respondent and Partner at the previous sweep was made available for interviewers. In some cases, the interviewer coded variable is discrepant with the derived variable indicating if the MCS2/3 Main/Partner is the same as at MCS1/2. There are derived variables indicating this (bdmsmi00, bdpsmi00, cdmsmi00, cdpsmi00). This implies that information was fed forward into the Main/Partner interview because the CAPI thought the respondents were the same but in fact the respondents were different.

## **10. Data Cleaning**

### **MCS1 Data Cleaning**

Details of coding and editing activities can be found in the Codebook and Edit Instructions prepared by NatCen, included in this deposit and their Technical Report on Fieldwork (NatCen 2004).

### **MCS2 Data Cleaning**

MCS2 data were received from NOP in SPSS format. The data went through an extensive process of restructuring to produce the current datasets.

Because the Household Grid information was not fed forward from MCS1, construction of the current household grids had to be carried out by a process of matching each individual recorded at MCS2 with that at MCS1. Twenty-eight per cent of individuals did not require matching because the family did not take part in MCS2, either they were new families entering the study for the first time, or younger siblings of the cohort member.

Of the remaining, 37 per cent matched on name, sex and date of birth. Cohort members matching on full name accounted for another 18 per cent (date of birth of cohort members was not re-collected unless there was a discrepancy with that fed forward). Of the remaining 17 per cent, 9 per cent were either new entrants or leavers from the household and 6 per cent matched on full name only. The remaining 3 per cent matched on less reliable measures. The full list of matches was checked by eye to reveal any discrepancies (false positives and false negatives). A fuller analysis of how this compared to the final cleaned data used in the deposited data will appear in due course.

Data for child assessments, child measurements and home observations also needed to be matched as their number in the household was not passed between instruments. As there are only a relatively small number this was done by hand.

### **MCS3 Data Cleaning**

Household grid information was fed forward from the interview at MCS1 and MCS2. Where responses conflicted, the value used was a majority where response from the previous two surveys were available or the latest where only one previous interview was available for comparison. Checks were also applied to investigate implausible or unlikely values, grandmothers under 30, natural siblings more than 30 years apart, etc.

## **MCS4 Data Cleaning**

Essentially the same strategy was employed as at MCS3, but with the addition of more checks on inter-family relationships with an emphasis on relationship to the cohort child and the main and partner respondents.

## **11. Coding and Editing**

### **a) MCS1**

Details of coding and editing activities can be found in the Codebook and Edit Instructions prepared by NatCen, included in this deposit and their Technical Report on Fieldwork (NatCen 2004). Special thanks to Professor Neville Butler who was tireless in developing coding frames for the open-text answers to health questions, and in supervising the ICD10 coding at CLS of responses on mothers' and fathers' longstanding illness.

### **b) MCS2**

Details of coding and editing activities can be found in the NOP Technical Report on Fieldwork (NOP 2006).

### **Recoding of occupation codes**

In 2007, ONS were commissioned to re-code the occupation variables for MCS2.

### **Coding Approach**

#### *Automated Coding*

In total, ONS received 52,868 records. The first stage of coding for ONS was to run the entire sample through its corporate automated coding tool "ACTR" (Automated Coding by Text Recognition). ACTR automatically coded 24,281 records, leaving 28,587 records.

#### *Manual Coding*

The 28,587 records not automatically coded were distributed equally between the coders in ONS, who were asked to make a variety of assumptions, as follows:

- Where the job title is non-descript, code using the job description.
- Where the job title and the job description differ, the record was coded to the job title.
- If the job title is not sufficiently detailed to assign a SOC 2000 code to the unit group (4-digit) level, code to the most detailed level possible.

- In the cases where there are 2 possible codes for the job title and a subjective approach was called for, the code assigned was always to the lowest level. For example, “Armed Forces” were coded to “Other rank” rather than “Officer”.

## Quality Checking

### *Automated Coding*

At present, ACTR is tuned to code an ONS survey, for which the accuracy has been adjudicated as 99.80 per cent. As the MCS is new to ACTR all records coded by ACTR were checked and found to be 98 per cent accurate, with incorrect records manually changed. The reduced quality for the MCS was due to ACTR not being tuned for the survey, as it was the first time ACTR had seen it. Information from the MCS will be used to tune ACTR so the quality of ACTR will be enhanced should the MCS be automatically coded in the future.

### *Manual Coding*

Once the manual coding was completed, a 10 per cent sample of the manually coded records was drawn by the ONS Methodology Division. The sample selected maintained the SOC code distribution, and was checked by someone other than the coder who initially coded the record. Coder accuracy was found to be 95 per cent, with queried records changed where appropriate.

Where there was ambiguity as to how to code a record it was decided that these would be coded after the majority of coding was completed in a “committee” format, all coders discussing and then coding the record together.

The final quality check involved grouping the job titles and showing all the different SOC codes associated with them. This allowed the coders to identify areas of inconsistency and make changes accordingly.

## **c) MCS3**

Details of coding and editing activities can be found in the NatCen Technical Report on Fieldwork (NatCen 2007).

## **d) MCS4**

Details of coding and editing activities can be found in the NatCen Technical Report on Fieldwork (NatCen 2010).

## 12. Weighting in MCS1-4

The sample of births selected for the first survey of the MCS was clustered, geographically, and disproportionately stratified to over-represent areas with high proportions of ethnic minorities in England, residents of areas of high child poverty and residents of the three smaller countries of the UK respectively. The distribution of the cases in the dataset across strata for each country is given in Table 39 below.

Table 39: MCS Cases by stratum and country

|               | England      |               | Wales       |               | Scotland    |               | N Ireland   |               | UK           |               |
|---------------|--------------|---------------|-------------|---------------|-------------|---------------|-------------|---------------|--------------|---------------|
| Strata        | <i>N</i>     | %             | <i>n</i>    | %             | <i>N</i>    | %             | <i>n</i>    | %             | <i>n</i>     | %             |
| Advantaged    | 4828         | 39.49         | 832         | 30.14         | 1145        | 49.04         | 723         | 37.69         | 7528         | 39.12         |
| Disadvantaged | 4806         | 39.31         | 1928        | 69.86         | 1191        | 50.96         | 1200        | 62.31         | 9125         | 47.42         |
| Ethnic        | 2591         | 21.19         | n/a         | n/a           | n/a         | n/a           | n/a         | n/a           | 2591         | 13.46         |
| <b>Total</b>  | <b>12225</b> | <b>100.00</b> | <b>2760</b> | <b>100.00</b> | <b>2336</b> | <b>100.00</b> | <b>1923</b> | <b>100.00</b> | <b>19244</b> | <b>100.00</b> |

The sample design weights or probability weights can be used to correct for MCS cases having unequal probabilities of selection that result from the stratified cluster sample design. The sample weights to be used depend on whether the analysis is confined to data relating to a single country, see Table 40 for country-specific weights, or whether the analysis covers all countries of the UK, see Table 41 for UK weights.

Table 40: MCS sample design weights by stratum and country (weight1)

| Strata        | England | Wales | Scotland | N Ireland |
|---------------|---------|-------|----------|-----------|
| Advantaged    | 1.32    | 1.77  | 1.23     | 1.41      |
| Disadvantaged | 0.71    | 0.65  | 0.75     | 0.76      |
| Ethnic        | 0.24    |       |          |           |

**Table 41: MCS sample design weights by stratum for the UK (weight2)**

| Strata        | England | Wales | Scotland | N Ireland |
|---------------|---------|-------|----------|-----------|
| Advantaged    | 2.00    | 0.62  | 0.93     | 0.47      |
| Disadvantaged | 1.09    | 0.23  | 0.57     | 0.25      |
| Ethnic        | 0.37    |       |          |           |

Further details are included in *The Millennium Cohort Study: Technical Report on Sampling*, 4th Edition. Plewis, I. (Ed.) July 2007.

Two variables have been included on the dataset to facilitate such weighting by providing the sample weights attached to each case. These are:

- **weight1:** This variable should be used when your analysis is within one country only.
- **weight2:** This variable should be used when your analysis covers the whole of the UK.

### **Weighting (including non-response adjustment) for MCS1 and MCS2**

One way of adjusting for possible biases generated by systematic unit non-response is to use non-response weights. Unit non-response in MCS1 and non-response from MCS1 to MCS2 was studied by Plewis (2007). The correlates of non-response for MCS1 and MCS2 were studied and used to produce non-response weights that can be used to adjust for non-response. For MCS2, there are three different types of weights to consider: 1) the sample design weights; 2) the non-response weights at wave 1 which when multiplied by the sample weights produce the overall weights at wave 1 (see Table 11.1 in *The Millennium Cohort Study: Technical Report on Sampling*, 4th Edition. Plewis, I. (Ed.) July 2007); and 3) the non-response weights at sweep 2 which when multiplied by the overall weights at sweep 1 produce the overall weights at sweep 2 (see Table 3 in Plewis (2007) for the mean and standard deviation of these weights by stratum for whole UK analyses as well as further technical details on their calculation). Note that the sample at sweep 2 was supplemented by New Families who were eligible at MCS1, but excluded because their addresses held by the Child Benefit Office were not up to date. For these new families, their non-response weight at sweep 2 is defined to be 1. There were 97 sweep 2 productive families that were not used to generate non-response weights due to missing data on the variables used in the response model. These 97 productive families were given a non-response weight of 1.

All family level weights and response level variables are in a file called:

mcs\_longitudinal\_family\_level\_information. (The user needs to link this file to other files.)

The relevant variable names and value labels are below, where s1, s2, s3 and s4 denote sweeps 1, 2, 3 and 4 respectively and 'inc nr adjustment' denotes including non-response adjustment:

|         |                                                                |
|---------|----------------------------------------------------------------|
| pttyp2  | stratum within country fieldwork point number inc. superwards  |
| weight1 | mcs weight to use on single country analyses                   |
| weight2 | mcs weight to use on whole uk analyses                         |
| aovwt1  | s1: overall weight (inc nr adjustment) single country analysis |
| aovwt2  | s1: overall weight (inc nr adjustment) whole uk analysis       |
| bovwt1  | s2: overall weight (inc nr adjustment) single country analysis |
| bovwt2  | s2: overall weight (inc nr adjustment) whole uk analysis       |
| covwt1  | s3: overall weight (inc nr adjustment) single country analysis |
| covwt2  | s3: overall weight (inc nr adjustment) whole uk analysis       |
| dovwt1  | s4: overall weight (inc nr adjustment) single country analysis |
| dovwt2  | s4: overall weight (inc nr adjustment) whole uk analysis.      |

### **Weighting (including non-response adjustment) for MCS3**

Weighting methods to compensate for attrition are available for monotone patterns of non-response. For a monotone pattern, a sequential weighting procedure is typically used. The longitudinal weight at sweep 1 is defined as the sample (design) weight. For each sweep thereafter, the longitudinal weight is the product of the longitudinal weight at the previous sweep multiplied by a non-response weight for the current sweep. Typically, at each sweep the non-response weight is the estimated inverse of the probability of responding based on a logistic regression model. These logistic models use data from previous sweeps to predict response at the current sweep. However, for non-monotone patterns of non-response, some cases have missing data for previous sweeps and therefore the standard approach cannot be easily applied. For MCS, 1,444 unproductive families at MCS2 were recovered at MCS3, thus yielding a non-monotonic pattern of non-response.

In order to calculate non-response weights for MCS3, multiple imputation was used to impute the required missing data at sweep 2 for the logistic regression model for the probability of responding. With the missing data 'filled in', the pattern of non-response was monotone and then the standard sequential weighting procedure could be used to estimate non-response weights. Note that imputation of missing values was only done for variables found in earlier non-response analyses to be related to non-response, not for all variables in the MCS2 with missing values. Multiple imputation was used to impute missing values at sweep 2 due to unit non-response for unproductive cases and item non-response for productive cases. For example, for the 1,444 unproductive families at MCS2 which were recovered at MCS3, missing housing tenure at MCS2 was imputed using their housing tenure at MCS1 and MCS3 along with other predictor variables in the imputation model. We expect that the imputation of missing values of housing tenure at MCS2 to be 'good'

as the imputation model ‘loosely speaking’ involves ‘interpolation’ of the values at MCS1 and MCS3. Further detail on the non-response predictor variables and imputation models used will be provided in the Second Edition of the Technical Report on Response.

At sweep 3 all families in the MCS ‘active’ sample, the 1,922 families had a non-response adjusted weight at sweep 2 and therefore we didn’t have to deal with missing weights at sweep 2. As a result of using multiple imputation, all 18,526 issued cases were used in the logistic modelling of response at sweep 3. Missing values were imputed 10 times and a logistic model of responding at sweep 3 was estimated 10 times, once for each imputed dataset. This yielded 10 estimated non-response weights at sweep 3 and the weights issued for sweep 3 were the average of the 10 weights. The overall weights, including non-response adjustment, for single country analysis and whole UK analysis are:

|        |                                                                |
|--------|----------------------------------------------------------------|
| covwt1 | s3: overall weight (inc nr adjustment) single country analysis |
| covwt2 | s3: overall weight (inc nr adjustment) whole uk analysis.      |

The minimum, mean, maximum non-response adjusted weight estimates for single country analysis and whole UK analysis for MCS1-3 are given in Tables 42- 43.

Although non-respondents were typically systematically different from respondents at sweeps 1 and 2, Plewis (2007) found that these differences in the probability to respond were small compared to the unequal selection probabilities built into the sample design. The logistic modelling of sweep 3 non-response also found that these differences in the probability to respond were small compared to the unequal selection probabilities built into the sample design. It is, therefore, unlikely that any weighting adjustment for wave 3 non-response would have a substantial effect on most analyses.

### **Weighting (including non-response adjustment) for MCS4**

At sweep 4 we used the same modelling approach and procedures as at sweep 3. In order to calculate non-response weights for MCS4, multiple imputation was used to impute the required missing data at sweep 3 for the logistic regression model for the probability of responding. With the missing data ‘filled in’, the pattern of non-response was monotone and then the standard sequential weighting procedure could be used to estimate non-response weights. Note that imputation of missing values was only done for variables found in earlier non-response analyses to be related to non-response, not for all variables in the MCS3 with missing values.

As a result of using multiple imputation to deal with missing data, all issued cases at sweep 4 were used in the logistic modelling of response at sweep 4. Missing values were imputed 10 times and a logistic model of responding at sweep 4 was estimated 10 times, once for each imputed dataset. This yielded 10 estimated non-response weights at sweep 4 and the weights issued for sweep 4 were the average of the 10

weights. The overall weights, including non-response adjustment, for single country analysis, and whole UK analyses are presented in Tables 42 and 43, respectively.

Table 44 shows design and non-response adjusted weights for the analysis of the Great Britain sample (i.e. excluding Northern Ireland) or for analyses comparing Great Britain to Northern Ireland. The Northern Ireland weights in this table are the same as in Table 41 and are provided in this table for completeness.

## Recommendations

Analysts wanting to estimate population quantities such as population proportions, quantiles, means and totals, should use weighted estimates. Weights with variables ending with a 1 are for UK country specific analyses and those ending with a 2 are for whole UK analyses. For sweep 1 datasets, use the weight variables beginning with 'a'; for sweep 2 datasets, use the weight variables beginning with 'b'; and for sweep 3 datasets, use the weight variables beginning with 'c', e.g. for sweep 3 datasets, use covwt1 for single country estimates or covwt2 for whole UK estimates. These weights take into account the unequal selection probabilities of wards and adjust for non-response. Analysts wanting to estimate coefficients of regression models should include the stratum design variable (pttype2) as dummy variables in the model and unweighted methods can then be used to estimate model parameters. In order to properly estimate standard errors, the clustering of the MCS sample should be taken into account in any analysis by using the STATA svy commands or robust cluster option with PSU identifier if clustering is a nuisance or using a multilevel (hierarchical) model if the clustering is of substantive interest.

**Table 42: MCS1 to MCS4 – Minimum, mean and maximum non-response adjusted weight estimates for country specific analyses\***

| Stratum        | weight1 |       | aovwt1 |       |       |       | bovwt1 |       |      |       | covwt1 |       |      |       | dovwt1 |       |      |       |
|----------------|---------|-------|--------|-------|-------|-------|--------|-------|------|-------|--------|-------|------|-------|--------|-------|------|-------|
|                | mean    | N     | min    | mean  | max   | n     | min    | mean  | max  | n     | min    | mean  | max  | n     | min    | mean  | max  | n     |
| England – ADV  | 1.32    | 4828  | 1.28   | 1.42  | 3.21  | 4617  | 1.08   | 1.31  | 3.27 | 4209  | 1.11   | 1.35  | 3.33 | 4069  | 0.977  | 1.34  | 4.29 | 3788  |
| England – DIS  | 0.71    | 4806  | 0.691  | 0.793 | 1.85  | 4522  | 0.591  | 0.791 | 2.58 | 3880  | 0.608  | 0.807 | 2.65 | 3759  | 0.556  | 0.911 | 6.12 | 3377  |
| England – ETHN | 0.24    | 2591  | 0.247  | 0.296 | 0.693 | 2394  | 0.213  | 0.323 | 1.25 | 1961  | 0.219  | 0.327 | 1.28 | 1889  | 0.198  | 0.399 | 2.09 | 1674  |
| Wales – ADV    | 1.77    | 832   | 1.7    | 1.84  | 3.73  | 832   | 1.45   | 1.75  | 4.88 | 692   | 1.49   | 1.79  | 5.01 | 669   | 1.14   | 1.65  | 5.23 | 622   |
| Wales – DIS    | 0.65    | 1928  | 0.637  | 0.725 | 1.52  | 1928  | 0.546  | 0.737 | 1.65 | 1569  | 0.561  | 0.752 | 1.7  | 1512  | 0.433  | 0.712 | 2.1  | 1396  |
| Scotland – ADV | 1.23    | 1145  | 0.681  | 1.32  | 2.37  | 1145  | 0.598  | 1.29  | 2.64 | 932   | 0.614  | 1.32  | 2.72 | 917   | 0.456  | 1.14  | 5.11 | 829   |
| Scotland – DIS | 0.75    | 1191  | 0.416  | 0.828 | 1.67  | 1191  | 0.354  | 0.887 | 2.22 | 882   | 0.363  | 0.896 | 2.2  | 897   | 0.282  | 0.853 | 2.49 | 799   |
| NI- ADV        | 1.41    | 723   | 0.83   | 1.66  | 3.39  | 723   | 0.712  | 1.65  | 3.4  | 586   | 0.731  | 1.67  | 3.5  | 594   | 0.465  | 1.28  | 6.19 | 534   |
| NI- DIS        | 0.76    | 1200  | 0.438  | 0.917 | 1.68  | 1200  | 0.399  | 1.01  | 2.26 | 879   | 0.41   | 1.03  | 2.32 | 940   | 0.26   | 0.818 | 2.85 | 838   |
| <b>Total</b>   | 0.902   | 19244 | 0.247  | 1.00  | 3.73  | 18552 | 0.213  | 0.989 | 4.88 | 15590 | 0.219  | 1.01  | 5.01 | 15246 | 0.198  | 1.00  | 6.19 | 13857 |

\* *weight1 and observed sample numbers are also provided.*

Note: aovwt1=sweep 1 non-response adjusted weight for analyses of the overall sample; bovwt1, covwt1 and dovwt1 are the corresponding variables for sweep 2, 3 and 4, respectively.

**Table 43: MCS1 to MCS4 – Minimum, mean and maximum non-response adjusted weight estimates for analyses of the whole of UK sample\***

| Stratum        | weight2 |       | aovwt2 |       |       |       | bovwt2 |       |       |       | covwt2 |       |       |       | dovwt2 |       |       |       |
|----------------|---------|-------|--------|-------|-------|-------|--------|-------|-------|-------|--------|-------|-------|-------|--------|-------|-------|-------|
|                | mean    | n     | min    | mean  | max   | n     | min    | mean  | max   | n     | min    | mean  | max   | n     | min    | mean  | max   | n     |
| England – ADV  | 2.0     | 4828  | 1.78   | 1.97  | 4.48  | 4617  | 1.50   | 1.83  | 4.56  | 4209  | 1.54   | 1.88  | 4.64  | 4069  | 1.24   | 1.71  | 5.45  | 3788  |
| England – DIS  | 1.09    | 4806  | 0.975  | 1.12  | 2.62  | 4522  | 0.835  | 1.12  | 3.64  | 3880  | 0.858  | 1.14  | 3.74  | 3759  | 0.716  | 1.18  | 7.88  | 3377  |
| England – ETHN | 0.37    | 2591  | 0.35   | 0.419 | 0.982 | 2394  | 0.302  | 0.461 | 1.77  | 1961  | 0.310  | 0.465 | 1.82  | 1889  | 0.256  | 0.518 | 2.69  | 1674  |
| Wales – ADV    | 0.62    | 832   | 0.549  | 0.593 | 1.20  | 832   | 0.469  | 0.564 | 1.57  | 692   | 0.481  | 0.578 | 1.61  | 669   | 0.387  | 0.558 | 1.78  | 622   |
| Wales – DIS    | 0.23    | 1928  | 0.207  | 0.236 | 0.496 | 1928  | 0.178  | 0.24  | 0.538 | 1569  | 0.183  | 0.244 | 0.553 | 1512  | 0.148  | 0.244 | 0.721 | 1396  |
| Scotland – ADV | 0.93    | 1145  | 0.473  | 0.916 | 1.65  | 1145  | 0.416  | 0.899 | 1.84  | 932   | 0.427  | 0.918 | 1.89  | 917   | 0.358  | 0.898 | 4.02  | 829   |
| Scotland – DIS | 0.57    | 1191  | 0.290  | 0.579 | 1.17  | 1191  | 0.247  | 0.62  | 1.55  | 882   | 0.254  | 0.626 | 1.54  | 897   | 0.223  | 0.674 | 1.97  | 799   |
| NI- ADV        | 0.47    | 723   | 0.255  | 0.509 | 1.04  | 723   | 0.218  | 0.507 | 1.04  | 586   | 0.224  | 0.513 | 1.07  | 594   | 0.186  | 0.513 | 2.48  | 534   |
| NI- DIS        | 0.25    | 1200  | 0.132  | 0.277 | 0.508 | 1200  | 0.121  | 0.306 | 0.683 | 879   | 0.124  | 0.311 | 0.702 | 940   | 0.102  | 0.322 | 1.13  | 838   |
| <b>Total</b>   | 0.998   | 19244 | 0.132  | 1.00  | 4.48  | 18552 | 0.121  | 1.01  | 4.56  | 15590 | 0.124  | 1.02  | 4.64  | 15246 | 0.102  | 1.00  | 7.88  | 13857 |

\* *weight2 and observed sample numbers are also provided.*

Note: aovwt2=sweep 1 non-response adjusted weight for analyses of the overall sample; bovwt2, covwt2 and dovwt2 are the corresponding variables for sweep 2, 3 and 4, respectively.

**Table 44: MCS1 to MCS4 – Minimum, mean and maximum non-response adjusted weight estimates for analyses of the Great Britain (GB) sample or for GB versus Northern Ireland comparisons**

| Stratum        | weightgb |          |      |       | bovwtgb |       |       |       | covwtgb |       |      |       | dovwtgb |       |       |       |
|----------------|----------|----------|------|-------|---------|-------|-------|-------|---------|-------|------|-------|---------|-------|-------|-------|
|                | min      | mean     | max  | n     | min     | mean  | max   | n     | min     | mean  | max  | n     | min     | mean  | max   | n     |
| England – ADV  | 1.78     | 1.78     | 1.78 | 4617  | 1.58    | 1.75  | 3.96  | 4209  | 1.40    | 1.71  | 4.29 | 4069  | 1.13    | 1.59  | 5.83  | 3788  |
| England – DIS  | 0.97     | 0.97     | 0.97 | 4522  | 0.86    | 1.03  | 2.29  | 3880  | 0.765   | 1.06  | 3.77 | 3759  | 0.63    | 1.12  | 6.98  | 3377  |
| England – FTHN | 0.33     | 0.33     | 0.33 | 2394  | 0.294   | 0.387 | 1.08  | 1961  | 0.265   | 0.426 | 1.71 | 1889  | 0.222   | 0.491 | 3.21  | 1674  |
| Wales – ADV    | 0.55     | 0.55     | 0.55 | 832   | 0.491   | 0.549 | 1.04  | 692   | 0.444   | 0.536 | 1.2  | 669   | 0.364   | 0.534 | 1.4   | 622   |
| Wales – DIS    | 0.20     | 0.20     | 0.20 | 1928  | 0.179   | 0.214 | 0.416 | 1569  | 0.161   | 0.219 | 0.75 | 1512  | 0.131   | 0.225 | 0.909 | 1396  |
| Scotland – ADV | 0.82     | 0.82     | 0.82 | 1145  | 0.735   | 0.848 | 1.55  | 932   | 0.665   | 0.828 | 1.80 | 917   | 0.560   | 0.826 | 3.23  | 829   |
| Scotland – DIS | 0.51     | 0.51     | 0.51 | 1191  | 0.457   | 0.582 | 1.19  | 882   | 0.414   | 0.593 | 1.88 | 897   | 0.355   | 0.658 | 2.82  | 799   |
| NI- ADV        | 1.41     | 1.41     | 1.41 | 723   | 0.712   | 1.65  | 3.40  | 586   | 0.732   | 1.68  | 3.5  | 594   | 0.465   | 1.28  | 6.19  | 534   |
| NI- DIS        | 0.76     | 0.76     | 0.76 | 1200  | 0.399   | 1.01  | 2.26  | 879   | 0.41    | 1.03  | 2.32 | 940   | 0.26    | 0.818 | 2.85  | 838   |
| <b>Total</b>   | 0.2      | 0.954914 | 1.78 | 18552 | 0.179   | 1.03  | 3.96  | 15590 | 0.161   | 1.03  | 4.29 | 15246 | 0.131   | 1.00  | 6.98  | 13857 |

\* Non-response adjusted weights at sweep 1 are not provided. Observed sample numbers are also provided.

Note: weightgb= design weights at sweep 1. bovwtgb, covwtgb and dovwtgb are the n on-response adjusted weights at sweep 2, 3 and 4, respectively.

### 13. Geographically Linked Data including IMD & Rural Urban Indicators

#### Index of Multiple Deprivation

The Index of Multiple Deprivation (IMD) measures relative levels of deprivation in small areas. As there is no unified definition for these measures across the UK, these are held as country specific variables. Whilst the IMD definitions are not directly equivalent, they could be broadly compared using the within country deciles.

The deciles were created using the rank for each sub-measure provided. As a practical example, in England there were 32,482 LSOA's, each decile containing 3,248 or 3,249 LSOA's. This data was then linked to address at interview at Lower Super Output Area Level. The IMD measures used were based on the following definitions:

**England:** ODPM Indices of Deprivation 2004

**Wales:** Welsh Assembly Index of Multiple Deprivation 2005

**Scotland:** Scottish Assembly Index of Multiple Deprivation 2004

**Northern Ireland:** NISRA Multiple Deprivation Measure 2005 May 2005.

The websites for ONS, Welsh Assembly, Scottish Assembly and NISRA have specific details:

**England:** [www.communities.gov.uk/documents/communities/pdf/131209.pdf](http://www.communities.gov.uk/documents/communities/pdf/131209.pdf)

**Wales:** <http://wales.gov.uk/topics/statistics/theme/wimd/2005/?lang=en>

**Scotland:** [www.scotland.gov.uk/Publications/2004/10/20089/45181](http://www.scotland.gov.uk/Publications/2004/10/20089/45181)

**Northern Ireland:** [www.nisra.gov.uk/archive/deprivation/nimdm2005fullreport.pdf](http://www.nisra.gov.uk/archive/deprivation/nimdm2005fullreport.pdf)

#### Rural Urban Indicators

MCS postcodes have been classified into different types of rural and urban areas. Again these are country specific. An overview is provided by ONS at <http://www.ons.gov.uk>.

The data for the Rural Urban measures were linked at Lower Super Output Area Level and used the following definitions:

**England and Wales:** ONS Rural Urban Classification (2005)

**Scotland:** Scottish Executive Urban Rural Classification (version 2005–2006)

**Northern Ireland:** (NISRA) Urban Rural Classification 2005.

The Birkbeck definition of Rural Urban in England is that used by DEFRA. More information on this is available from ONS at the above URL.

## **PART EIGHT: ETHICAL CONSIDERATIONS**

### **1. MREC for MCS1**

The process of gaining medical research ethical approval proved a major hurdle. As had been the practice with the previous cohort studies, medical research ethical clearance was sought from the National Health Service Ethical Authority (in February 2001, MREC/01/6/19). This was as a general precaution for future health data collection and was specifically required because of the proposal to involve Health Visitors. Any research involving NHS staff needs to be given such clearance. We were directed to the South West Multi-Centre Research Ethics Committee in March 2001, who felt that opt-out sampling could be coercive and might fail to obtain properly informed consent. They did, however, accept that written opt-ins would tend to exclude vulnerable people, so procedures were devised in consultation with the Committee to give potential respondents more information before they committed themselves for interview. Advance letters introducing the interviewer were sent shortly before her/his first visit and they were asked to arrange interviews generally after their first visit, whose main purpose should be to give information. A simplified information sheet was produced, and translated into several languages.

### **2. MREC for MCS2**

For MCS2, ethical approval was again sought for the pilot and main surveys – on this occasion from the London Multi-Centre Research Ethics Committee. Following their deliberations, the members of the Committee sought additional information on various aspects of the survey, commented on aspects of tracing procedures adopted for families discovered to have moved, and requested that a number of specific changes be made to information leaflets and consent forms. Ethical approval was given in September 2004, MREC/03/2/022).

### **3. MREC for MCS3**

Both pilot surveys and the main survey of MCS3 were considered by the London Multi-Centre Research Ethics Committee of the NHS. Their letter granting a favourable ethical opinion for the *Economic and Social Research Council Millennium Cohort Study Third Survey 2005: Dress Rehearsal and Main Survey 2nd amendment* (12 December 2005) was granted on 15 December 2005, with the REC Reference No. 05/MRE02/46.

### **4. MREC for MCS4**

Both pilot surveys and the main survey of MCS4 were considered by the Northern and Yorkshire Multi-Centre Research *Ethics Committee of the NHS*. Their letter granting a favourable ethical opinion for the *Economic and Social Research Council Millennium Cohort Study Fourth Survey: Dress Rehearsal and Main Survey 2nd amendment* (3 January 2008) was granted on 5 February 2008, with the REC Reference No. 07/MRE03/32.

## 5. Codes of Practice

In order to support our assurances of confidentiality to informants, ethics committees, and government agencies to whose records links are being made, the CLS extended the Cohort Studies Code of Practice to cover all those working with MCS data and developed a Data Security Policy, setting out the secure, isolated computing environment which handles any named data files within CLS.

## 6. Consents

At each sweep of the survey a series of consents were asked of the respondents. These are detailed below.

**Table 45: Consents at each sweep of MCS**

| Survey      | Consent | Who from | Elements                                                                                                                                                                                                                                                                                                              | Document                                                                                                                                                                                                                                                                                                                                                       |
|-------------|---------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MCS1</b> | Verbal  | Parent   | Interview                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                |
|             | Written | Mother   | Maternity & birth linkage<br><br>Linkage to child's National Health Service registration                                                                                                                                                                                                                              | <a href="#">MCS1 Technical Report</a>                                                                                                                                                                                                                                                                                                                          |
| <b>MCS2</b> | Written | Parent   | Child assessments & measurements<br>Older sibling's questionnaire (Aged 10-15)<br><br>Linkage to Child's Records of school performance & attendance<br><br>Linkage to Older Siblings Records of school performance & attendance (Aged 5+)<br><br>Linkage to NHS Medical records (birth to age 7)<br><br>Saliva Sample | <a href="#">MCS2 Technical Report - appendices</a>                                                                                                                                                                                                                                                                                                             |
| <b>MCS3</b> | Written | Parent1  | Main interview & self-completion<br><br>Child assessments & measurements<br><br>Linkage to NHS medical records and accidents (birth to age 7 – IF NOT GOT AT 3)<br><br>Linkage to Foundation Stage Profile (England)<br><br>Teacher postal survey (S, W & NI)                                                         | <a href="#">MCS3 Technical Report on Fieldwork - appendices</a><br>Consent 1: data collection parent 1 and cohort child<br>Consent 1: data collection parent 1 and cohort child<br>Consent 3: cohort child health records<br><br>Consent 4: E cohort child school admin data England<br><br>Consent 4: NSW cohort child school data Ireland<br>Wales Scotland1 |

| Survey      | Consent | Who from    | Elements                                                                                                                                                                                                                                                                                                                                                                                                | Document                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------|---------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             |         |             | Older siblings questionnaire (England)                                                                                                                                                                                                                                                                                                                                                                  | Consent 5: E older sibling questionnaire placement                                                                                                                                                                                                                                                                                                                                                                                                                             |
|             |         |             | Linkage to older siblings school records (Age 7-16)                                                                                                                                                                                                                                                                                                                                                     | Consent 6: E older sibling school records parents                                                                                                                                                                                                                                                                                                                                                                                                                              |
|             | Written | Parent 2    | Partner Interview & self-completion                                                                                                                                                                                                                                                                                                                                                                     | <a href="#">MCS3 Technical Report on Fieldwork - appendices</a><br>Consent 2: data collection parent 2                                                                                                                                                                                                                                                                                                                                                                         |
| <b>MCS4</b> | Written | Main Parent | Main interview & self-completion<br>Linkage to health & economic records<br>Child assessments & measurements<br>Child self-completion questionnaire<br>Child physical activity monitor<br>Class teacher postal survey<br>Child's health records (birth to age 14)<br>Child's educational records (to age 16)<br>Sibling's health records (birth to age 14)<br>Sibling's educational records (to age 16) | <a href="#">MCS4 Technical Report on Fieldwork: appendices</a><br>MCS4_Consent 1: Main respondent<br>MCS4_Consent 1: Main respondent<br>MCS4 Consent 2 Cohort child data<br>MCS4 Consent 3: Cohort child linkage<br>MCS4 Consent 3: Cohort child linkage<br>MCS4 Consent 3: Cohort child linkage<br>MCS4 Consent 3: Cohort child linkage<br>MCS4 Consent 3: Cohort child linkage<br>MCS4 Consent 3: Cohort child linkage<br>MCS4_Consent 4_Siblings<br>MCS4_Consent 4_Siblings |
|             | Written | Partner     | Partner interview & self-completion<br>Linkage to health & economic records                                                                                                                                                                                                                                                                                                                             | <a href="#">MCS4 Technical Report on Fieldwork - appendices</a><br>MCS4 Consent 5: Partner respondent<br>MCS4 Consent 5: Partner respondent                                                                                                                                                                                                                                                                                                                                    |
| <b>MCS5</b> | Written | Main Parent | Main interview & self-completion<br>To approach the child to complete assessments & measurements & child questionnaire (England and Wales – includes permission to approach child's class teacher)<br>To approach the child to complete assessments & measurements & child questionnaire (Scotland and Northern Ireland)                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| Survey | Consent | Who from | Elements                                                                                                                                                                                                                       | Document |
|--------|---------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|        |         |          | Linkage to DWP records                                                                                                                                                                                                         |          |
|        | Written | Partner  | Partner interview & self-completion & linkage to DWP records                                                                                                                                                                   |          |
|        | Verbal  | Child    | Child assessments & measurements<br><br>Child questionnaire<br><br>Approach class teacher for postal survey (England & Wales)<br><br>Child assessments & measurements<br><br>Child questionnaire (Scotland & Northern Ireland) |          |

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